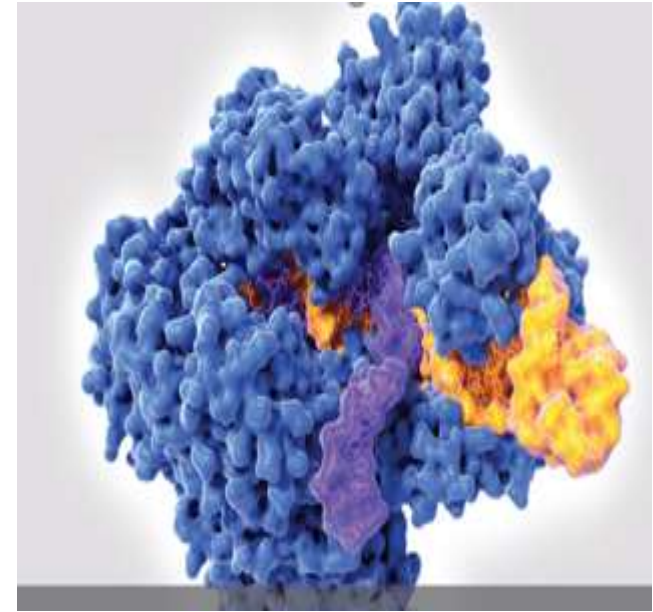


ENZYMES



By: Tamrat Nida
MSc (Medical Biochemistry)
March 2023

Lecture outline

- (1) Introduction to Enzymes
- (2) Classification of enzymes
- (3) How Enzymes Work
- (4) Properties of enzymes
- (5) Enzyme kinetics
- (6) Use of Enzymes in drug synthesis
- (7) Enzymes in clinical diagnosis

General characteristics of enzymes

- ENZYME – Usually a protein, acting as a catalyst in specific biochemical reaction
- Most enzymes are globular proteins

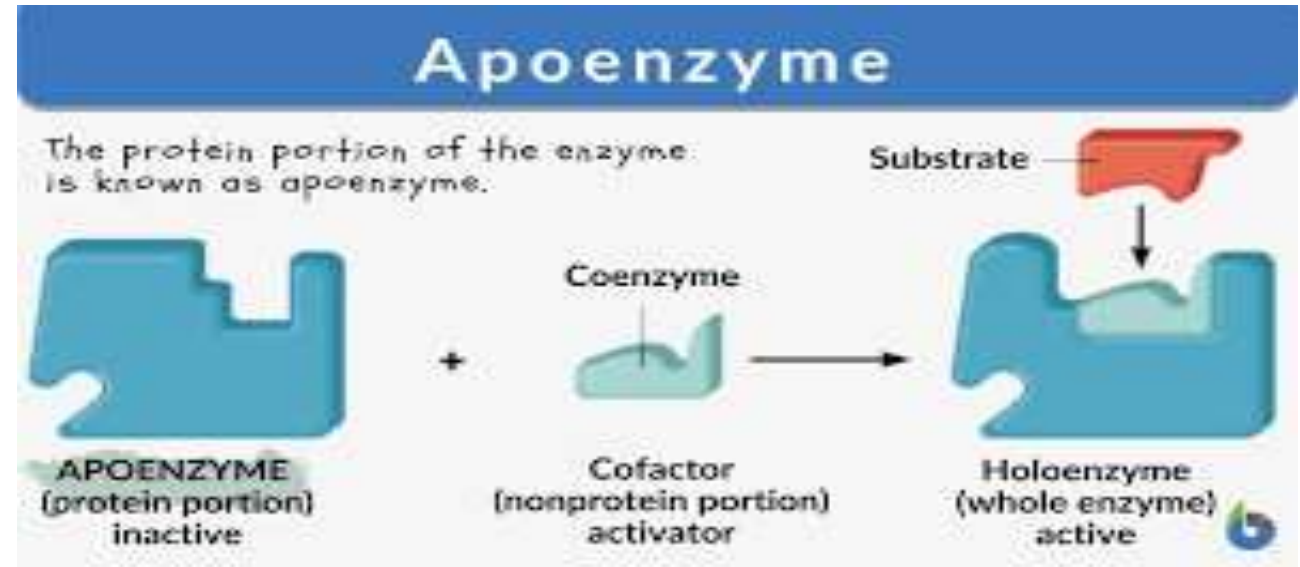
→ A few enzymes are made of RNA, RNA enzyme called a ribozyme → for RNA splicing, Catalyze biochemical reactions involving nucleic acids

Enzymology is the study of enzymes and their **properties, function, regulation, and kinetics**

- Enzymes undergo all the reactions of proteins
- Enzymes denaturation due to pH or temperature change

- **Catalyst**: a substance that **increases the rate** of a reaction by **lowering the activation energy** **without itself being consumed** (changing)
- **Enzymes**: are biological catalysts (protein), **not altered** or consumed during the reaction (**reusable**)
- Each cell in the human body contains **1,000s of different enzymes** –
Every reaction in the cell requires its own specific enzyme

Enzyme structure



- SIMPLE ENZYMES

- Composed only of protein

- CONJUGATED ENZYMES

Composed of:

- Apoenzyme

- Conjugate enzyme without its cofactor

- Protein part of a conjugated enzyme

- Coenzyme (Cofactor)

- Non-protein part of a conjugated enzyme

- The apoenzyme can't catalyze its reaction without its cofactor

- The combination of the apoenzyme with the cofactor makes the conjugated enzyme functional

- Holoenzyme = apoenzyme + cofactor

- The biochemically active conjugated enzyme

➤ Simple enzymes consist of **only proteins** and catalyze their **substrate-specific reactions**. Protein digestive enzyme, **chymotrypsin**, Urease is an enzyme that is only made up of protein and it catalyzes the hydrolysis of urea to ammonia and carbon dioxide (measured as BUN, no direct measurement)

Some enzymes require co-factors (e.g. Hexokinase requires Mg^{2+}) conjugate

- Coenzyme – a complex organic or metalloorganic molecule (may contain a vitamin as a component) e.g. NAD, FAD, TPP(B1), PPL (B6)
- Prosthetic group – coenzyme or metal ion that is very tightly bound (sometimes covalently to the enzyme)
- Apoenzyme – the protein part of an enzyme
- Holoenzyme – enzyme with a cofactor

Coenzymes and cofactors

- Coenzymes provide additional chemically reactive functional groups besides those present in the amino acids of the apoenzymes
 - Are either small organic molecules or inorganic ions
- Metal ions often act as additional cofactors (Zn^{2+} , Mg^{2+} , Mn^{2+} & Fe^{2+})
 - A metal ion cofactor can be bound directly to the enzyme or to a coenzyme
- COENZYME
 - A small organic molecule, acting as a cofactor in a conjugated enzyme
 - Coenzymes are derived from vitamins or vitamin derivatives
 - Many vitamins act as coenzymes, esp. B-vitamins

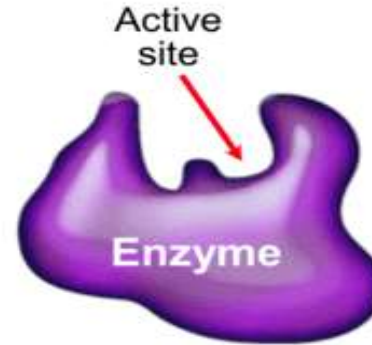
- Two types of co-factor (required for catalysis) → termed **biochemical helper molecules**. (based on the strength of attachment)

1) Co-ENZYME:- the nonprotein components loosely bound to **apoenzyme** by noncovalent bond e.g. – vitamins or compounds derived from vitamins or additional metals

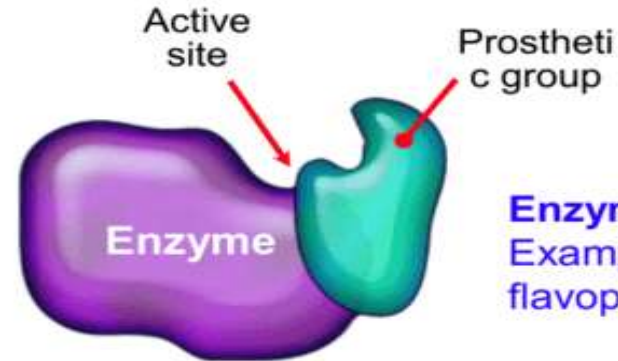
2) Prosthetic group:- the nonprotein components tightly bound to **apoenzyme** by covalent bond

Enzyme Cofactors

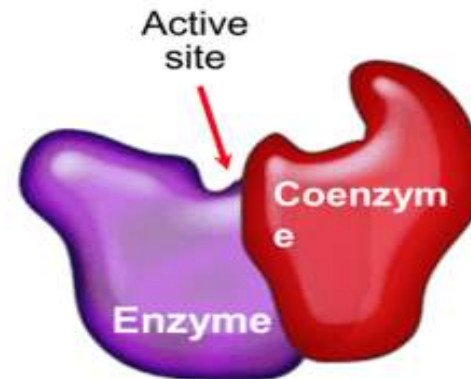
- Some enzymes require **cofactors** to be active.
- Cofactors are a **nonprotein** component of an enzyme. Cofactors can be:
 - organic molecules** (coenzymes).
 - inorganic ions** (e.g. Ca^{2+} , Zn^{2+}).
- Cofactors may be:
 - Permanently attached**, in which case they are called **prosthetic groups**.
 - Temporarily attached coenzymes**, which detach after a reaction, and may participate with another enzyme in other reactions.



Enzyme is protein only
Example: lysozyme

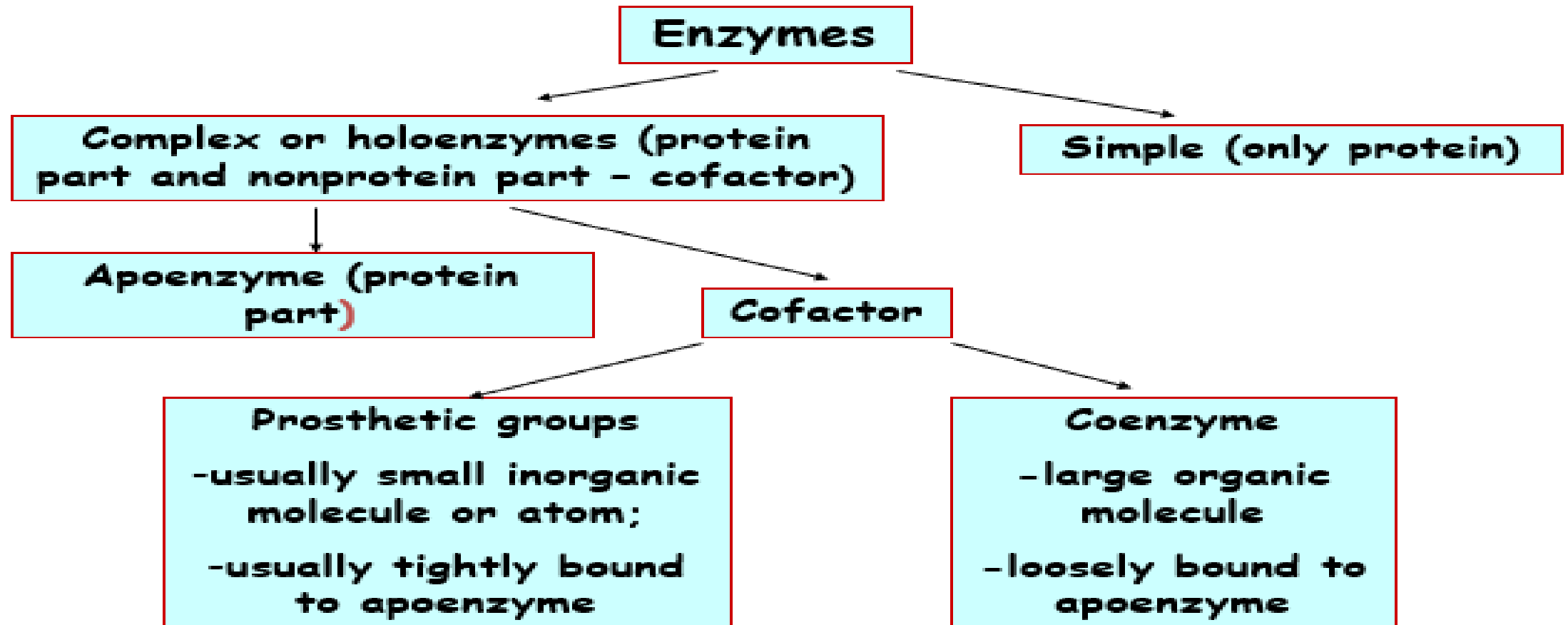


Enzyme + prosthetic group
Example:
flavoprotein + FAD



Enzyme + coenzyme
Example:
dehydrogenases + NAD

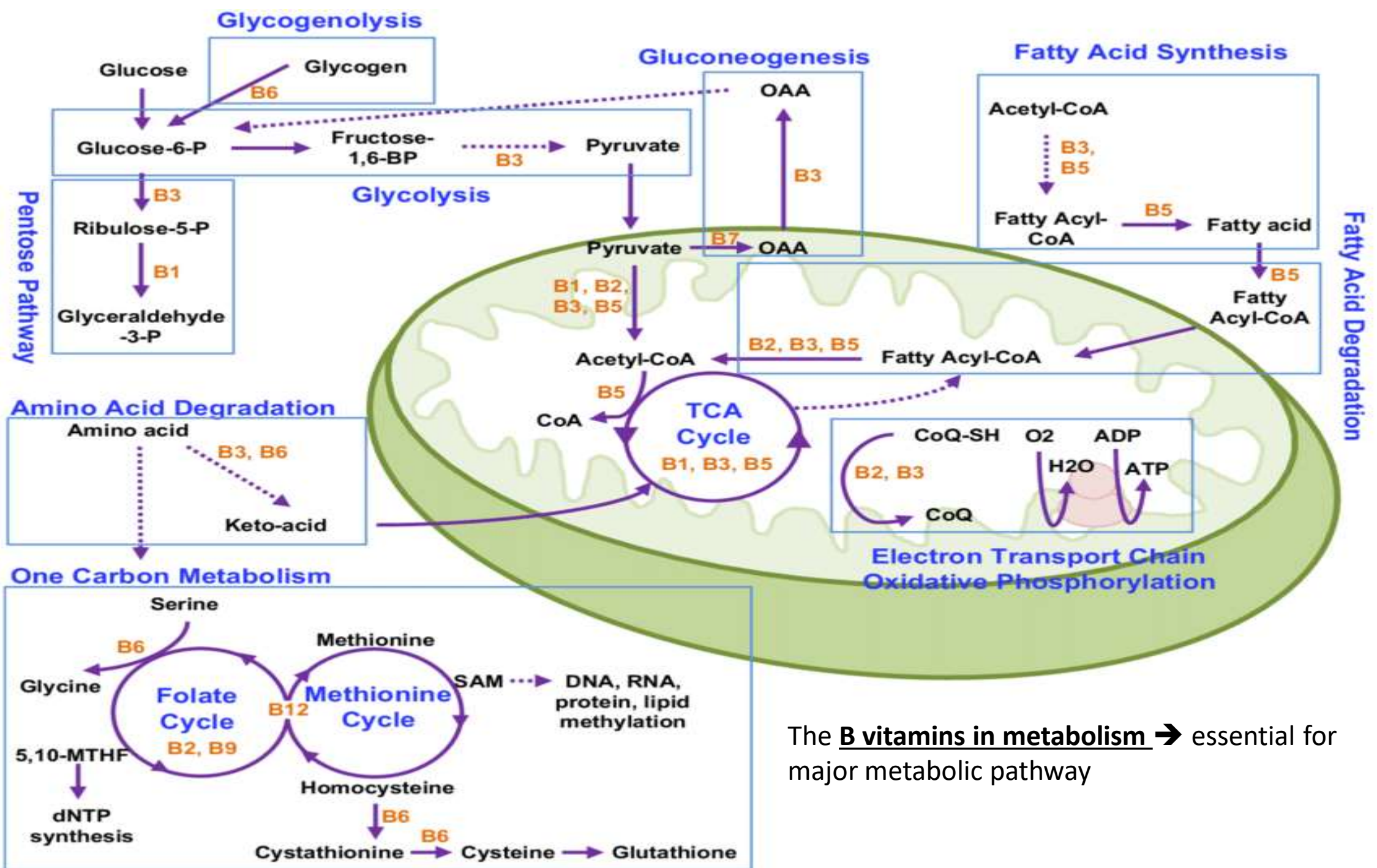
Term	Definition
Enzyme (simple)	Protein only enzyme that facilitates a chemical reaction
Coenzyme	Compound derived from a vitamin (e.g. NAD ⁺) that assists an enzyme in facilitating a chemical reaction
Cofactor	Metal ion (e.g. Mg ²⁺) that that assists an enzyme in facilitating a chemical reaction
Apoenzyme	Protein only part of an enzyme (e.g. isocitrate dehydrogenase) that requires an additional coenzyme to facilitate a chemical reaction (not functional alone)
Holoenzyme	Combination of the apoenzyme and coenzyme which together facilitating a chemical reaction (functional)



Selected important Coenzymes in which B vitamins are present

B Vitamin	Coenzymes	Groups Transferred
thiamin	thiamin pyrophosphate (TPP)	aldehydes
riboflavin	flavin mononucleotide (FMN) flavin adenine dinucleotide (FAD)	hydrogen atoms
niacin	nicotinamide adenine dinucleotide (NAD ⁺) nicotinamide adenine dinucleotide phosphate (NADP ⁺)	hydrogen atoms
pantothenic acid	coenzyme A (CoA)	acyl groups
vitamin B ₆	pyridoxal-5-phosphate (PLP) pyridoxine-5'-phosphate (PNP) pyridoxamine-5'-phosphate (PMP)	amino groups
biotin	biotin	carbon dioxide (carboxyl group)
folate	tetrahydrofolate (THF)	one-carbon groups other than CO ₂
vitamin B ₁₂	methylcobalamin	methyl groups, hydrogen atoms

- Thiamine (Vit B1), Riboflavin (B 2), Niacin (B3)
- Pantothenic acid (also known as vitamin B5 used in the synthesis of CoA and acyl carrier protein
- Biotin (vitamin B7) → from diet or produced by normal flora GI, Raw egg whites contain avidin, a type of sugar protein, which tightly, binds biotin (complex formed) preventing absorption it, required for **fat** and **CHO** breakdown for carboxylation rxn → Pyruvate Carboxylase reaction to form Oxaloacetate from pyruvate, **heme** biosynthesis
- Folate (B-9) → nucleotide synthesis and methionine (SAM)(a.a) methyl group 1C metabolism, for synthesis of thiamine from uracil
- VB-12 → use for brain and nerve, synthesis of RBC (Pernicious anemia)
- NADP → fatty acid synthesis reduction of HMG CoA to mevalonic acid (MVA) by HMG CoA reductase “step inhibited by statin drug”



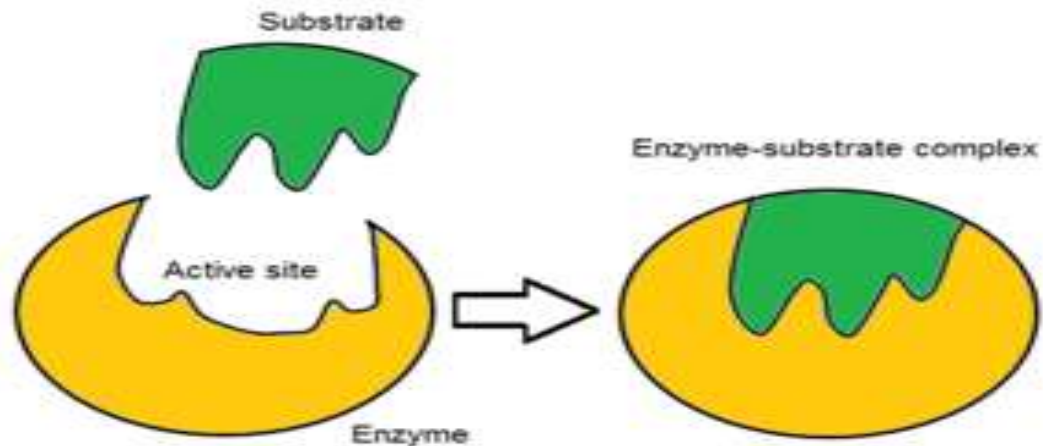
The B vitamins in metabolism → essential for major metabolic pathway

Importance

- Play an important role in metabolism, diagnosis, and therapeutics
- All biochemical reactions are enzyme-catalyzed in the living organism
- Levels of enzymes in the blood are of diagnostic importance: e.g. Good indicators in diseases such as myocardial infarction
- Enzymes can be used therapeutically such as digestive enzyme **supplementation** used clinically in “digestive and absorption disorders”.

Enzyme Nomenclature

- Enzymes are named according to the type of reaction they catalyze and/or their substrate
- Substrate = the reactant upon which the specific enzyme acts
 - Enzyme physically binds to the substrate



- Suffix of an enzyme –ase
 - Lactase, amylase, lipase, or protease
- Some digestive enzymes have the suffix –in
 - Pepsin, trypsin & chymotrypsin
- Prefix denotes the type of reaction the enzyme catalyzes
 - Oxidase: redox reaction
 - Hydrolase: Addition of water to break one component into two parts
- Substrate identity is often used together with the reaction type
 - Pyruvate carboxylase, lactate dehydrogenase

Enzyme Nomenclature

IUBMB → International union of biochemistry and molecular biology, 4-digit commission number

- use E.C.N (Enzyme Commission Number)
- systematic naming by IUBMB

Eg: ATP + D-glucose ^{HK} → ADP + D-glucose-6-P

ATP: glucose phosphotransferase

Trivial name: Hexokinase

E.C. 2.7.1.1 (Commission number for Hexokinase)

(2) – denotes class name-transferase

(7) – denotes subclass-phosphotransferase

(1) – denotes phosphotransferase with a hydroxyl group as an acceptor

(1) denotes D-Glucose as the phosphoryl group acceptor

IUBMB Enzyme List	
EC 1	Oxidoreductases
EC 2	Transferases
EC 3	Hydrolases
EC 4	Lyases
EC 5	Isomerases
EC 5.6	Macromolecule Isomerases
EC 6	Ligases
EC 7	Translocases

new

Hexokinase

Enzyme Commission

EC 2.7.1.1

Class: Transferase

Sub – Class: Transfer of Phosphate

Specific name: ATP,D-Hexose-6-Phosphotransferase (hexokinase)

Sub SubClass: Alcohol group is phosphate acceptor

All type of ENZYME has their own number

Classification of Enzymes

6 Classes of Enzymes are recognized by IUBMB

1) **Oxidoreductases**: transfer of electrons between substrates.

E.g Dehydrogenases, peroxidases, catalase

ALCOHOL DEHYDROGENASE



2) **Transferases**: Catalyze transfer of groups between substrates during biological reactions.

Eg. Kinases, acyltransferases, hexokinase



3) **Hydrolases**: Hydrolysis reactions. E.g. hydroxylases in steroid hormone synthesis e.g. fumarate to malate by a fumarase enzyme

Classification of Enzymes...

4) **Lyases**: Addition of groups to double bonds or formation of double bonds by group removal. E.g. citrate lyase, ATP citrate lyase is responsible for catalyzing the conversion of citrate and Coenzyme A (CoA) to acetyl-CoA and oxaloacetate

5) **Isomerases**: Transfer of groups within a molecule to yield isomeric forms

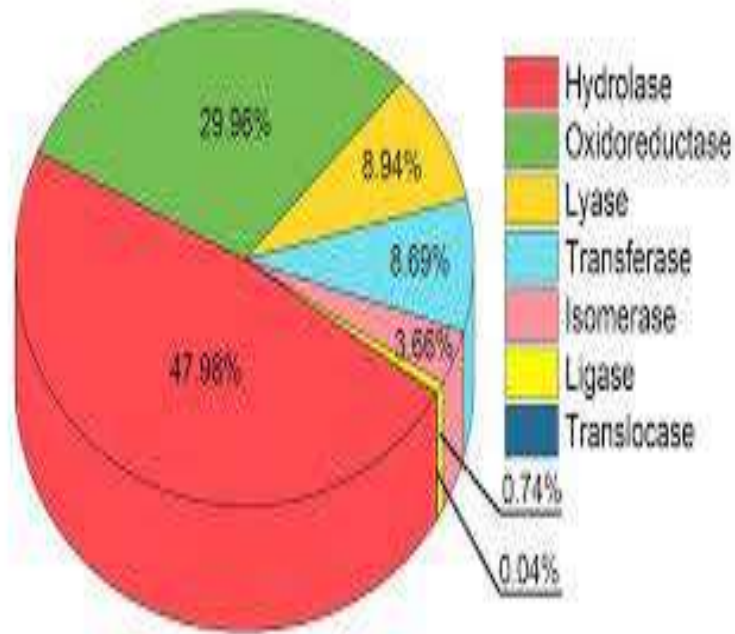
E.g. phosphohexose isomerase

6) **Ligases**: Bond formation C-C, C-O, C-S, C-N

e.g. Glutamine **synthetase**, glutathione **synthetase**, Succinate thiokinase, and..

Acetyl CoA carboxylase (is a **biotin-dependent** enzyme that catalyzes the **irreversible carboxylation** of **acetyl-CoA** to produce **malonyl-CoA**)

7. **Translocase** → the **7th class**, is a protein **that assists in moving another molecule**, usually **across a cell membrane**. These enzymes **catalyze the movement of ions** or **molecules across membranes** or their separation within membranes. (**mitochondrial membrane**)



Group	Reaction Catalyzed	Typical Reaction	Enzyme example with trivial name
EC 1 Oxidoreductases	To catalyze oxidation/reduction reactions; transfer of H and O atoms or electrons from one substance to another.	$AH + B \rightarrow A + BH$ (reduced) $A + O \rightarrow AO$ (oxidized)	Dehydrogenase, oxidase
EC 2 Transferases	Transfer of a functional group from one substance to another. The group may be methyl-, acyl-, amino- or phosphate group	$AB + C \rightarrow A + BC$	Transaminase, kinase
EC 3 Hydrolases	Formation of two products from a substrate by hydrolysis	$AB + H_2O \rightarrow AOH + BH$	Lipase, amylase, peptidase
EC 4 Lyases	Non-hydrolytic addition or removal of groups from substrates. C-C, C-N, C-O or C-S bonds may be cleaved.	$RCO_2COOH \rightarrow RCOH + CO_2$ or $[X-A-B-Y] \rightarrow [A=B + X-Y]$	Decarboxylase
EC 5 Isomerases	Intramolecule rearrangement, i.e. isomerization changes within a single molecule.	$ABC \rightarrow BCA$	Isomerase, mutase
EC 6 Ligases	Join together two molecules by synthesis of new C-O, C-S, C-N or C-C bonds with simultaneous breakdown of ATP.	$X + Y + ATP \rightarrow XY + ADP + Pi$	Synthetase

TABLE 12-1 CLASSIFICATION OF FUNCTIONALLY QUANTITATED ENZYMES

CLASS	RECOMMENDED NAME	COMMON ABBREVIATION	STANDARD ABBREVIATION	EC CODE NO.	SYSTEMATIC NAME
Oxidoreductases	Lactate dehydrogenase	LDH	LDH	1.1.1.27	L-Lactate:NAD ⁺ oxidoreductase
	Glucose-6-phosphate dehydrogenase	G-6-PDH	G-6-PD	1.1.1.49	D-Glucose-6-phosphate:NADP ⁺ 1-oxidoreductase
	Glutamate dehydrogenase	GLD	GLD	1.4.1.3	L-glutamate:NAD(P) oxidoreductase, deaminase
Transferases	Aspartate amino-transferase	GOT (glutamate oxaloacetate transaminase)	AST	2.6.1.1	L-Aspartate:2-oxaloglutarate aminotransferase
	Alanine amino-transferase	GPT (glutamate transaminase)	ALT	2.6.1.2	L-Alanine:2-oxaloglutarate aminotransferase
	Creatine kinase	CPK (creatine phosphokinase)	CK	2.7.3.2	ATP:creatine <i>N</i> -phosphotransferase
	γ-Glutamyl-transferase	GGTP	GGT	2.3.2.2	(5-Glutamyl)peptide: amino acid-5-glutamyltransferase
	Glutathione-S-transferase	α-GST	GST	2.5.1.18	Glutathione transferase
	Glycogen phosphorylase	GP	GP	2.4.1.1	1,4-α-D-Glucan: orthophosphate α-D-glucosyltransferase
	Pyruvate kinase	PK	PK	2.7.1.40	Pyruvate kinase
Hydrolases	Alkaline phosphatase	ALP	ALP	3.1.3.1	Orthophosphoric monoester phosphohydrolase (alkaline optimum)
	Acid phosphatase	ACP	ACP	3.1.3.2	Orthophosphoric monoester phosphohydrolase (acid optimum)
	α-Amylase	AMY	AMS	3.2.1.1	1,4-D-Glucan glucanohydrolase
	Cholinesterase	PCHE	CHE	3.1.1.8	Acylcholine acylhydrolase
	Chymotrypsin	CHY	CHY	3.4.21.1	Chymotrypsin
	Elastase-1	E1	E1	3.4.21.36	Elastase
	5-Nucleotidase	NTP	NTP	3.1.3.5	5'-Ribonucleotide phosphohydrolase
	Triacylglycerol lipase		LPS	3.1.1.3	Triacylglycerol acylhydrolase
Lyases	Trypsin	TRY	TRY	3.4.21.4	Trypsin
	Aldolase	ALD	ALD	4.1.2.13	D-D-Fructose-1,6-bisdiphosphate D-glyceraldehyde-3-phosphate-lyase
Isomerases	Triosephosphate isomerase	TPI	TPI	5.3.1.1	Triose-phosphate isomerase
Ligase	Glutathione Synthetase	GSH-S	GSH-S	6.3.2.3	Glutathione synthase

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Properties of Enzymes

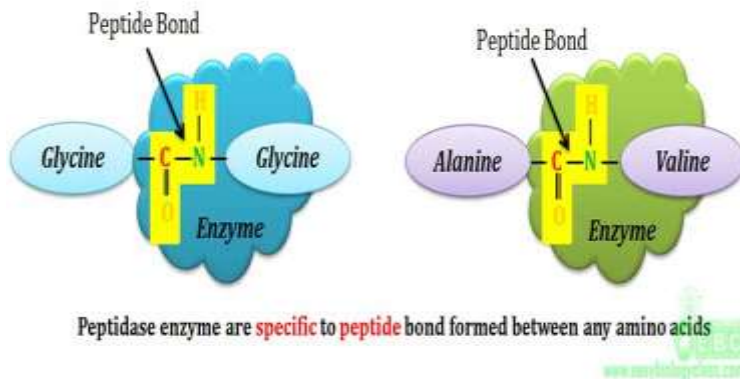
1) **Specificity**: Enzymes are very specific to their substrates (Broad or absolute)

E.g. Hexokinase has broad specificity b/c it acts on hexose sugars (Glucose, Mannose, Fructose)

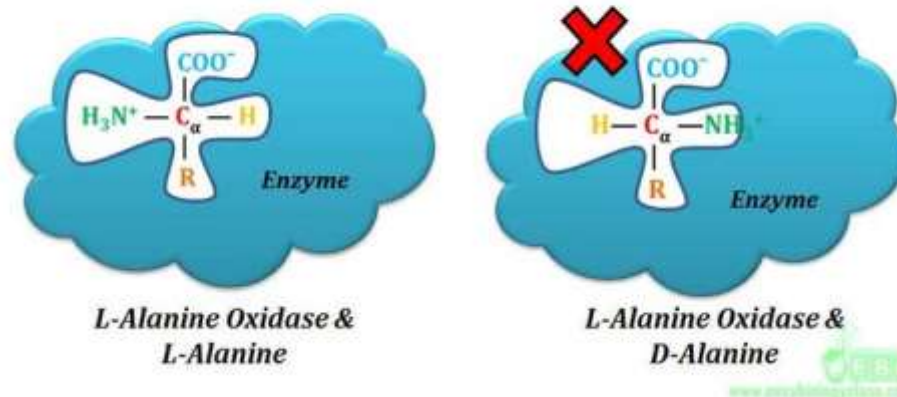
- Glucokinase is specific for only Glucose

Enzymes are also stereospecific and hence may act on only one stereoisomer

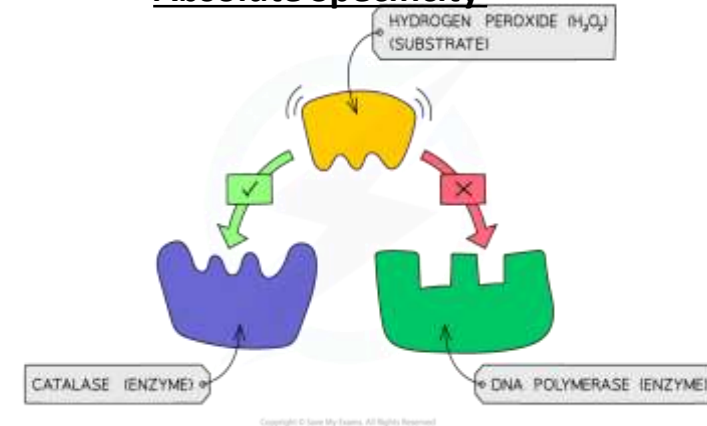
Bond Specificity of Enzyme



Stereo specificity of Enzymes



Absolute Specificity



1. Absolute Specificity – catalyze a particular reaction for only one substrate – “**Most restrictive**” of all specificities •

Not common

e.g. Catalase has **absolute specificity** for hydrogen peroxide (H_2O_2) – *Urease* catalyzes only the hydrolysis of urea - *Glucokinase*

2. Group Specificity – act only on similar substrates that have a **specific functional group**

✓ *Carboxypeptidase* cleaves amino acids one at a time from the carboxyl end, *Hexokinase* adds a phosphate group to hexoses

3. Linkage Specificity – acts on a particular type of chemical bond, “**The most general**” of the enzyme specificities,

e.g. Chymotrypsin (peptide bond), Phosphatases hydrolyze **phosphate–ester bonds** in all types of phosphate esters

4. Stereochemical Specificity – can **distinguish between stereoisomers** – **Chirality is inherent** in an active site

e.g. L-Amino-acid oxidase catalyzes reactions of L-amino acids (common form) but **not of D-amino acid**

Properties of Enzymes...

2) **Turnover number**: Enzymes have a high turnover number i.e. an enzyme molecule can act on many substrate molecules in a reaction

- This simply refers to their catalytic power
- Enzymes do not themselves take part in biological reactions and hence are reproducible and can be recycled during catalysis
- A single enzyme can catalyze millions of d/t rxn, not consumed as a product of “catalytic power” reproducibility
- Reusable (recycled) after completing the rxn

Properties of Enzymes....

3) Regulation: Enzyme activities can be regulated in many ways

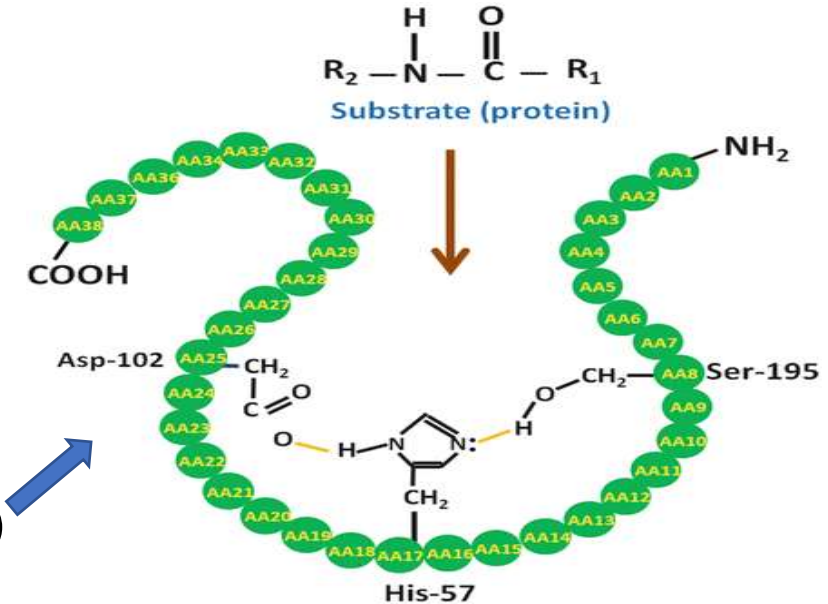
- **Multiple mechanisms** are available to **control** the **activity** of enzymes
- Enzymes can be **regulated** by **other molecules** that **either increase or reduce** their **activity**
 - a. Allosteric regulation
 - b. Feedback inhibition
 - c. Zymogen activation
 - d. Covalent modification
 - e. Induction or repression of enzymes

Properties...

4) Active site: a cleft or pocket formed as the enzyme folds

- Contains amino acid side chains that form complementary structures to the substrate
- Site at which substrate binds
- Mostly polar amino acids like Cys, Ser, Thr, Tyr are found at the active site

- Sometimes amino acids triads are found (Serine proteases)



- E.G. Serine protease → cleave peptide bond, catalytic triad common set of **3 amino acids** found in the active site, Histidine, Aspartate, and serine, cleave peptide bond
- Serine → essential role serves as the nucleophilic amino acid at the (enzyme's) active site, has a "nucleophile OH group" (initiate reaction) (protonation and deprotonation)
- A nucleophile is a reactant that donates a pair of electrons to form a new bond (can form a temporary covalent bond)

Properties...

5) **Distribution**: enzymes are found distributed in different tissues

- ✓ They are found in specific organelles and compartmentalization helps specific reactions to be undergone without interference in a controlled way
- ✓ Some enzymes are tissue specifically expressed, which makes them markers for disease diagnosis
- Within a cell's **compartmentalized organelles**, there are mitochondrial, **cytosol** (glycolytic enzymes), and **Endoplasmic reticulum** (**G-6-phosphatase**) in liver
E.g. **Heme biosynthesis** (8 steps) (**ALA synthase 1st** and **ferro chelatase** (**last**) in **mitochondria**, and **ALA dehydratase 2nd** step form PBGN(in the **cytosol**)
- **ALA synthase** in mitochondria → Condense **succinyl CoA** and **glycine** as a substrate to form **ALA**
- Some are **tissue-specific** (liver enzyme ALT), and used as a diagnostic marker (**Cardiac marker CK-MB**) in MI, also LDH 1 and 2 IE

Properties...

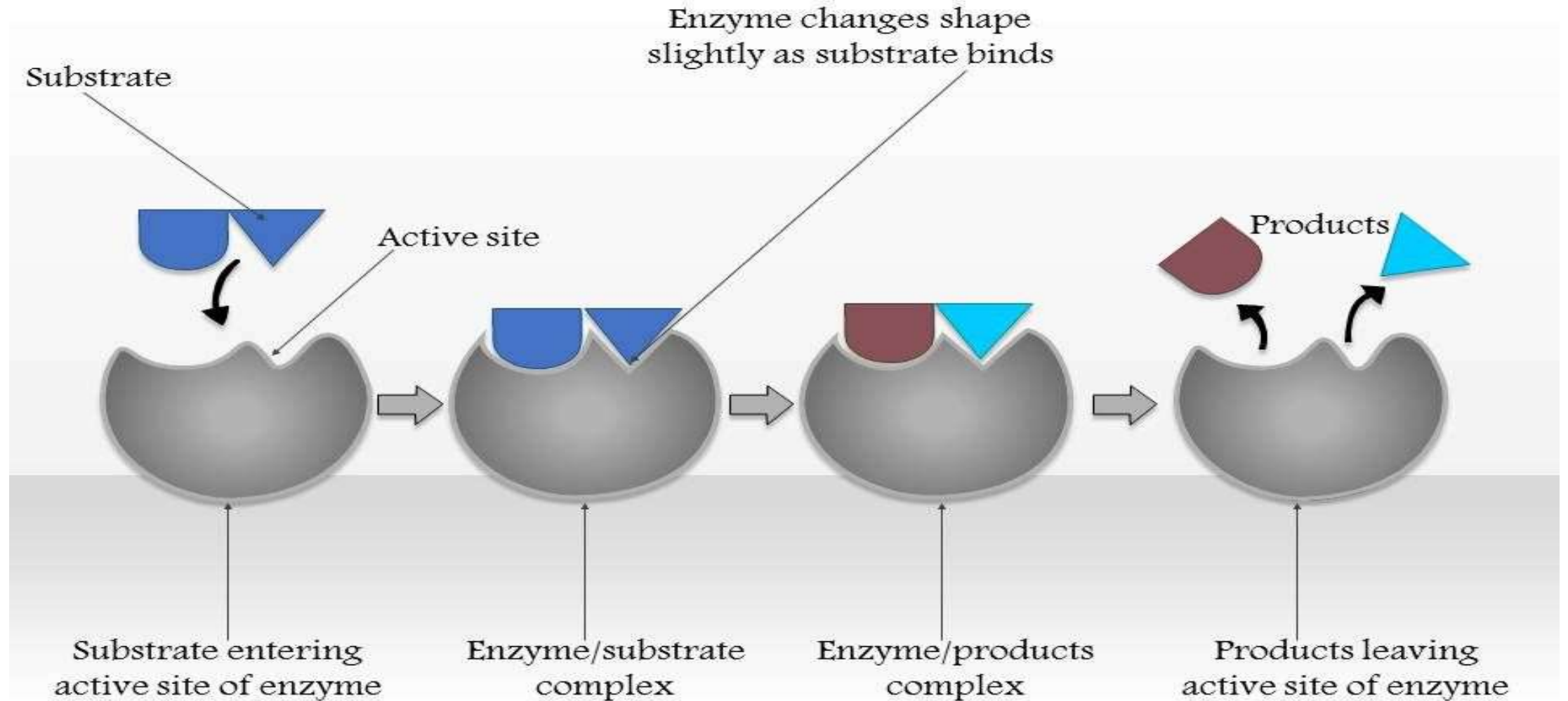
6) **Cofactors**: Enzymes require cofactors needed for catalytic activity. These cofactors may be inorganic metal ions (Mn, Mg, Fe, Se, Zn, etc.) or organic groups called coenzymes

✓ Coenzymes are derived from vitamins [E.g. TPP from thiamine (B1), PLP (B6), NAD (Niacin), FAD (Riboflavin), etc.]

7) **Isoenzymes**: Some enzymes are distributed between tissues as isoenzymes (enzymes that catalyze the same type of reaction)

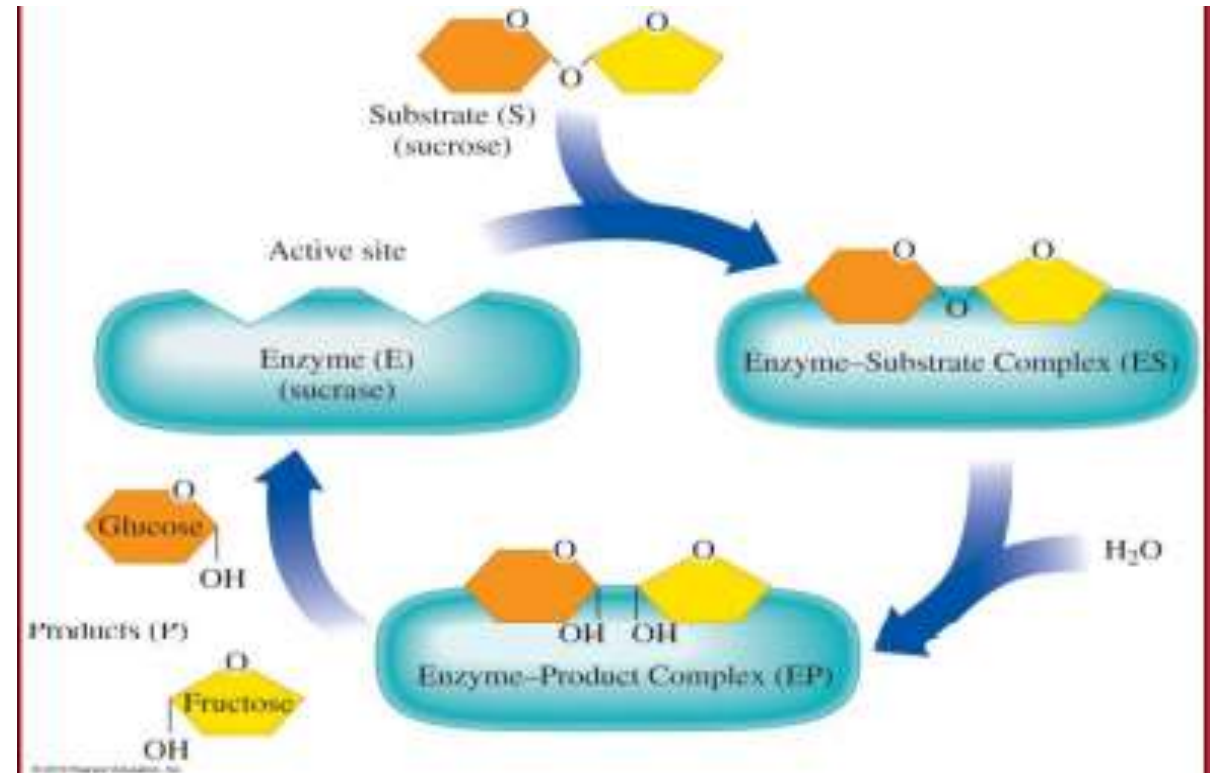
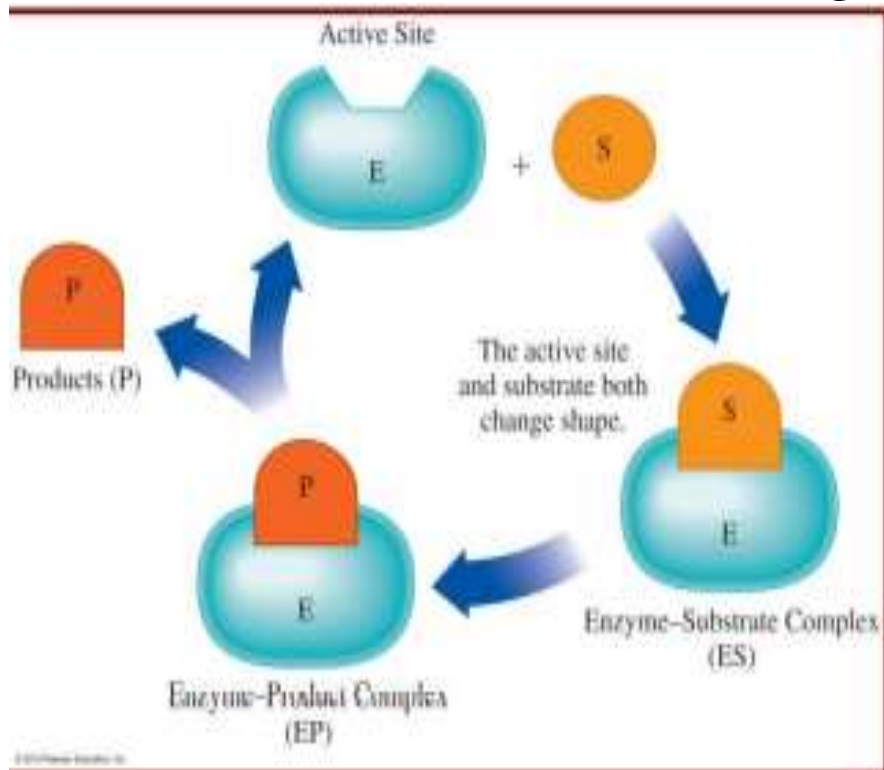
- multiple forms of enzymes differ in locations (tissue) and an amino acid sequence
- Different chemical and physical properties, kinetic (RATE) parameters, or are regulated differently

The Mechanism of enzyme action



Enzyme – Substrate Complex

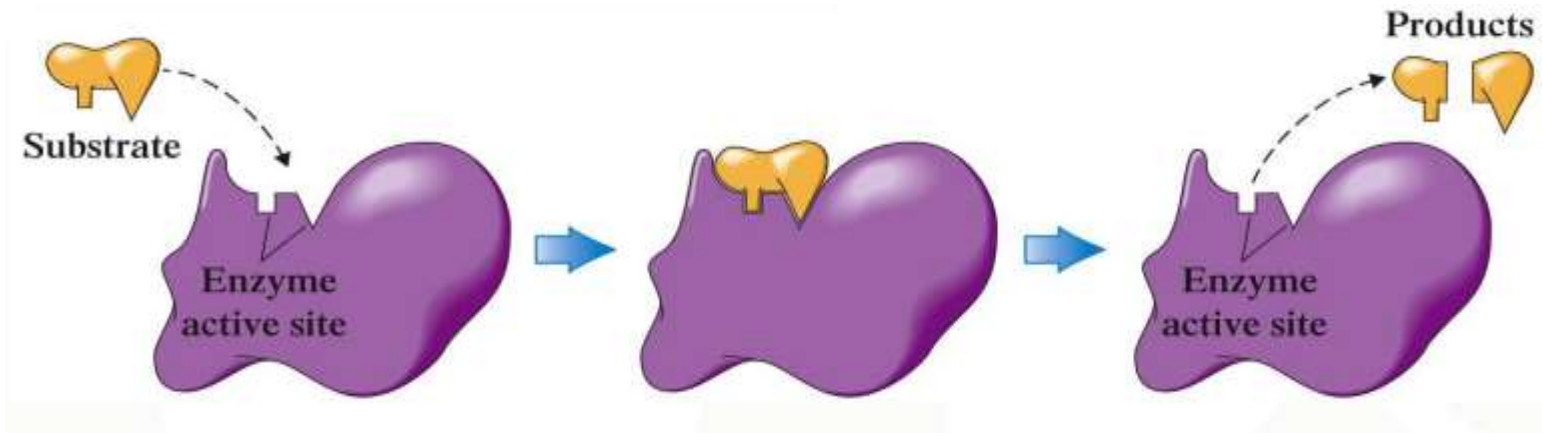
- When the substrate binds to the enzyme active site an Enzyme-Substrate Complex is formed temporarily
 - Allows the substrate to undergo its chemical reaction much faster



Models of Enzyme-Substrate interaction

Lock & Key Model of Enzyme Action

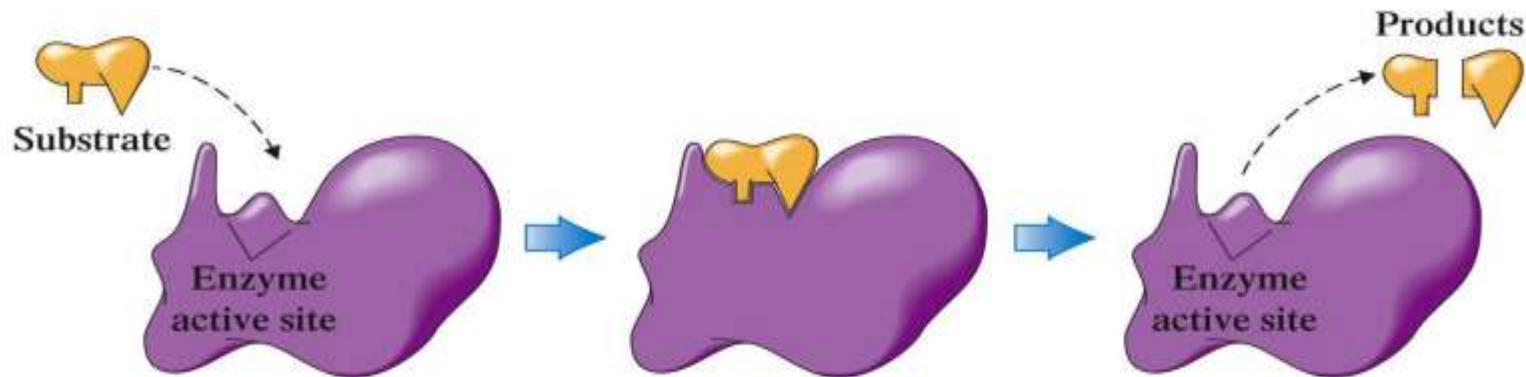
- The active site is fixed, with a rigid shape (LOCK) (similar structure with the substrate)
- The substrate (KEY) must fit exactly into the rigid enzyme (LOCK)
- Complementary shape & geometry between enzyme and substrate
 - Key (substrate) fits into the lock (enzyme)
- Upon completion of the chemical reaction, the products are released from the active site, so the next substrate molecule can bind



Models.....

Induced Fit Model of Enzyme Action

- Many enzymes are flexible & constantly change their shape
 - The shape of the active site changes to accept & accommodate the substrate
 - Conformation change in the enzyme's active site to allow the substrate to bind
 - Analogy: a glove (enzyme) changes shape when a hand (substrate) is inserted into it



This model better explains the **properties of allosteric enzymes** (B/C binding of regulator cause a conformational change to the active site)

Activation energy or Energy of Activation:

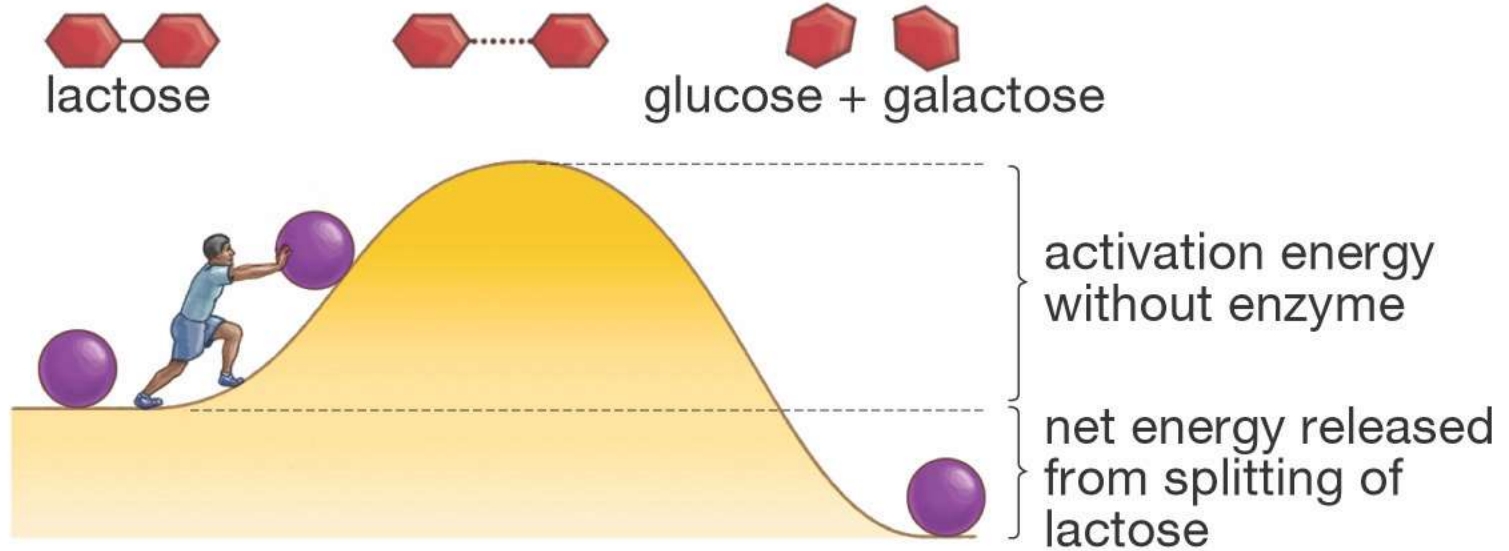
- All chemical reactions require some amount of energy to get them started
OR
- It is the First push to start the reaction

This energy is called activation energy

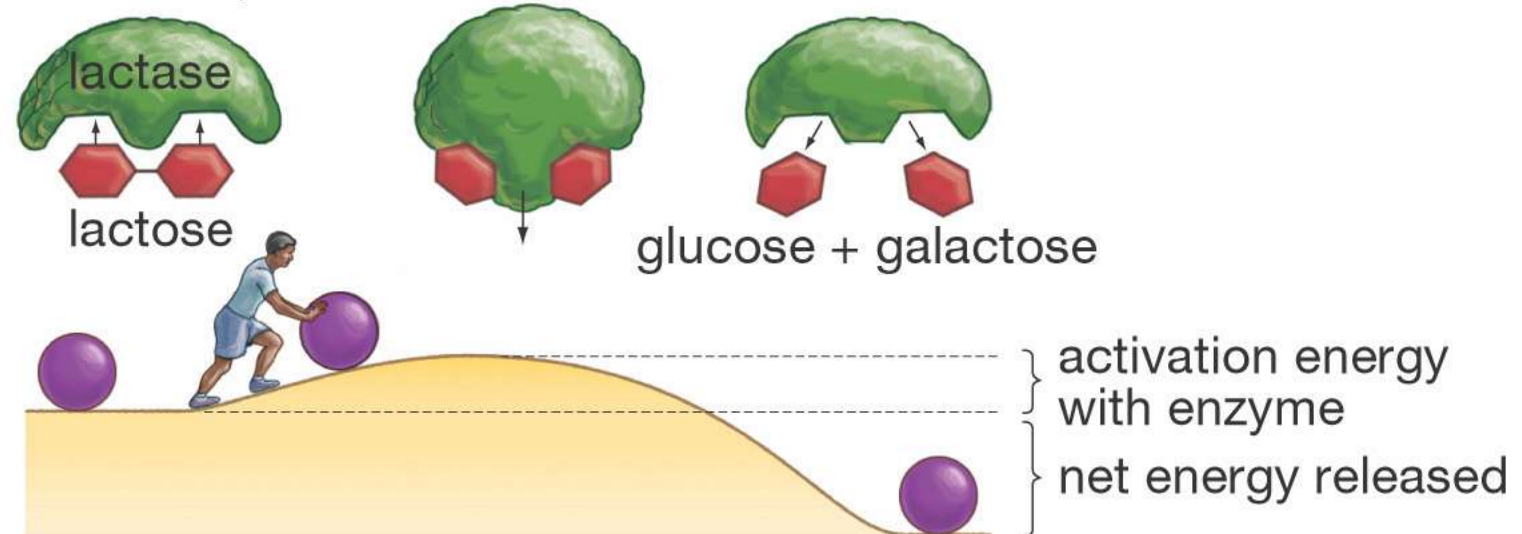


Enzymes Lower a Reaction's Activation Energy

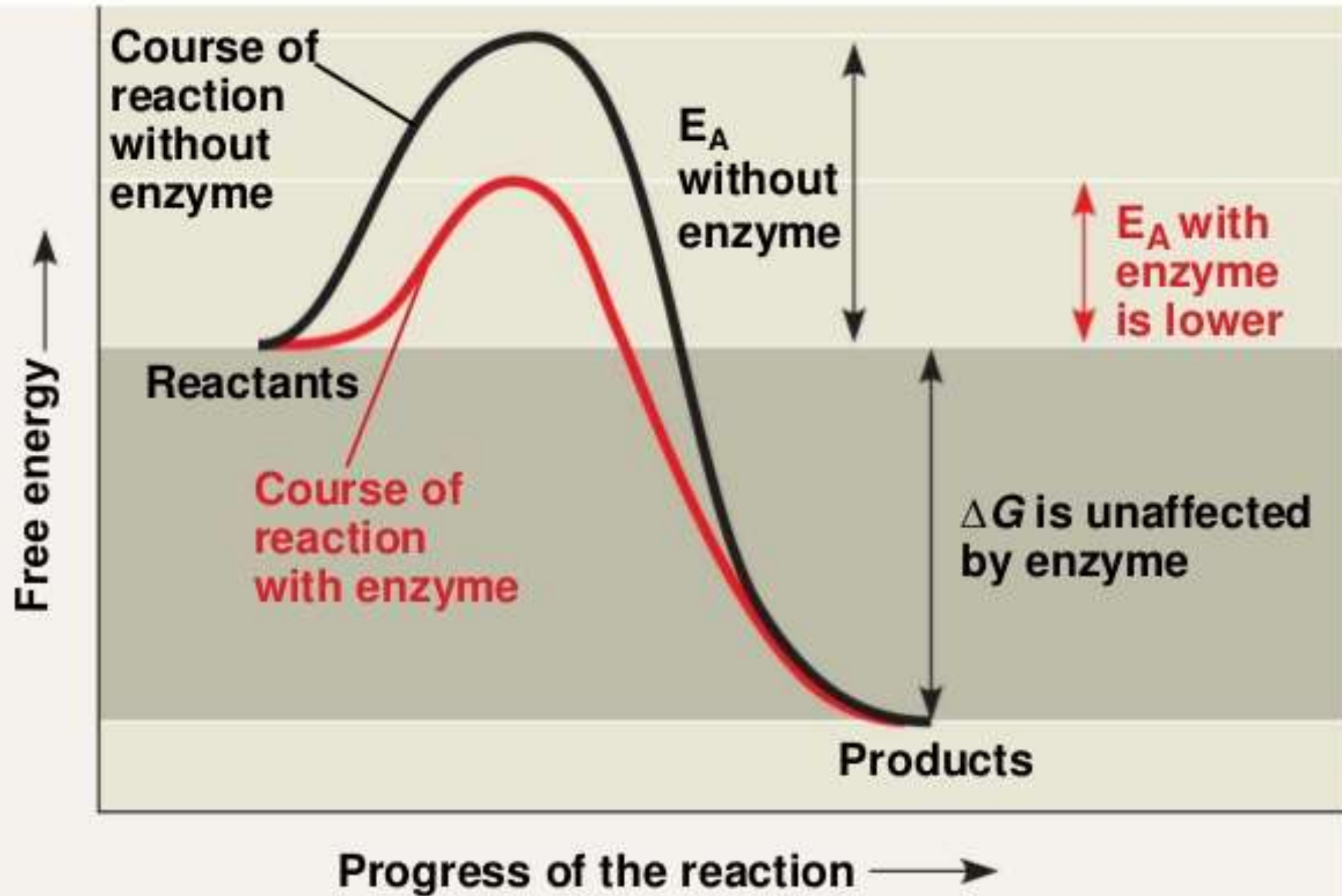
(a) Without enzyme



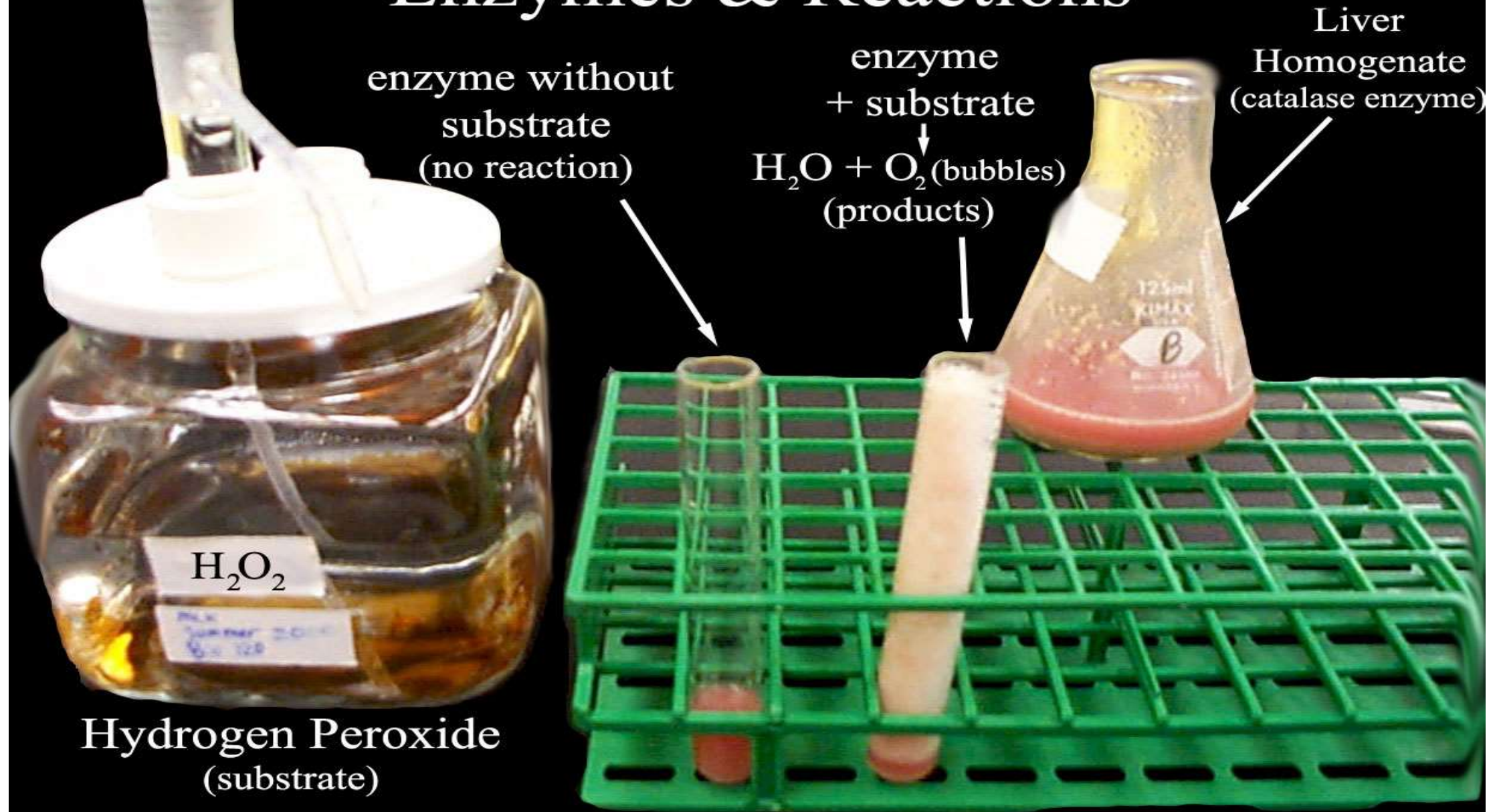
(b) With enzyme



- Net energy release at the product side is similar in catalyzed and uncatalyzed rxn
- Similar product rate but with increased minimum energy



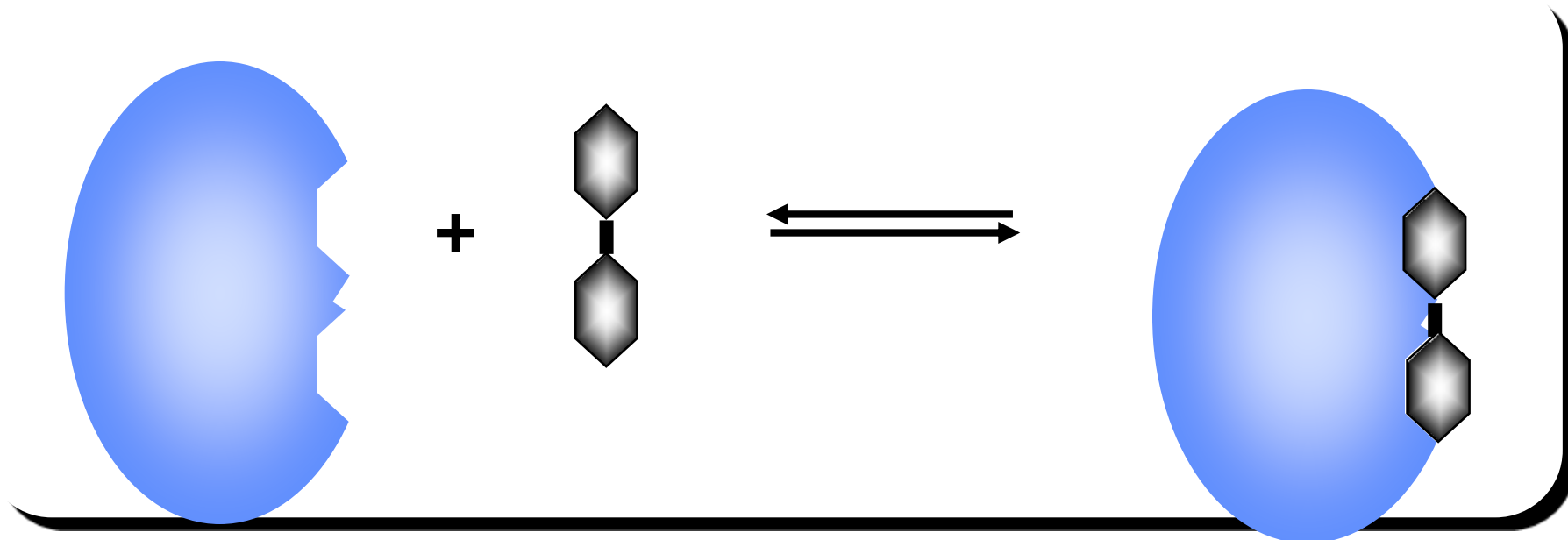
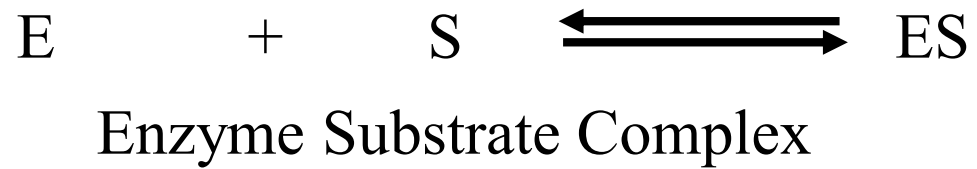
Enzymes & Reactions



Enzyme-substrate complex

Step 1:

- Enzyme and substrate combine to form complex



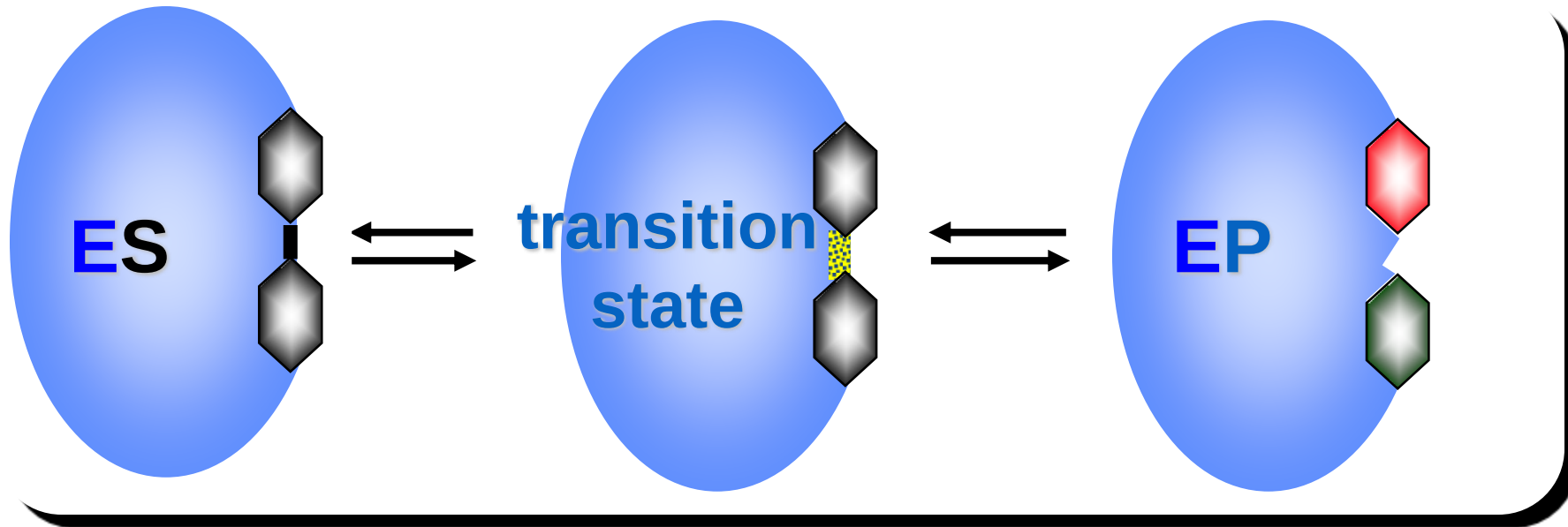
Enzyme-product complex

Step 2:

- An enzyme-product complex is formed.



Transition state → before product formation, the **reaction occurs** to convert E substrate to E product (EP):

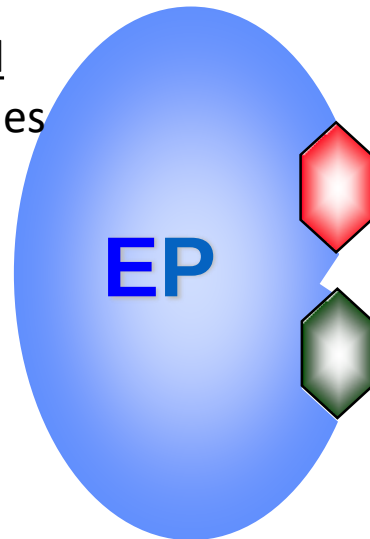


Product

- The enzyme and product separate

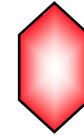


- The **products** are then **released, allowing another substrate molecule to bind** to the enzyme
- this cycle can be **repeated millions** (or even more) times **per minute**



Enzyme is
ready
for
another
substrate.

The product
is made



Factor affecting enzyme activity

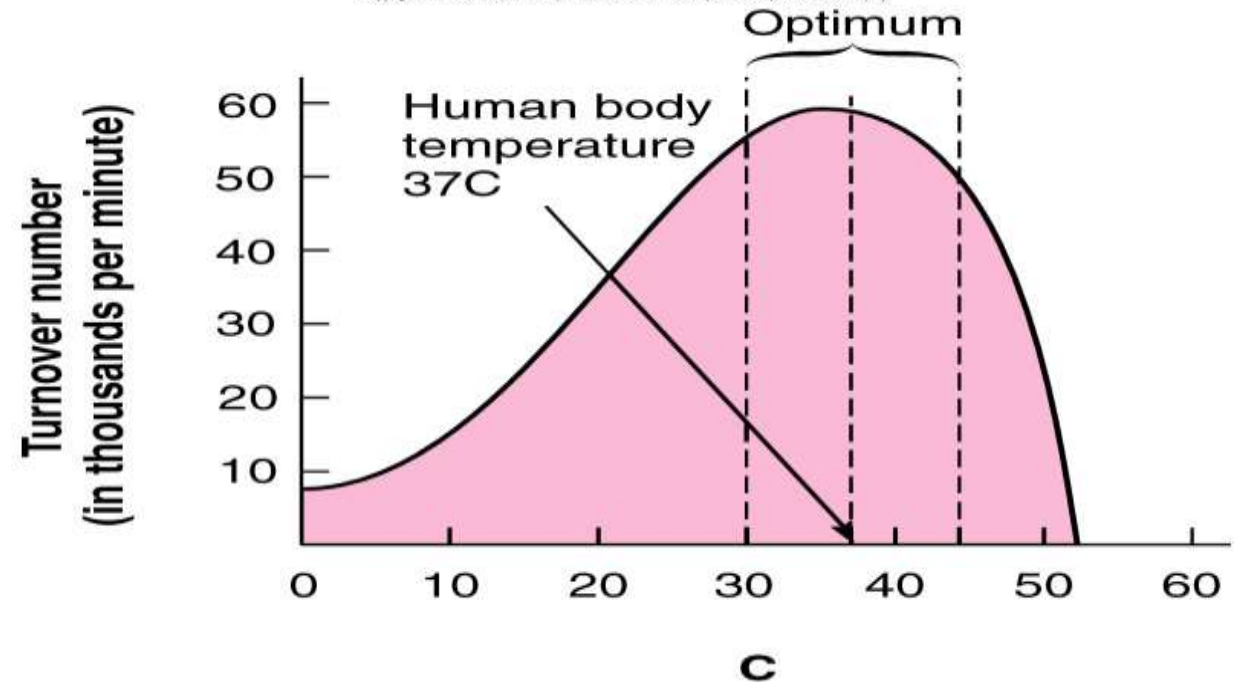
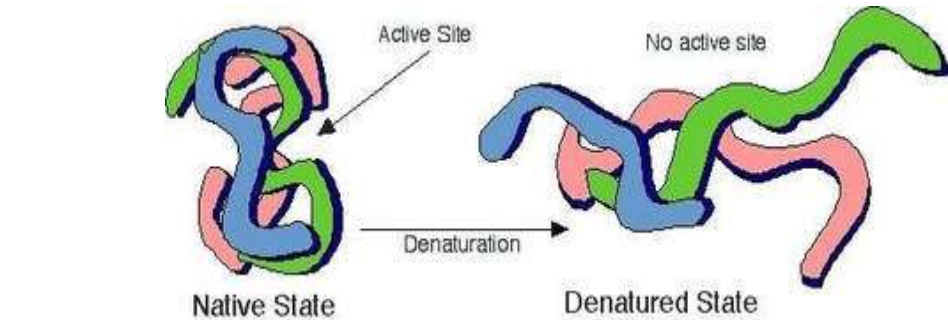
➤ Enzymes can be isolated from cells and their properties studied under different conditions

Three factors:

- 1. Environmental Conditions**
- 2. Cofactors and Coenzymes**
- 3. Enzyme Inhibitors**

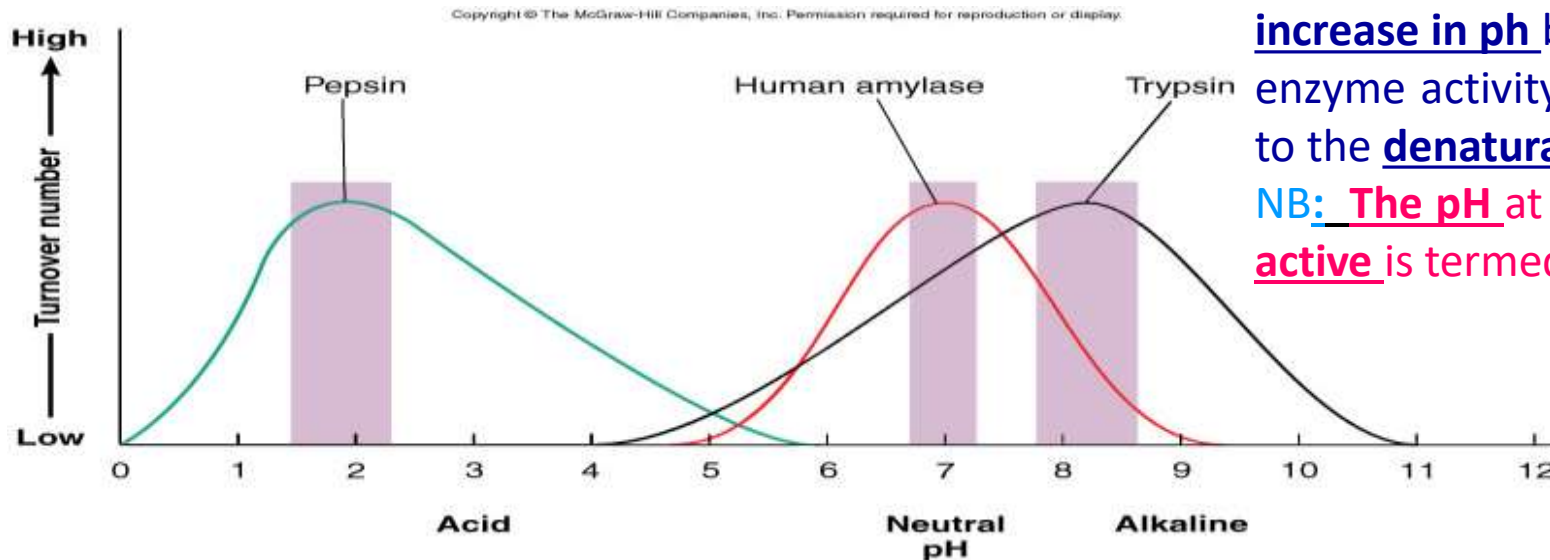
Environmental factors

- ***Optimum temperature*** The temp at which enzymatic reaction occurs fastest
- The temp at which enzymatic reaction occurs fastest is called **Optimum temperature** (the **enzyme is 100% active**)
- An increase in T increases the reaction rate to reach maximum activity **due to the increased number of molecules having sufficient energy**
- **but further increase lowers the rate** due to **denaturation of the enzyme** (**loss of native conformation** hence **loss of activity**)



Environmental factors

- pH also affects the rate of enzyme-substrate complexes
- Most enzymes have an optimum pH of around 7 (neutral)
- However, some prefer acidic or basic conditions



pH: Same as T, Optimum pH is required for enzyme activity but a further increase in pH brings in a sharp decline in enzyme activity hence reaction rate. due to the denaturation of the enzyme

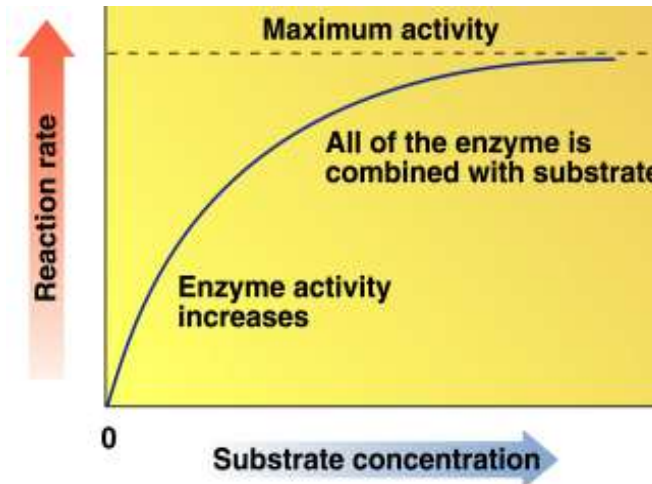
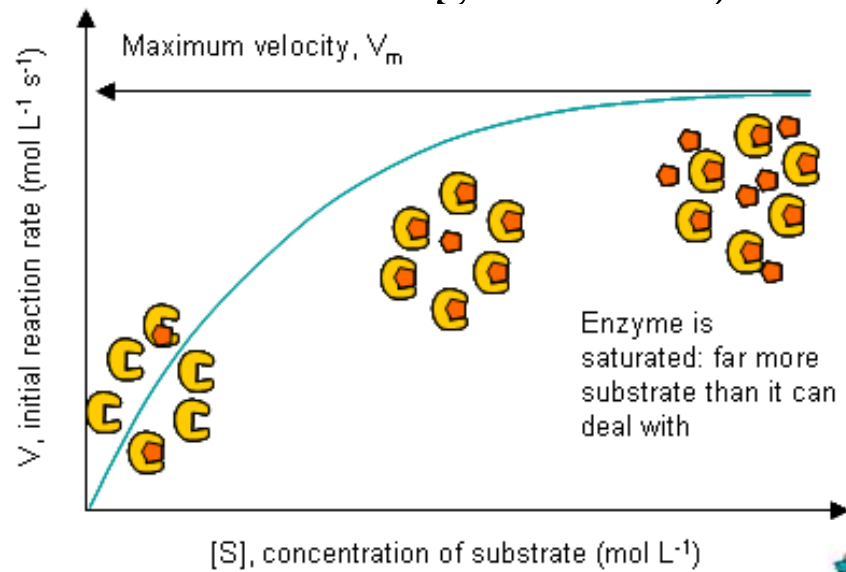
NB: The pH at which an enzyme is 100% active is termed the optimum pH.

- The optimum pH varies for different enzymes. Some are active at neutral pH some at acidic pH and some at basic Ph

e.g. pepsin (a stomach enzyme) functions best at a low (acidic) pH. At pH 1

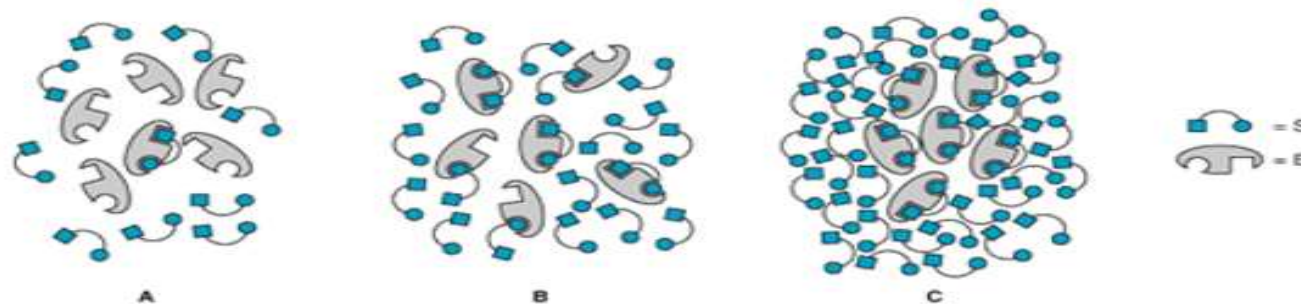
Substrate Concentration and Reaction Rate

- The rate of reaction increases as substrate concentration increases (at constant enzyme concentration)
- Maximum activity occurs when the enzyme is saturated (when all enzymes are binding substrate)



After $V_{\max}$ is reached increasing the $[S]$ will not have any effect on the reaction rate due to "saturation" (active sites are occupied by reactants forming ES).

Hence enzymes follow saturation kinetics



Enzyme Kinetics

Is The field of Biochemistry studies the quantitative measurement of the rates of enzyme-catalyzed reactions and Also, the systematic study of factors that affect these rates

Michaelis-Menten equation

The michaelis-Menten equation describes how reaction velocity (V) varies with substrate concentration [S]

- The following equation is obtained after suitable algebraic manipulation
- Tells you about the Rates of rxn Vs substrate concentration

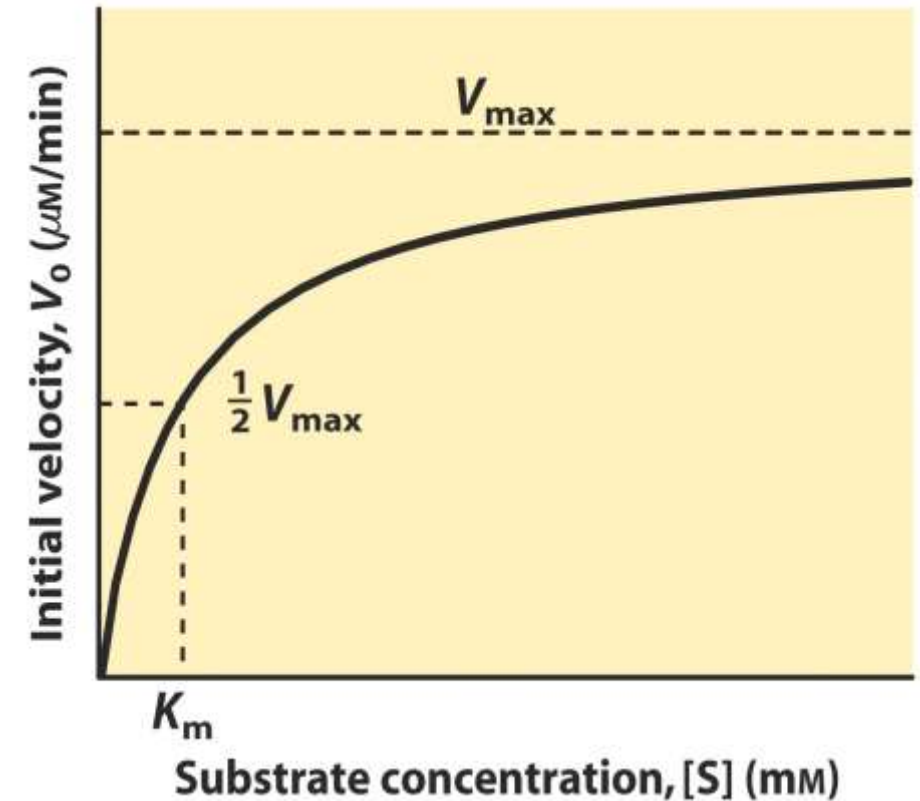
$$V = V_{\max} \frac{[S]}{[S] + K_M} \quad \text{Note: } V \text{ means } V_0$$

K_m : Michaelis constant

$$K_m = (k_2 + k_3)/k_1$$

Michaelis Menten Curve

- The “**basic idea**” is to measure the **rate** of reaction at **different substrate concentrations**.
 - $V_{\max}$ is important because it is part of the definition of the K_M : **the [S] that produces a rate that is $\frac{1}{2} V_{\max}$** .
 K_m is the [S] that permits the enzyme to achieve $\frac{1}{2}$ of $V_{\max}$.
 - The curve is a **rectangular hyperbola**...never really reaching $V_{\max}$ the maximum velocity. **0 order kinetics**
 - **V_0 is an active reaction enzyme with its substrate (1^{st} order rxn)** here, **rate (velocity) is proportional to substrate concentration**
 - **$V_{\max}$ is 0 order rxn (at saturation)** no rxn, Platou point, all E is occupied by S
- Low K_m → Enzyme higher affinity of the enzyme to substrate (E work on low [S] concentration)**



Enzyme kinetics

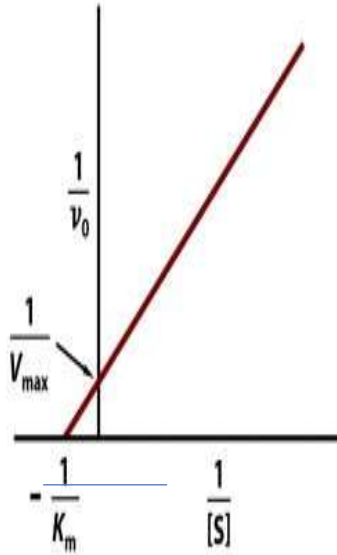
- Small K_m value reflects a **high affinity** of the enzyme for substrate and high K_m reflects a low affinity
- The rate of the reaction is directly proportional to the enzyme concentration at all substrate concentrations
- Only initial reaction velocities (v_o) are used in the analysis of enzyme reactions
- When $[S] \ll [K_m]$ the reaction velocity is **proportional** to the $[S]$, the rate depends on $[S]$ and the order of the reaction is the **first order**
- When $[S] \gg [K_m]$ $v_o = V_{max}$ the reaction velocity becomes **independent** of $[S]$ and the reaction is in zero-order
- K_m is inversely proportional to enzyme affinity to its substrate

Lineweaver-Burk Plots

NB: We can get this equation by taking the reciprocal of the Michaelis-Menten equation

$$V_o = \frac{V_{\max} [S]}{K_m + [S]}$$

$$\frac{1}{V_o} = \frac{K_m}{V_{\max} [S]} + \frac{1}{V_{\max}}$$



Lineweaver-Burk equation:

$$\frac{1}{v_0} = \left(\frac{K_m}{V_{\max}} \right) \frac{1}{[S]} + \frac{1}{V_{\max}}$$

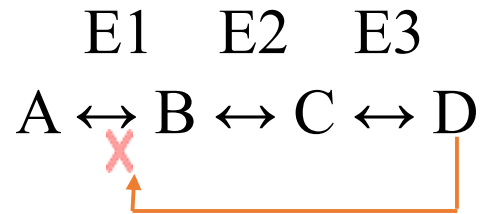
- because K_m the velocity curve is hyperbolic, it can be difficult to determine the $V_{\max}$ and the halfway point that determines K_m .
- These days, a computer program involves plotting the “reciprocal of the velocity” against the “reciprocal of the $[S]$ ”, which gives a linear curve instead of a hyperbolic one

“Used in describing the property of different enzyme inhibitors” and their effect on E action

K_m (hyperbolic) has an infinite $[S]$ at $V_{\max}$, difficult to determine them (require a plot), K_m and $V_{\max}$ values of an enzyme can be determined from a plot ($Y=MX+B$) needs a linear graph

Regulation of Enzyme Activity

1) **Feedback inhibition:** some enzymes are inhibited by their own final product



Note here that the final product D inhibits the first enzyme E1 and hence the whole reaction is slowed down

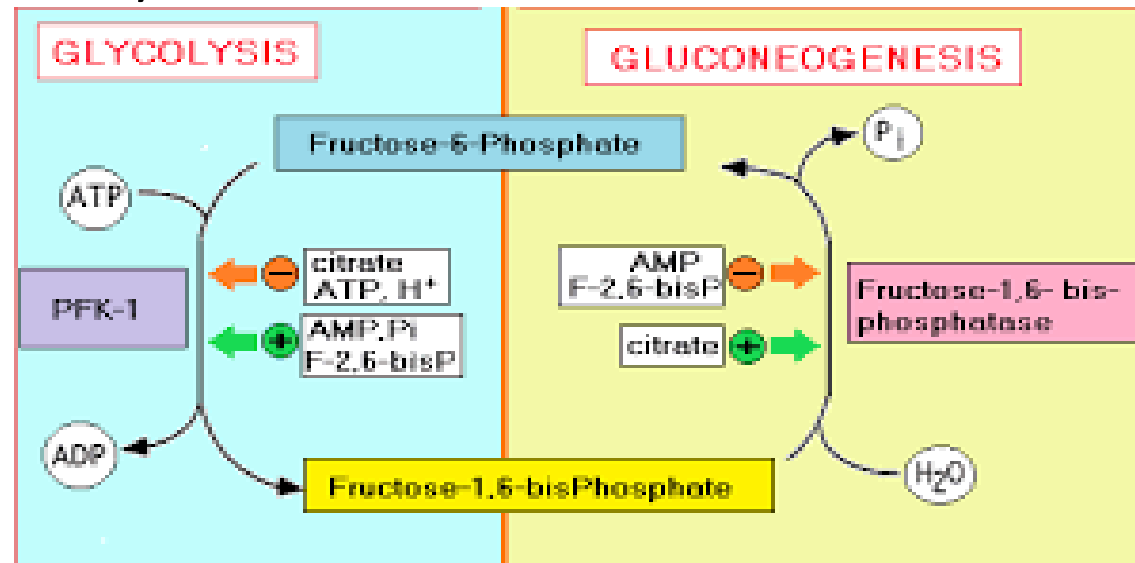
Eg: Conversion of L-threonine to L-Isoleucine where the 1st enzyme threonine dehydrogenase is inhibited by the final product Isoleucine (IN A BACTERIAL ENZYME SYSTEM) **Bacteria** can synthesize **isoleucine** from **threonine** in **5 step reaction** pathway but not human (essential amino acid)

Regulation of Enzyme....

2) **Allosteric regulation:** activity of some enzymes is regulated by molecules **other than** their substrates called allosteric modifiers

- ✓ Such molecules alter the conformation of active sites of allosteric enzymes
- ✓ Such enzymes have another site different from the active site in their folded structure called the **allosteric site**

E.g. **PFK 1** is allosterically regulated by ADP, AMP, ATP, and Citrate. The former two activate it but the latter two inhibit it



Regulation of Enzyme...

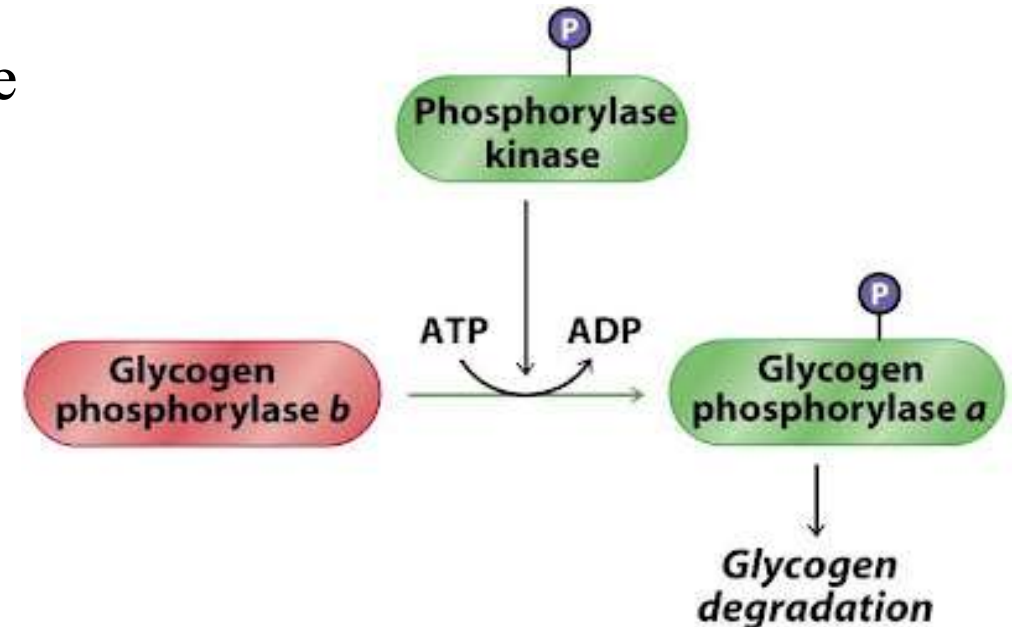
3) **Covalent modification:** some enzymes change their activities through a reversible covalent modification of their structure

Eg: Phosphorylation dephosphorylation of glycogen synthase and glycogen phosphorylase

NB: A phosphorylated form of glycogen phosphorylase is active but the phosphorylated form of glycogen synthase is inactive

Hormonal regulation of glycogen metabolism (phosphorylated Protein Kinase, phosphorylate G phosphorylase, make it activated)
G synthase inhibited

- Glucagon (low glucose state) → activates protein kinase A to phosphorylate Glycogen Phosphorylase (require B1 TPP) lysis activated
- Insulin does the reverse.. (dephosphorylate them)



Regulation of Enzyme...

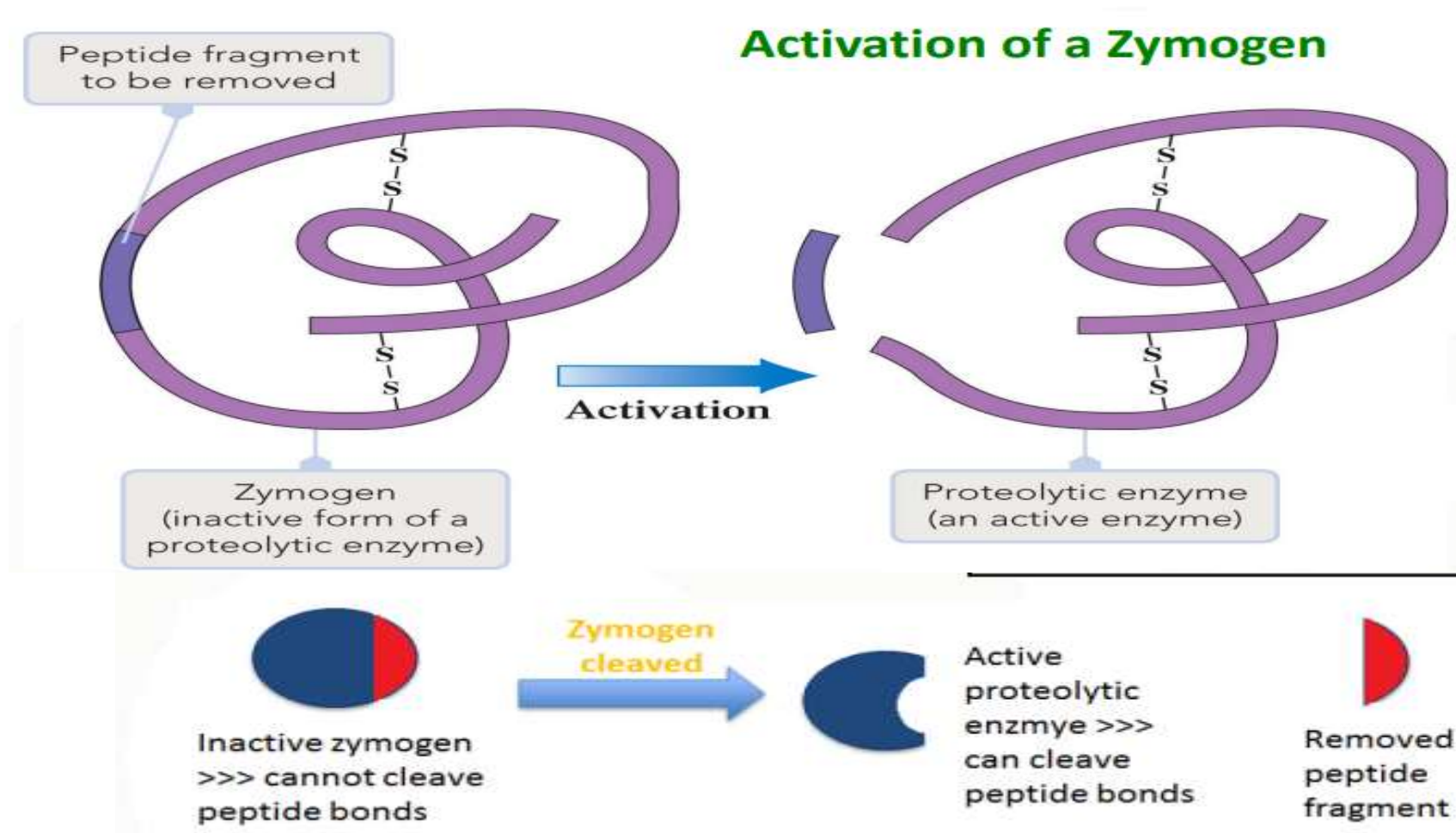
4) **Zymogen activation:** Proteolytic enzymes are produced in their inactive precursor form called zymogens but after losing part of their peptide they become active enzymes

- To **prevent** these enzymes from destroying the tissues, that produced them, they are released in an inactive form = **ZYMOGENS (contain peptide)**
- Zymogen needs to be activated to perform
- Most digestive (proteolytic), blood-clotting enzymes

Eg: Pepsinogen, Chymotrypsinogen, and Trypsinogen are inactive zymogen forms of the active enzymes Pepsin, Chymotrypsin, and Trypsin

- ✓ Some proteins are also produced in their inactive preprotein form like proinsulin. Others are the blood-clotting enzyme thrombin and fibrin (Prothrombin, fibrinogen)

- Activation of a zymogen requires the removal of a peptide fragment from the zymogen structure



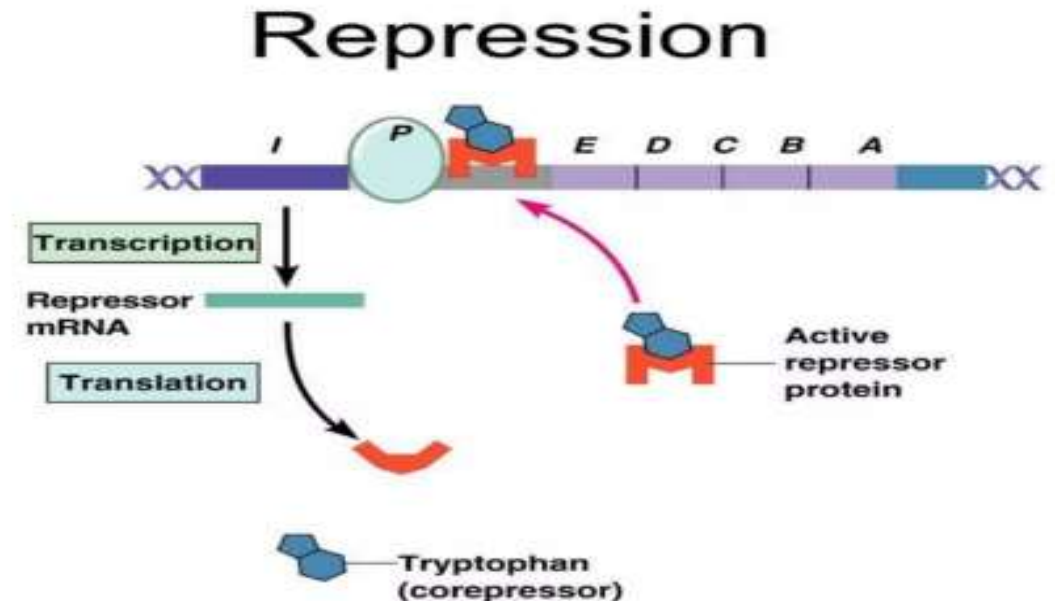
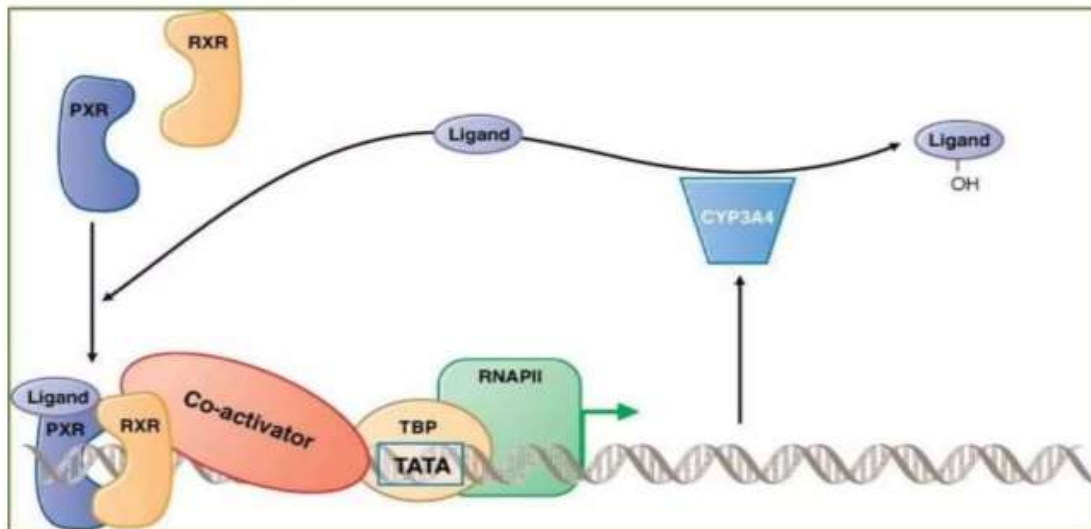
Regulation of Enzyme...

5) Induction or Repression of Enzymes: Cells can regulate the amount of enzyme present by altering the rate of enzyme synthesis

- ✓ The increased (induction) or decreased (repression) of enzyme synthesis leads to alteration in the total amount of active sites
- ✓ Induction and repression are slow processes that may be elicited by hormone release

E.g. Insulin increases the synthesis of key glycolytic enzymes

Mechanism of enzyme induction:



- Synthesis regulation (Hormonal) by activating transcription and translation (using operon)
- Insulin is considered to be an anabolic hormone in that it promotes the synthesis of protein and glycogen and it inhibits the degradation of these compounds in muscle tissue. “(induction operate in catabolic pathway)”
- The insulin increases the synthesis of glucokinase or hexokinase (gene expression) and promotes glucose uptake by the muscle (catabolism)
- “Slower process” than allosteric regulation, and covalent modification (phosphorylation)

Regulation of Enzyme...

Induction	Repression
1. It turns the operon on.	1. It turns the operon off.
2. It starts transcription and translation.	2. It stops transcription and translation.
3. It is caused by a new metabolite, which needs enzymes to get metabolised.	3. It is caused by an excess of existing metabolite.
4. It operates in a catabolic pathway.	4. It operates in an anabolic pathway.
5. Repressor is prevented by the inducer from joining the operator gene.	5. Aporepressor is enabled by a co-repressor to join the operator gene.

Operon → A group of genes or a segment of DNA that functions as a single transcription unit. It is comprised of an operator, a promoter, acts as a unit of transcription,

- The importance of enzyme induction → can be applied to stimulate the conversion of drugs into a more polar metabolite, treat hyperbilirubinemia, analyze drug interaction, study toxicity

Enzyme inhibition

- **Inhibitors** are molecules that resemble the substrate(s) or product(s) and bind to the **active site** thus they **interfere with catalysis** slowing or halting enzymatic reactions
- Many drugs are **reversible** enzyme inhibitors
- They have their physiological effect by decreasing the activity of a specific enzyme
- For example, aspirin (acetylsalicylate) inhibits the enzyme that catalyzes the first step in the synthesis of prostaglandins compounds involved in many processes including some that produce pain (irreversible)
- **Aspirin (Irreversible)**:- nonsteroidal anti-inflammatory drugs (**NSAIDs**) are **used to treat pain**, inflammation, and migraines, and reduce the risk of developing CVD. **Aspirin** work as a **blood thinner** (prevent clotting) one of the side effect (due to **PG not being formed**)

Enzyme inhibition...

- Concentration of inhibitor needed to inhibit the enzyme depends on **how tightly the inhibitor binds** to the enzyme
- **Inhibition constant (K_i)** is used to describe how tightly an inhibitor binds to an enzyme

Types of Inhibitors

There are **two broad classes** of enzyme inhibitors:

- Irreversible
- Reversible

Enzyme inhibition...

Irreversible

➤ irreversible inhibitors are those :

❖ that **bind covalently** with enzymes or

❖ **Destroy a functional group** on an enzyme that is essential for the enzyme's activity, or

❖ that form a particularly **stable noncovalent association**

➤ Formation of a **covalent link** between an irreversible inhibitor and an enzyme is **common, the** enzyme becomes **poisoned** and recovery becomes very difficult (irreversible) **suicide** inhibitors

➤ For example the reaction of chymotrypsin with diisopropylfluorophosphate (DIFP) **irreversibly inhibits the** enzyme by binding with Ser195 in the active site of chymotrypsin

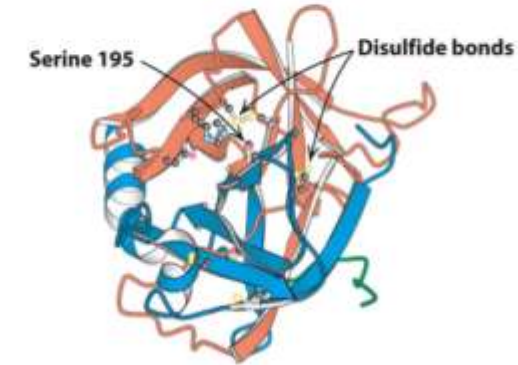
➤ DIFP is an **organophosphorus insecticide**

Enzyme inhibition... Diisopropylfluorophosphate as irreversible inhibitor of chymotrypsin

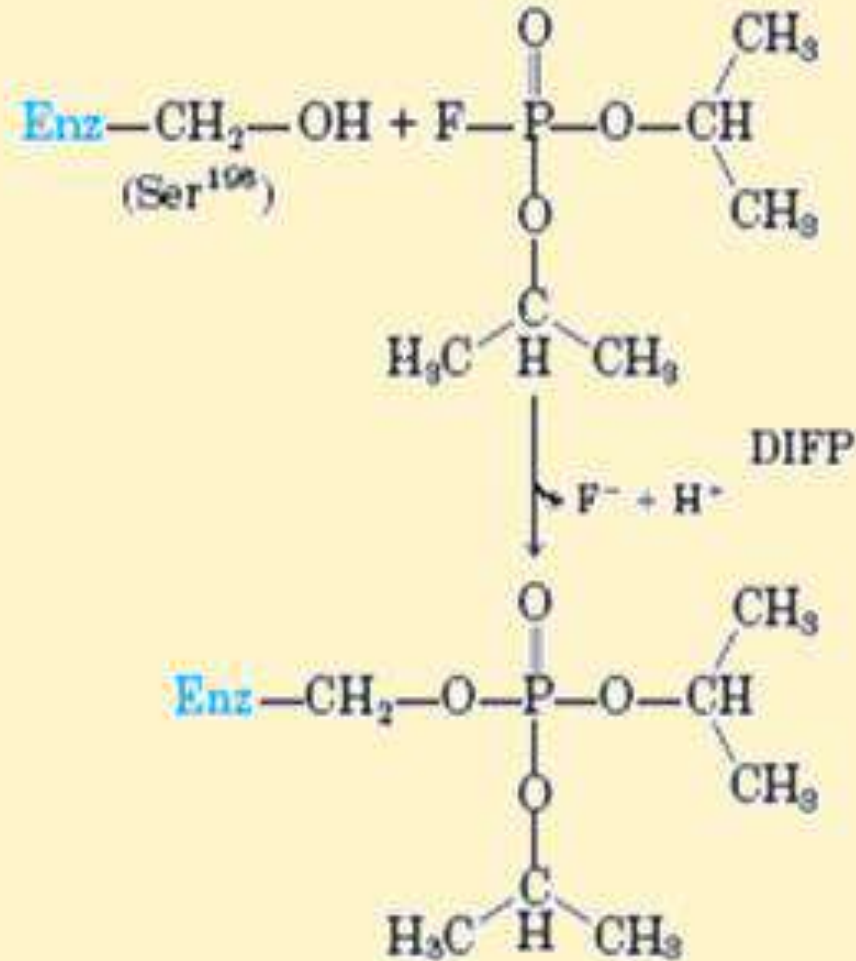
Chymotrypsin is synthesized as single polypeptide (chymotrypsinogen) and activated (cleaved) into three polypeptide chains connected by a disulfide bridges

The active center includes catalytic triad

Ser195 His 57 and Asp 102



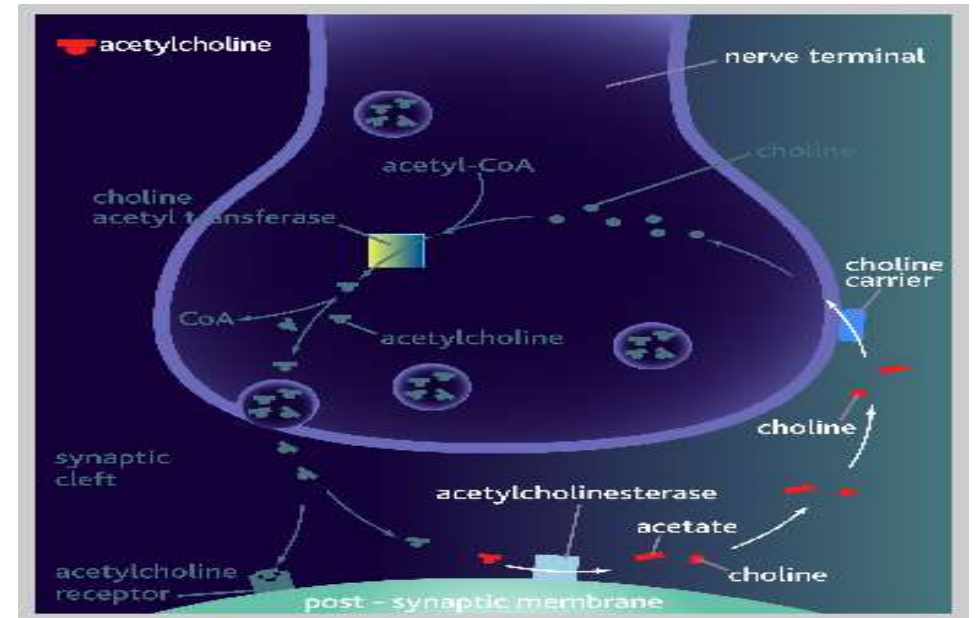
Chymotrypsin → serine protease is made by the pancreas that hydrolyzes the peptide bonds of tryptophan, leucine, tyrosine, and phenylalanine.



Sarine (Nerve gas)

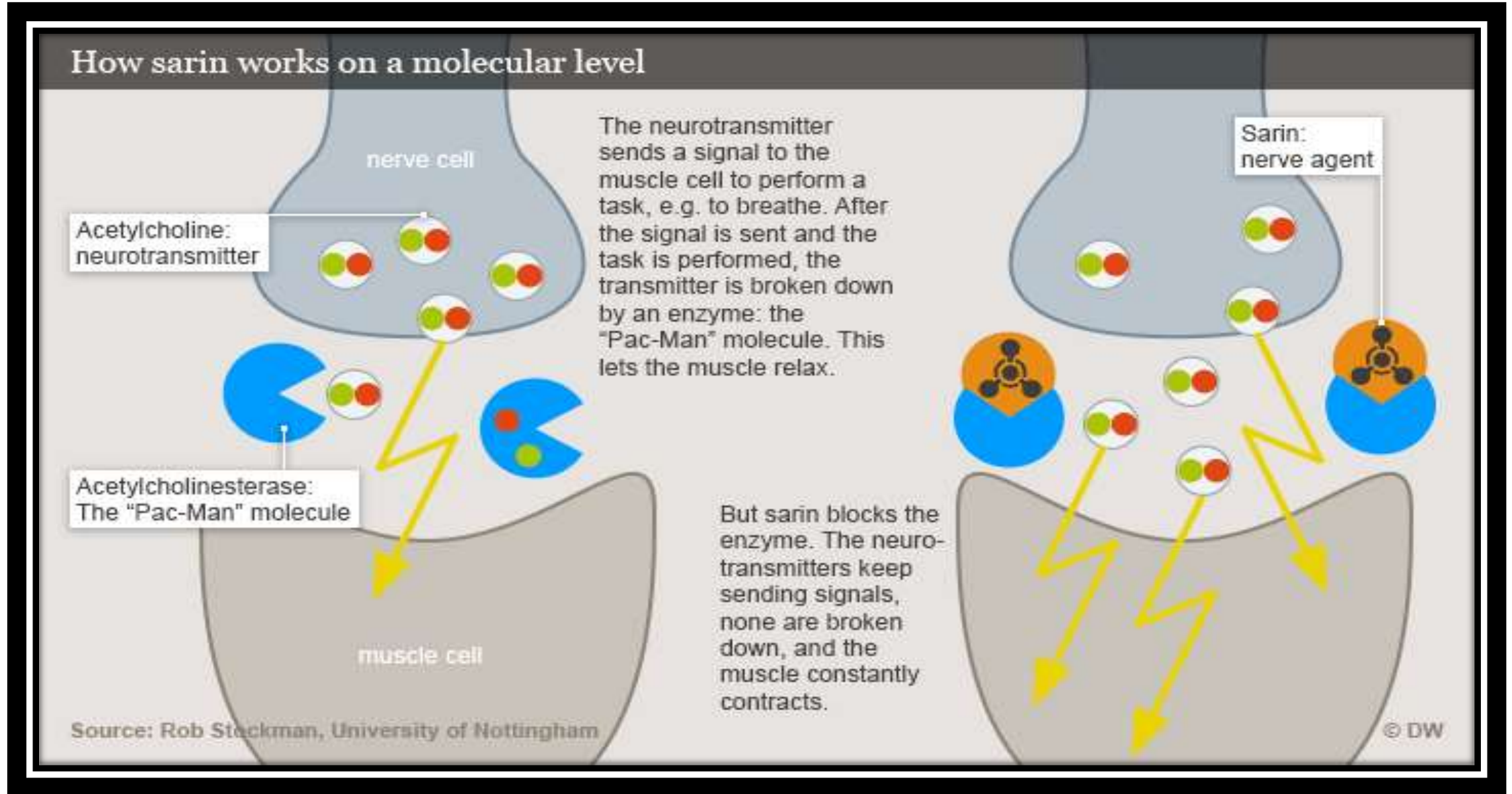


- **Sarine (Nerve gas, agent)** NATO, as a chemical weapon of mass destruction, inhibits Acetylcholinesterase (AChE) (irreversible), inhalation low conc. Lethal (1 -10 min), unless antidotes (atropine) are quickly administered,
- **“Organophosphate”** insecticide:- damage acetylcholinesterase. Keep send signal and Muscle constantly contracting, This enzyme is critical for controlling nerve signals in the body (AChE) is a cholinergic enzyme primarily found at postsynaptic neuromuscular junctions, especially in muscles and nerves, cleave Ach into choline and acetic acid,
- **Antidote** → removing it from the enzyme



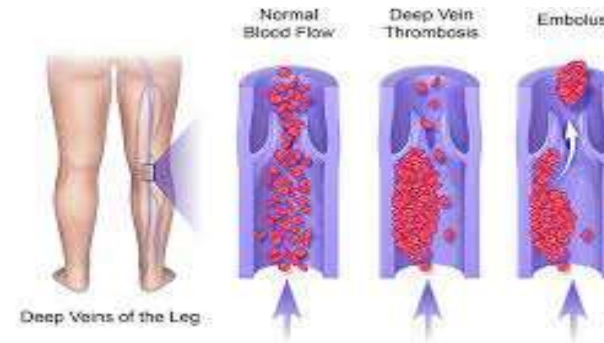
Enzyme inhibition...

Irreversible inhibition

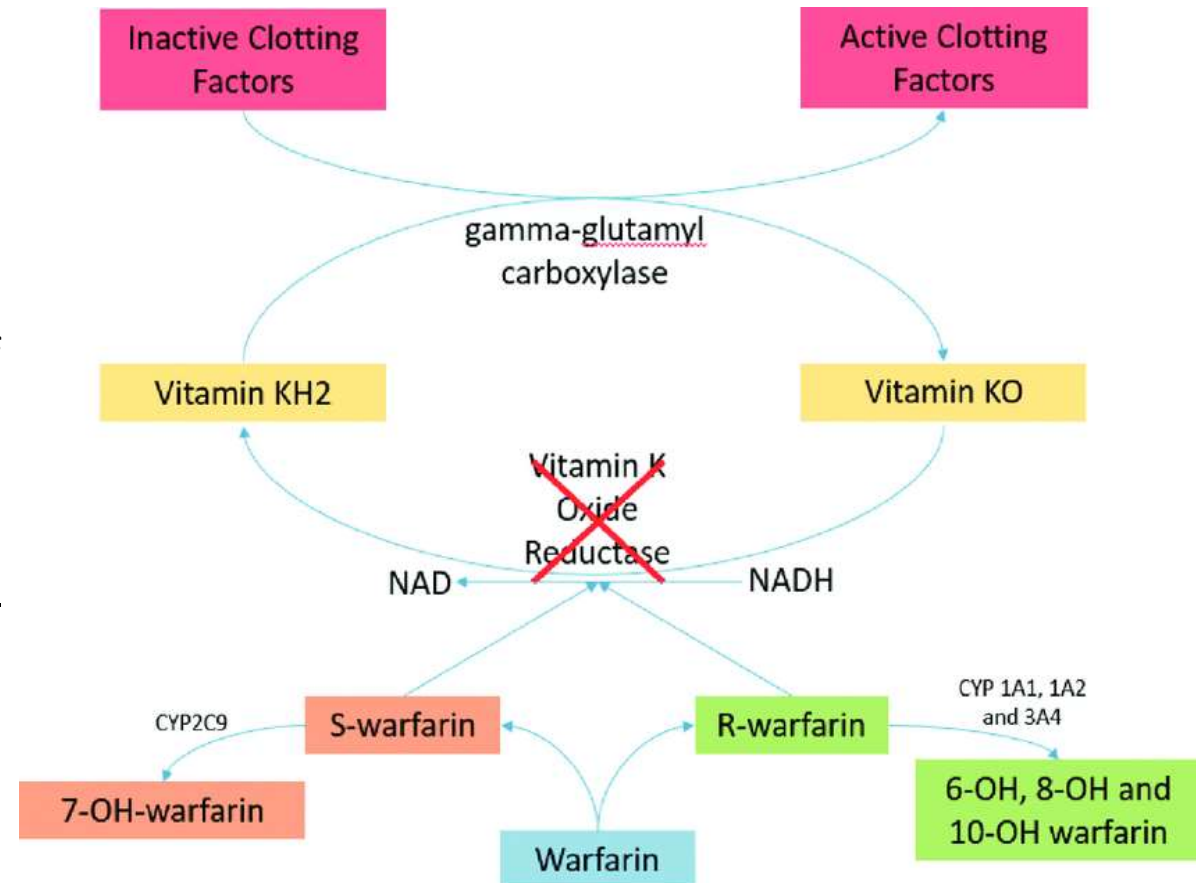


Warfarin

- **Warfarin** is used to prevent blood clots, **anticoagulant effects** (prescribe in the **treatment** of **venous thrombosis** and thromboembolic events.) irregular Hb, prosthetic (mechanical) heart valve, for the treatment of **heart attack (stroke)**, act as an **irreversible inhibitor** of “**prothrombin**” (cause **death** by **internal bleeding**)
- **Warfarin** and related **vitamin K antagonists** (VKAs) **block** the function of the **vitamin K epoxide reductase** complex in the **liver**, leading to **depletion** of the **reduced form** of **vitamin K** that serves as “a **cofactor**” for **gamma-carboxylation** of vitamin K-dependent coagulation factors
- **Warfarin -clinically available** as an **equal mixture** of **R** and **S enantiomers** – (right and left form)



Warfarin is used to prevent blood clots



Enzyme inhibition...

Reversible

- This type of inhibition involves equilibrium between enzyme and inhibitor **equilibrium constant** (K_i) being the measure of the affinity of the inhibitor for the enzyme. Most drugs are irreversible..
- This inhibition is further classified into **three categories**:
 - ❖ **Competitive**
 - ❖ **Uncompetitive**
 - ❖ **Noncompetitive**

Enzyme inhibition...

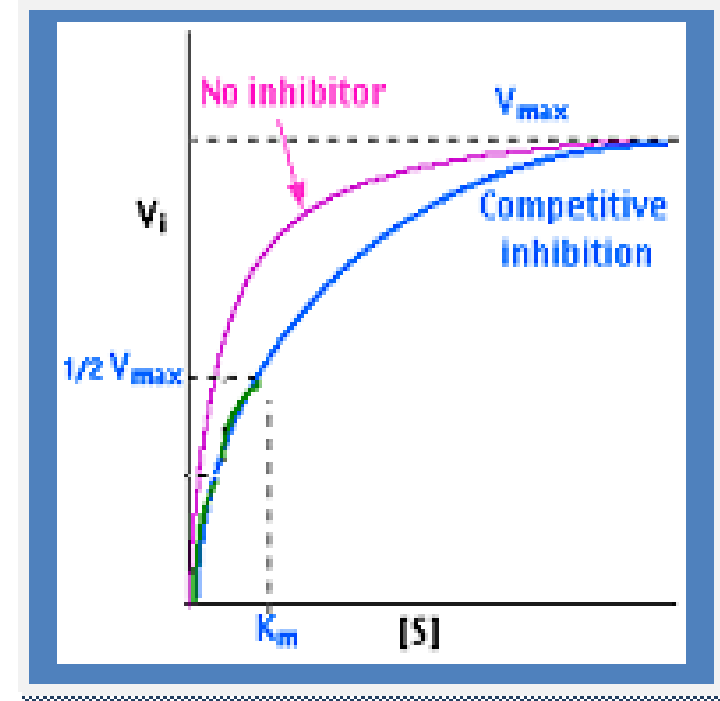
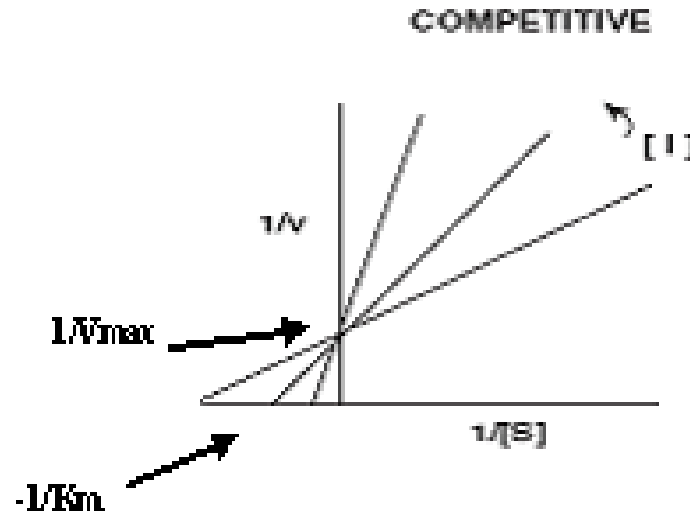
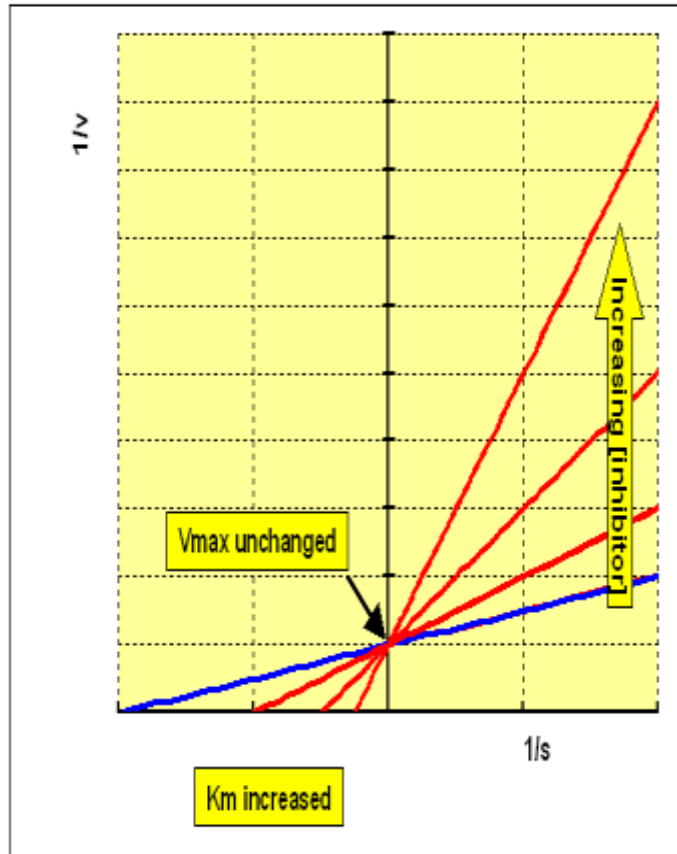
Competitive Inhibition

- Competitive inhibitors bind **only** to free enzymes and to the **same site** as the substrate
- Competitive inhibitors are molecules that usually look like the substrate but can't undergo the reaction
- At an infinite concentration of the substrate competitive inhibitor cannot bind to the enzyme since substrate concentration is high enough that there is virtually no free enzyme present

Compet Con'd...

- The Inhibitor (I) competes with Substrate (S) for the Enzyme (E) active site (**structural similarity**), with no product formation, **Only** with the **free E active site**,
- Attachment of the inhibitor is **reversible** and hence the effect can be **abolished** by **increasing [S]**, to **overcome** the **I affinity**
- In **high concentrations of substrate** Inhibitors **competing** for an **active site** are **very much minimized**.
- **requires free E** ➔ If no free enzyme (at **saturation**) and **no inhibition**, **B/C** the **active site**, is **already occupied**

Enzyme inhibition...



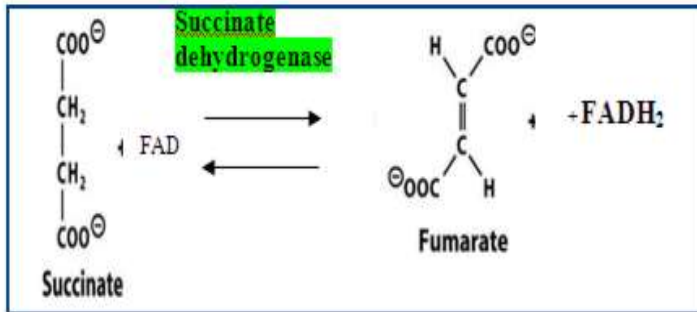
- Under *competitive inhibition*
- Affect enzyme-substrate affinity (affect K_m only)
 - **“ V_{max} remains unchanged; K_m increases”**
 - Increased K_m means “low affinity” of the enzyme to the substrate

Enzyme inhibition...

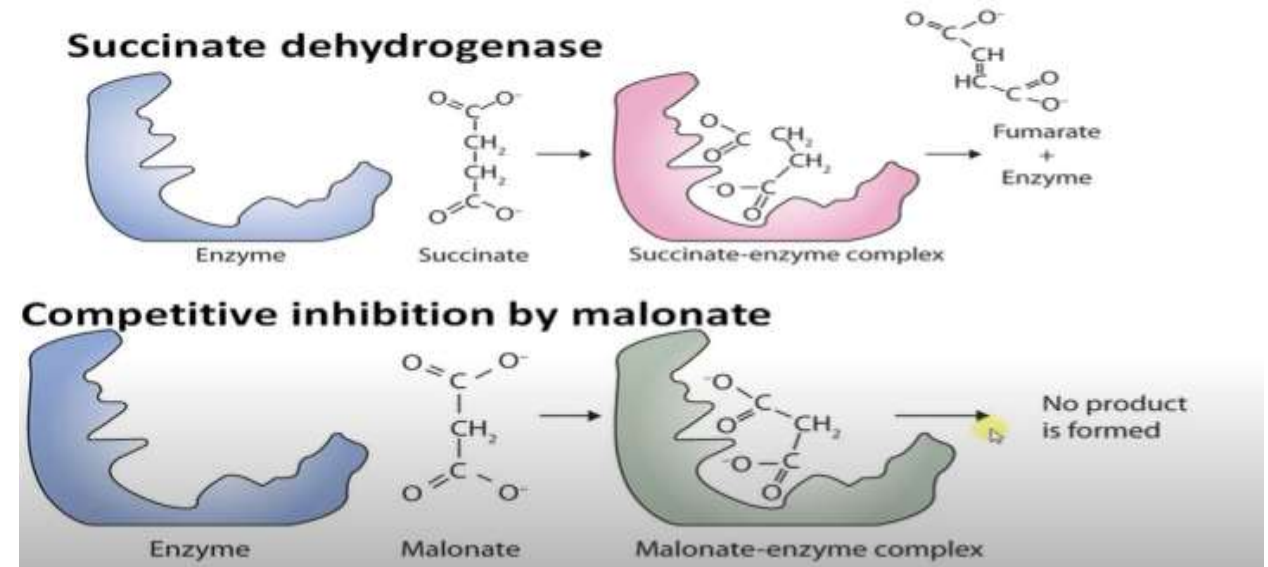
Under competitive inhibition

V_{max} remains unchanged; K_m increases

Example :



1. Competitive Inhibition or Isosteric inhibition



- Malonate is a competitive inhibitor of succinate dehydrogenase
- The enzyme **uses** succinate as its substrate but **is inhibited** by malonate which is structurally similar to succinate and differs in having one rather than two methylene groups



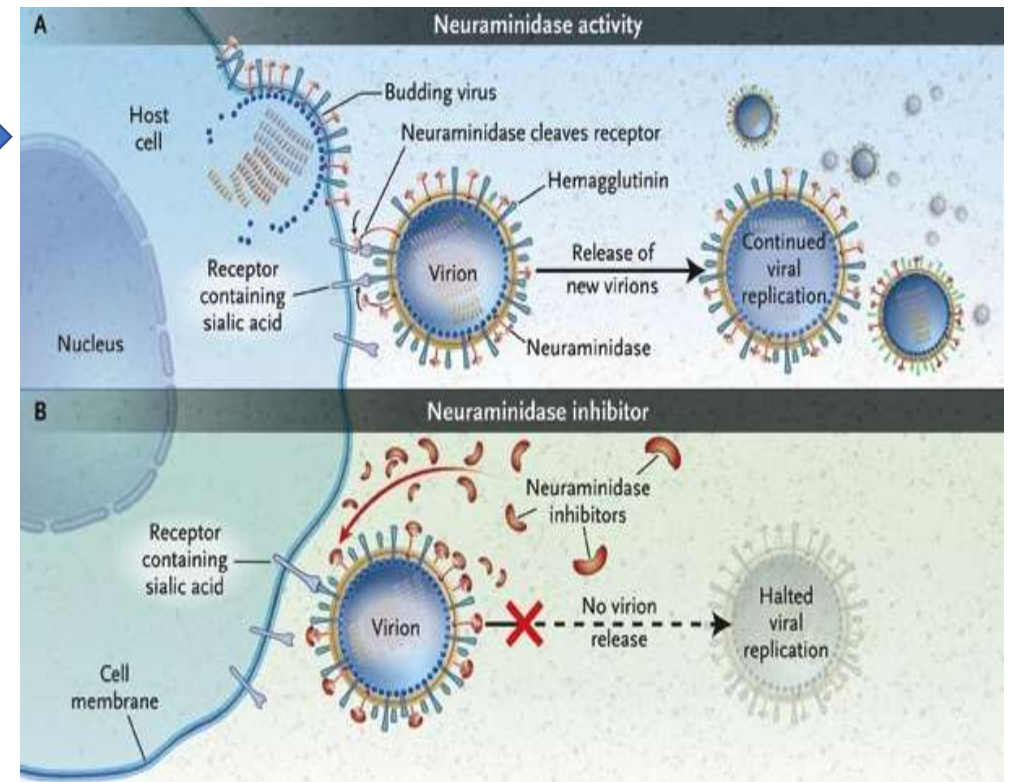
Enzyme inhibition...

Compet Con'd...

- E.g. Remember the TCA cycle.... Structural similarity, Succinyl CoA is a competitive feedback inhibitor of citrate synthase due to its structural similarity with acetyl CoA

E.g. Relenza (generic Zanamivir) is a synthetic drug for influenza: it competitively binds to the “neuraminidase” active site and prevents the cleavage of hemagglutinin (G.p found at the surface) used for binding to a target cell (fusion protein) to prevent fusion of the virus. Its Called a neuraminidase inhibitor

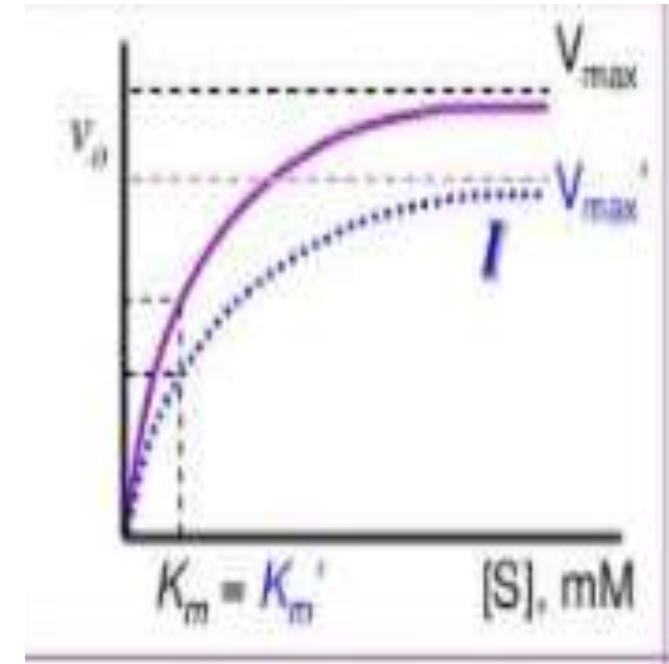
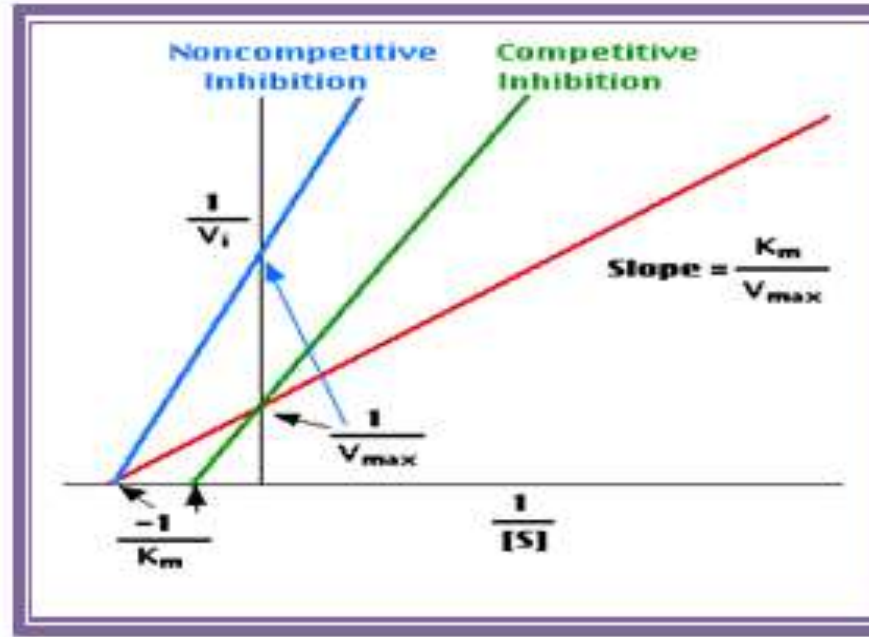
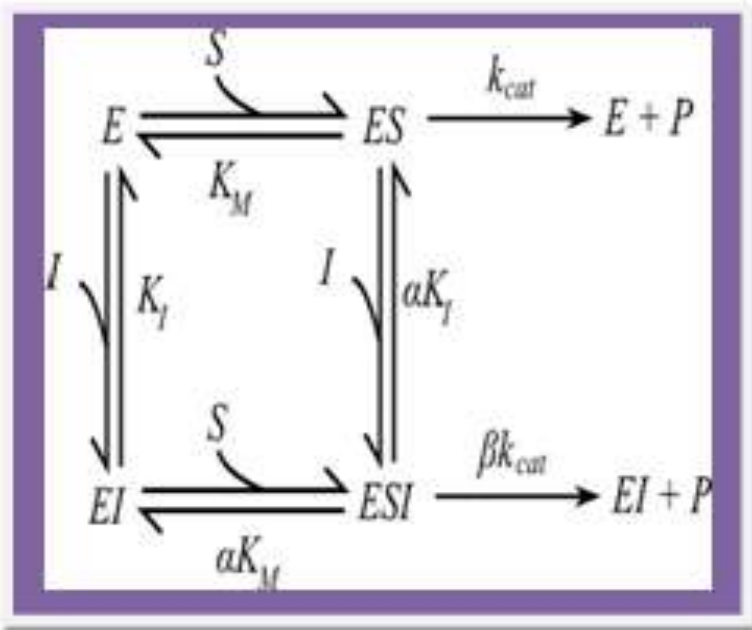
Consequently, virions are not released from infected cells, “preventing the spread of the influenza virus”



Enzyme inhibition...

Non-competitive Inhibition

- Compounds that **reversibly** bind with either the enzyme or the enzyme-substrate complex are designed as noncompetitive inhibitors and the following reaction describe these events



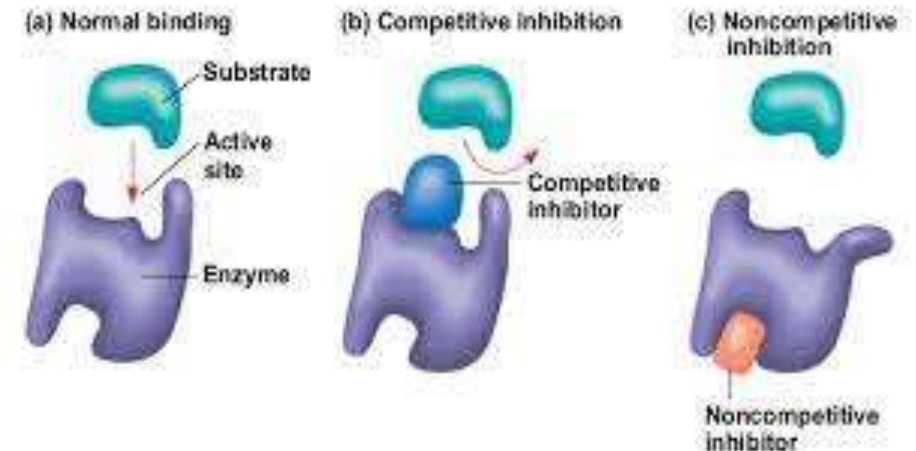
Enzyme inhibition...

Non-Compet I Con'd...

- The substrate and the inhibitor can both be bound to the enzyme “at the same time”,
- The inhibitor binds at an “allosteric site” not on an active site.
- I Can bind with **free E or ES complex** (“reduce the rate” of rxn by reducing V_{max})
without affecting the affinity
- This reduces the effective V_{max} but since substrate binding is not affected the K_m value remains the same
- Do not interfere with substrate binding, but the bound inhibitor renders the enzyme inactive (Never reaches V_{max})

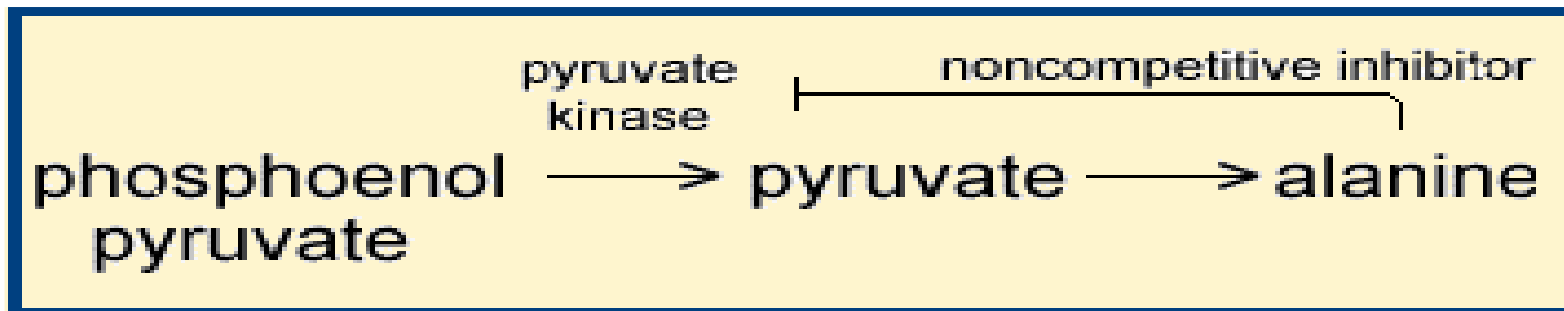
Enzyme inhibition...

- **Noncompetitive inhibition**, therefore, differs from competitive inhibition in that inhibitor can combine with ES, and S can combine with EI to form in both instances **EIS**
- This type of inhibition is not completely reversed by high substrate concentration since a closed sequence will occur regardless of the substrate concentration
- Since the inhibitor **binding site** is not identical to nor does it modify the active site directly K_m is not altered but V_{max} is decreased



Enzyme inhibition...

- For example amino acid **alanine** non-competitively inhibits the enzyme pyruvate kinase
- Alanine is one product of a series of enzyme-catalyzed reactions the first step of which is catalyzed by **pyruvate kinase**
- Remember in gluconeogenesis....End product **Alanine** is a substrate for **gluconeogenesis (Glucose from Non -CHO source)**, so should inhibit **PK** **allosterically** (inhibit glycolysis)



Enzyme inhibition...

Non-Compet I Con'd...

- E.g. **Cyanide(ETC inhibitor)** is a **chemical** that contains a carbon-nitrogen (CN) bond. Sources include: **Gas(HCN)** or **crystal (NACN)** (KCN) or Organic compounds like acetonitrile and Pharmaceuticals like nitroprusside
 - causing poisoning by **preventing ATP** production via **aerobic** respiration, leading to **eventual death**
 - CN binds to an **allosteric site** on **cytochrome oxidase (COMPLEX 4)**, and can **no longer pass electrons** to the final acceptor (**oxygen**)
 - Antidotes(treatments for cyanide toxicity) for CN poisoning include:-
 - Hydroxocobalamin** (that complexes with cyanide forming cyanocobalamin)
 - Sodium thiosulfate**(combine with CN and produce harmless compound Thiocyanate).

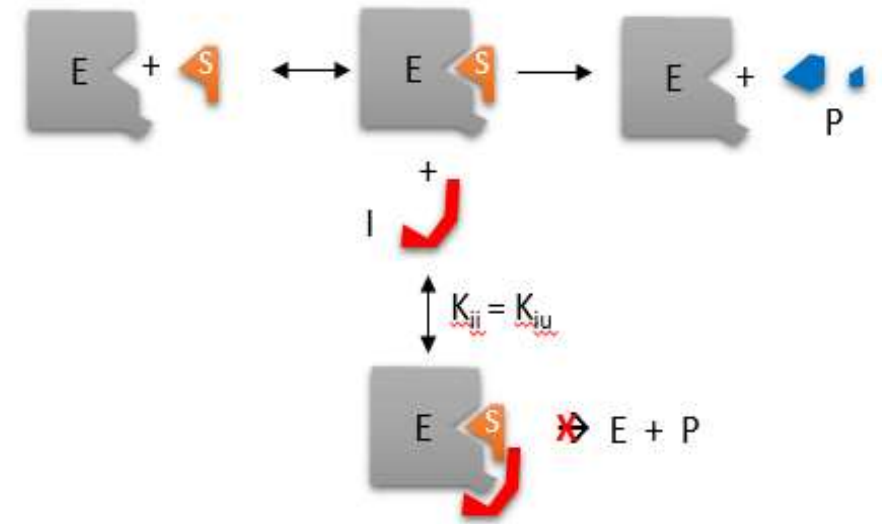
E.g. Heavy metal (**Hg**, **Pb**, also binds the **Sulfhydryl (SH)** group of the allosteric site INHIBITION) chelating treatment by **EDTA** **reverses the effect**



Enzyme inhibition...

Uncompetitive Inhibition

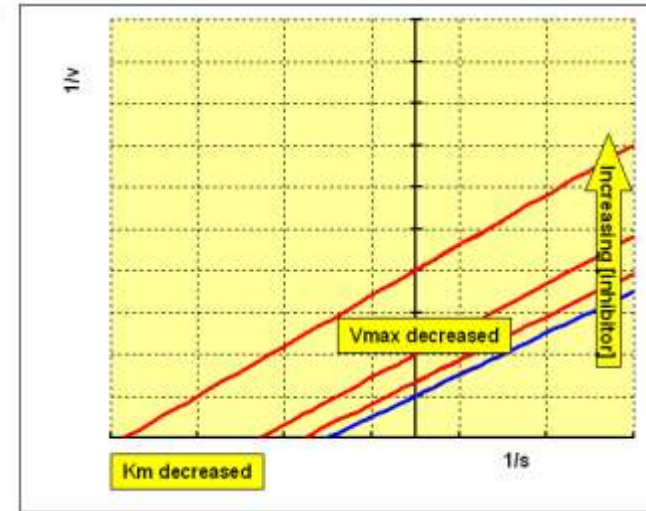
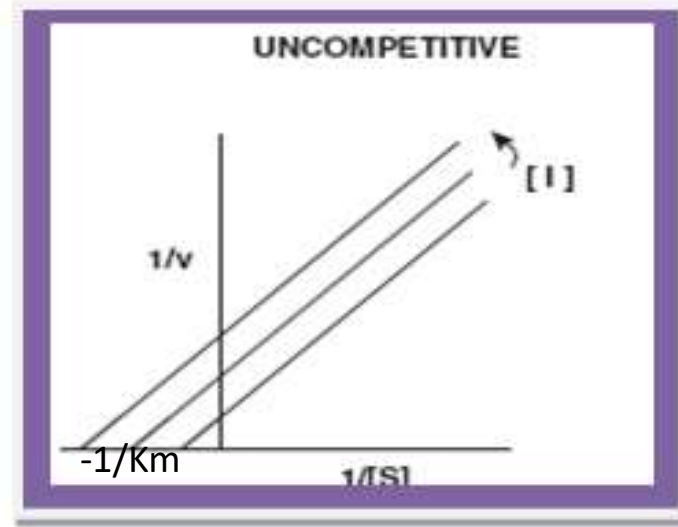
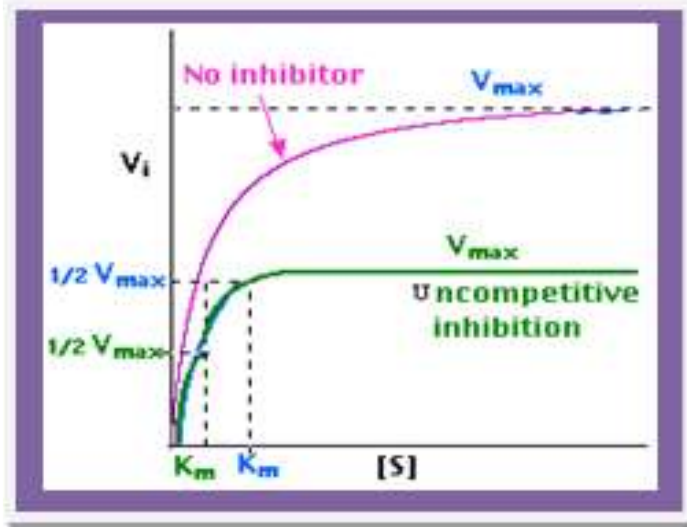
- If the inhibitor combines only with ES (and not E) inhibitor exerts its effect only at **high concentrations** of the substrate at which there is lots of ES around
- This means that the increasing substrate concentration (S) doesn't prevent the binding of the inhibitor
- ✓ Binds at a site distinct from the substrate active site (allosteric but only to ES complex), only when the enzyme is active
- ✓ , Unlike non-competitive inhibitors, it binds to that only the **ES complex**. Not free E
- ✓ Not affected by increased substrate concentration, increased S
- ✓ Uncompetitive inhibitors alter both $V_{\max}$ and K_m



Enzyme inhibition...

Uncompetitive Inhibition

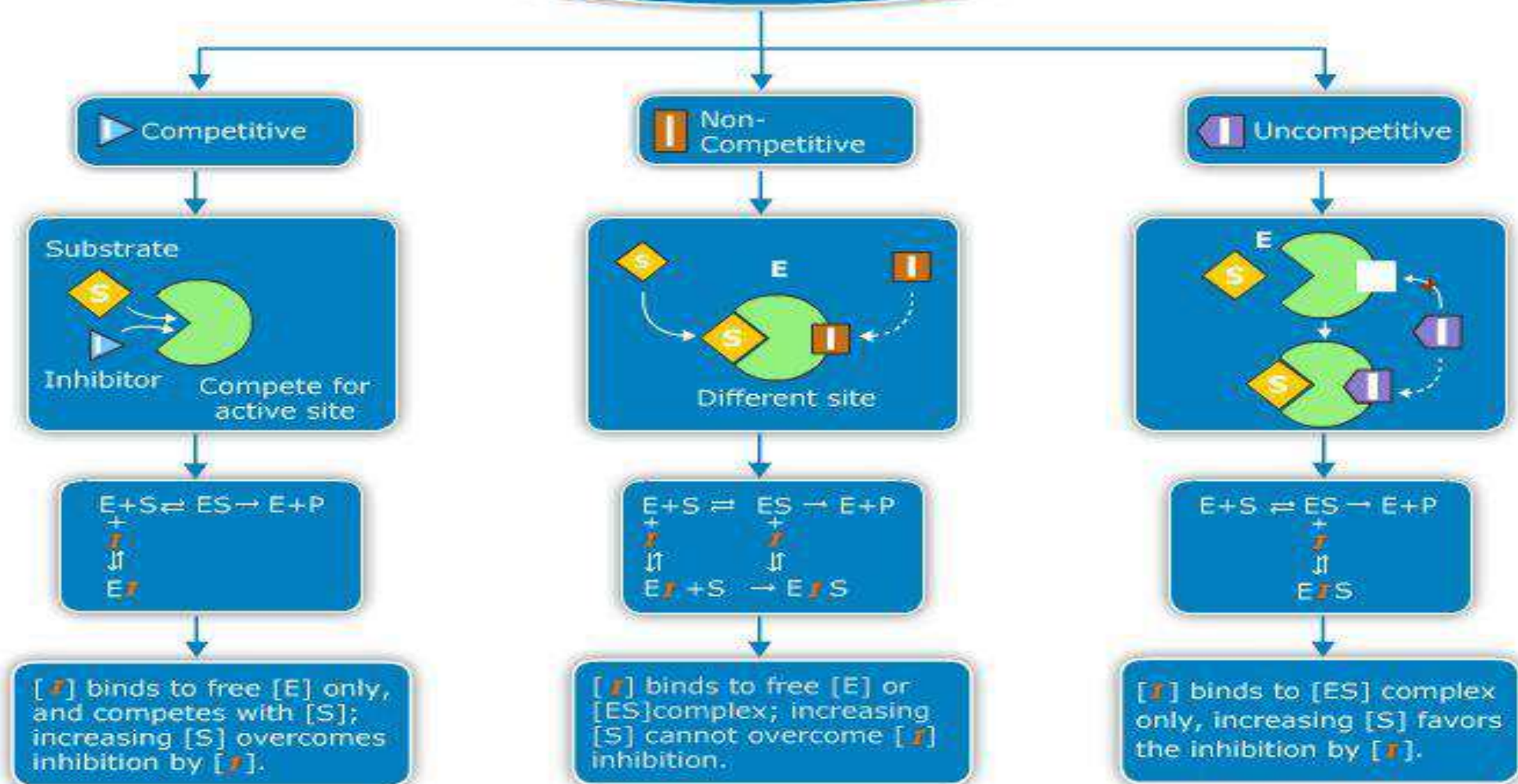
- The inhibitor does combine at the active site but does not prevent the binding of one of the substrates (and vice versa)
- In this type of inhibition **V_{max} as well as K_m both are decreased**



- block processes beyond ES formation, the inhibitor binds to and stabilizes the ES complex, it makes it “**more difficult** for **S to dissociate**” or be converted to product, increasing enzyme affinity for S and so reducing substrate K_m
- This **mode of action** is attractive for drug design as the inhibitors bind to the active enzyme target (substrate present)

Enzyme inhibition...

Enzyme Inhibition (Mechanism)



Enzyme Inhibition (Plots)

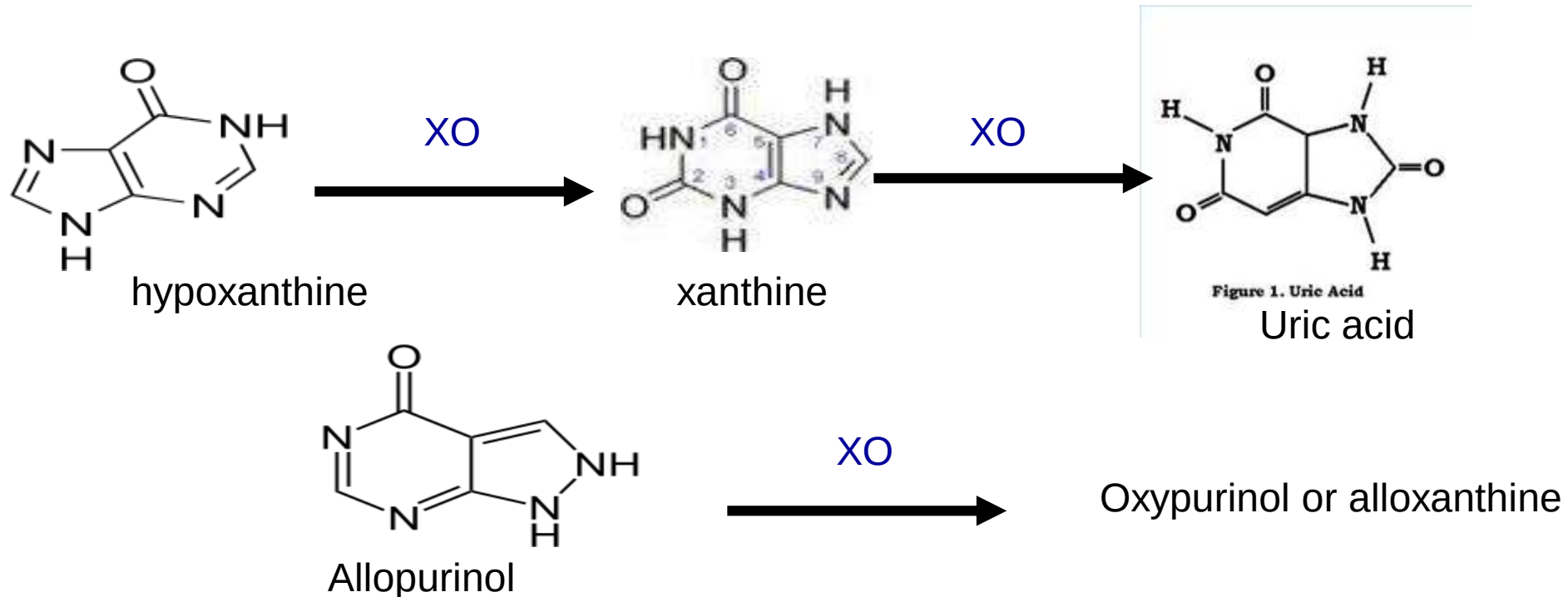
	▶ Competitive	■ Non-competitive	▣ Uncompetitive
Direct Plots	v_o V_{max} K_m K_m' $[S], \text{mM}$	v_o V_{max} V_{max}' $K_m = K_m'$ $[S], \text{mM}$	V_{max} V_{max}' K_m' K_m $[S], \text{mM}$
Double Reciprocal	V_{max} unchanged K_m increased Intersect at Y axis $1/v_o$ $1/V_{max}$ $1/K_m$ $1/[S]$	V_{max} decreased K_m unchanged Intersect at X axis $1/v_o$ $1/V_{max}$ $1/K_m$ $1/[S]$	Both V_{max} & K_m decreased Two parallel lines $1/v_o$ $1/V_{max}$ $1/K_m$ $1/[S]$

Application of Enzyme Inhibition

- This principle is applied in drug synthesis and at least half of modern drugs act as enzyme inhibitors
- Competitive inhibition is applied in the treatment of gout
- The drug **allopurinol** is structurally similar to **hypoxanthine**, which is a substrate of the enzyme *Xanthine oxidase*
- Gout is caused due to increased oxidation of purine bases that produce uric acid, which forms an insoluble sodium salt that precipitates between joints and causes inflammation

Treatment of gout

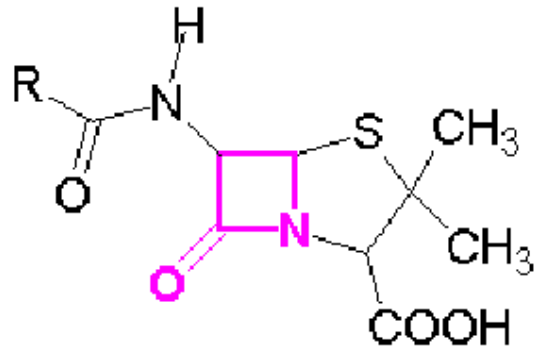
- Allopurinol is a metabolic analog of hypoxanthine. It, therefore, competes on the active site of xanthine oxidase (**XO**) and effectively lowers the rate of uric acid formation
- The allopurinol produces a soluble product **oxypurinol** blocking the formation of uric acid which forms a non-soluble **sodium urate** that precipitates between joints causing the disease **gout**



Mode of action of penicillin

- Penicillin kills susceptible bacteria by specifically inhibiting a transpeptidase that catalyzes the final step in cell wall biosynthesis
- Penicillin is a structural analog of the acyl-D-alanyl-D-alanyl terminus of newly synthesized peptidoglycan, that forms the cell wall of bacteria
- Penicillin **irreversibly acetylates** the **active site (serine)** of the cell wall synthesizing enzyme **transpeptidase**
- It has selective toxicity to bacteria without affecting human cells because protein synthesis in eukaryotes and prokaryotes is different

Structure of penicillin and its derivatives

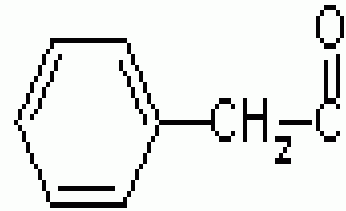


- Most important part of all penicillins is the β -lactam- ring

- The derivatives have different substituents (-R) with specific functions

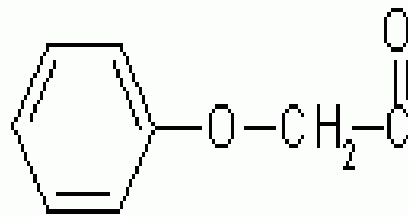
Antibiotic penicillin:

“**Transpeptidase inhibition**”.
causes the **cell wall to lose strength** and the **bacteria to burst**.



→ **Penicillin G:**

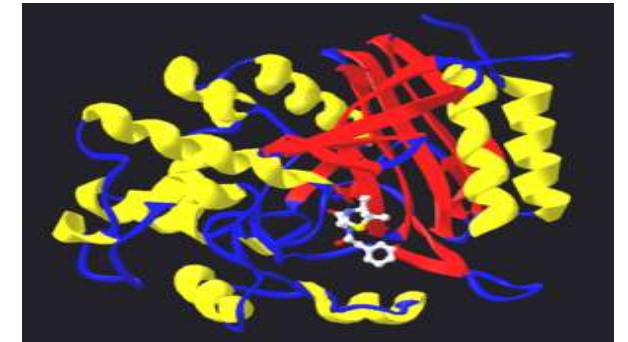
- given parenterally
- treatment: meningitis, syphilis, pneumonia



→ **Penicillin V:**

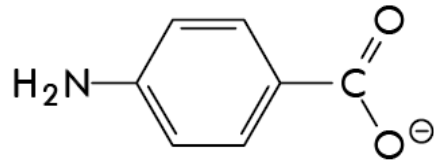
(more acid-stable),

- The orally active form
- less active than penicillin G
- treatment: tonsillitis, skin infections

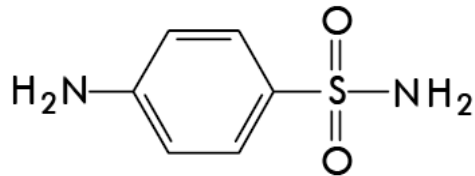


Examples: Sulpha Drugs

- Bacteria must synthesize folic acid from p-aminobenzoate for growth and division



p-Aminobenzoate $\longrightarrow$ folic acid (DHF)



Dihydropteroate Synthase

Sulfanilamide $\longrightarrow$ No product

Humans obtain folic acid from foods: cereals and vegetables. No requirement for **Dihydropteroate Synthase**, Sulfanilamide is very toxic to bacteria, mildly toxic to the patient

Examples: Anti tumor Drugs

- **Methotrexate** an anti-tumor drug, is an analog of dihydrofolate (DHF). It, therefore, inhibits dihydrofolate reductase catalyzed reaction in the formation of tetrahydrofolate (THF)

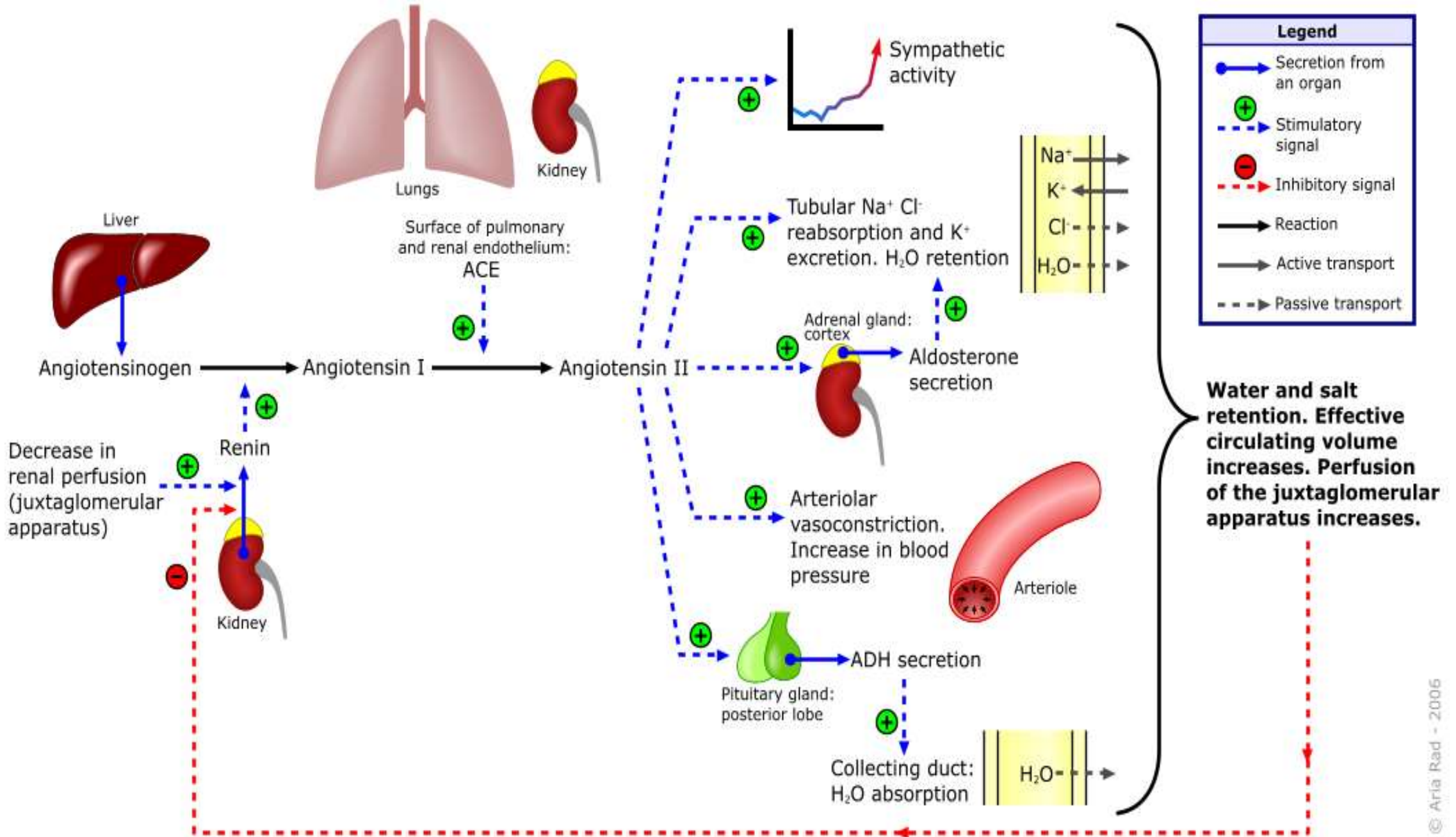


- THF is necessary for the conversion of dUMP (uridine monophosphate) to TMP (thymidine monophosphate) in DNA biosynthesis.
- In the presence of methotrexate, the fast-growing tumor (cancer) cells die out

Mechanism of action of antihypertensive drugs

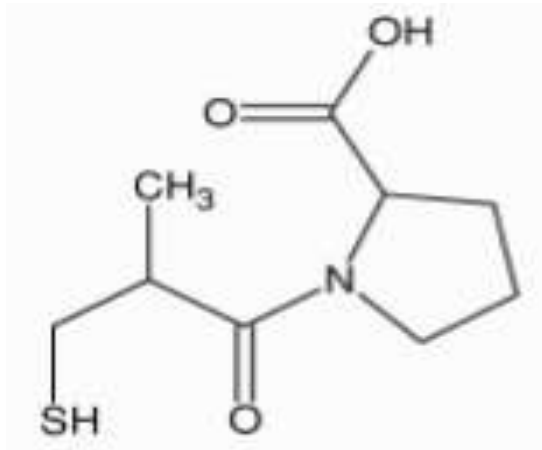
- Angiotensin-converting enzyme (ACE) inhibitors like **captopril**, **enalapril**, and **lisinopril** lower blood pressure by blocking the enzyme that cleaves angiotensin I to form the potent vasoconstrictor, angiotensin II
- Renin released from kidneys, which helps convert angiotensinogen (released from the liver) to be converted to angiotensin I
- Angiotensin I is then converted to angiotensin II by ACE released by the lungs and kidneys
- This signals adrenal glands to produce aldosterone which is activated in response to falling in blood pressure

Renin-angiotensin-aldosterone system



Mechanism of action of ACE inhibitors

- With ACE inhibitor use, the effects of angiotensin II are prevented, leading to decreased blood pressure
- This action of ACE inhibitors is utilized in the prevention of HTN renal failure



Structure of captopril

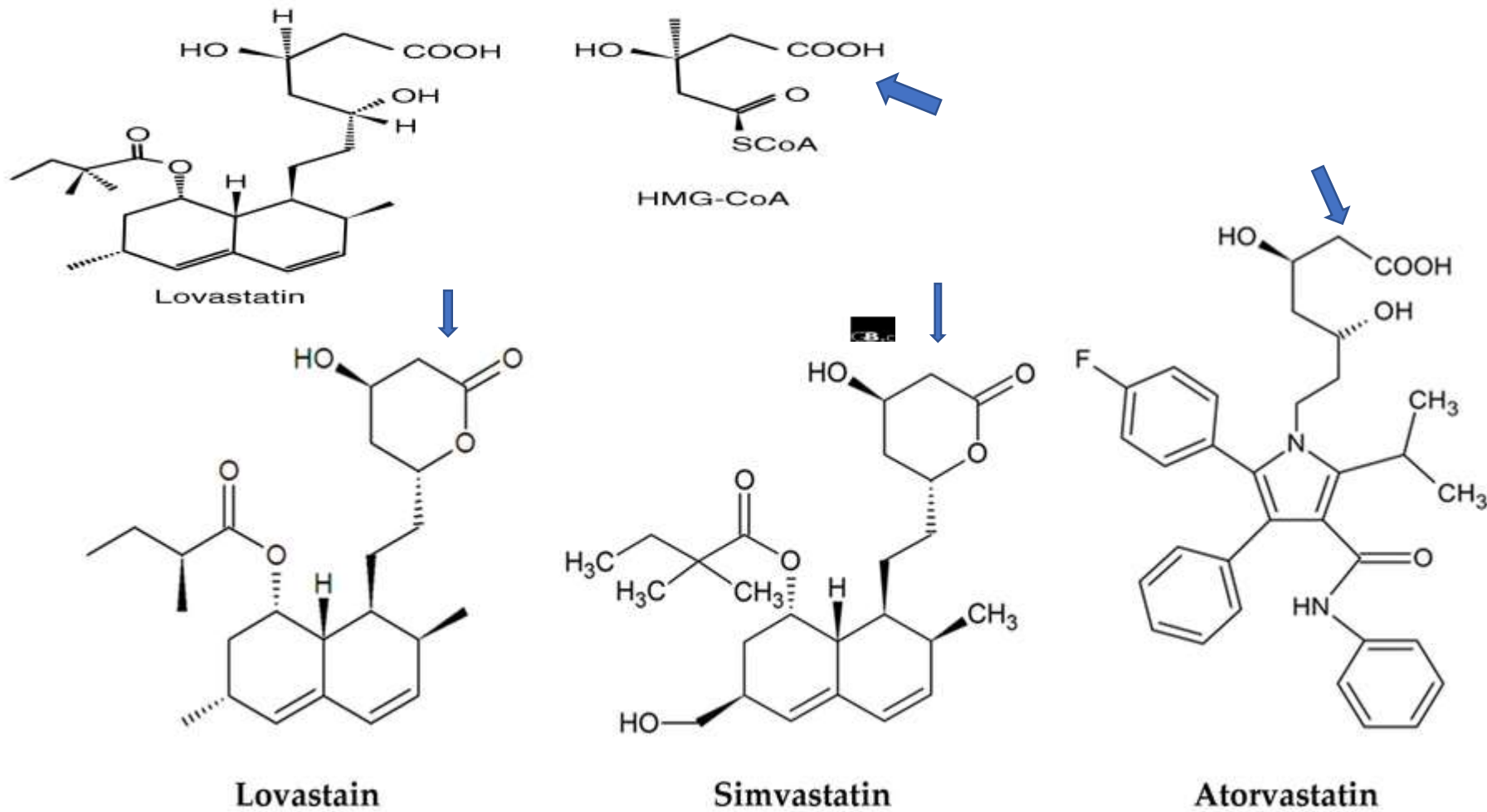
- **Captopril** competes with angiotensin I for binding at the angiotensin-converting enzyme, blocking the conversion of angiotensin I to angiotensin II
- As angiotensin II is a **vasoconstrictor** and a negative feedback mediator for renin activity, lower angiotensin II levels result in a decrease in blood pressure

Cholesterol lowering drugs and mode of action

- **Statins** are a class of drugs that lower the level of cholesterol in the blood by reducing the production of cholesterol by the liver
- Statins block the enzyme in the liver that is responsible for making cholesterol
- Cholesterol is critical to the normal function of every cell in the body. However, it also contributes to the development of atherosclerosis, a condition in which cholesterol-containing plaques form in arteries
- These plaques block the arteries and reduce the flow of blood to the tissues resulting in ischemia and heart attack

Cholesterol lowering drugs and mode of action

- Statins are used for preventing and treating atherosclerosis that causes chest pain, heart attacks, strokes, and intermittent claudication in individuals who have or are at risk for atherosclerosis
- Competitive inhibitors like statin drugs Lovastatin, atorvastatin (Lipitor), and simvastatin (Zocor) **inhibit HMG-CoA reductase-** the enzyme catalyzing the committed step in cholesterol synthesis
- These drugs **reduce the production** of bad cholesterol by the liver (**LDL-Cholesterol**)



Due to structural analogy with HMG-CoA, these compounds competitively inhibit the enzyme HMG-CoA reductase

Application of Enzyme Inhibition in poisoning

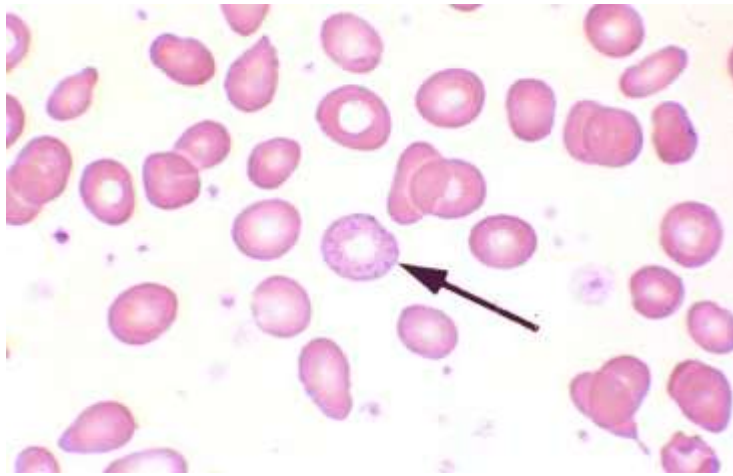
- **Organophosphate** insecticides like *malathion*, *parathion*, *diazinon*, etc are mostly derived nerve gases and can damage and inhibit useful enzymes
- Some **insecticides** that have a **neurotoxic effect** irreversibly bind to the active site of the enzyme **acetylcholinesterase** (an enzyme that cleaves the neurotransmitter acetylcholine)
- Noncompetitive inhibitors like Pb (heavy metal) inhibit the enzyme **ferrochelatase** that chelates Fe^{2+} to protoporphyrin to form heme
- The Pb attaches to the sulfhydryl residues on the enzyme protein and inhibits the enzyme

Application of Enzyme Inhibition in poisoning

- **Organo Phosphorus** → Insecticide binds **irreversibly to “acetylcholinesterase”**
(**interrupt neurotransmission**)
 - ✓ **SARIN (nerve gas)** → is an **extremely toxic** synthetic organophosphorus compound that is an **irreversible inhibitor of ACEs**, a “**chemical weapon**”
toxic at low concentration
 - ✓ It affects **acetylcholine** and **inhibits its degradation** at the **neuromuscular junction** (acetic acid and choline) treated with **an antidote**

Lead poisoning

- Laboratory testing reveals an elevated free erythrocyte protoporphyrin (FEP) level, (inhibit rRNA degradation aggregated, basophilic stippling of RBCs)
- Ferrochelatase allosterically inhibited by Pb, is the “last reaction in heme biosynthesis” used to incorporate Fe^{2+} to pyrrole ring, also ALA dehydratase
- caused by occupational and environmental lead exposure and can cause damage to various body systems (through ROS) anemia
 - ✓ children < 3 years at greatest risk and most likely to put items containing lead into their mouths (neurotoxin), cognitive, memory, and Bhr problem
- Medication may involve chelation therapy, The “chelation agents” succimer and penicillamine are “given orally”, whereas dimercaprol and edetate (EDTA) calcium disodium ($CaNa_2$ EDTA) are administered “parenterally (IV)”. They increase urinary excretion of Pb.



basophilic stippling of RBCs



lead lines (mouth) on gingivae at the base of the teeth, peripheral neuropathy

Lead poisoning

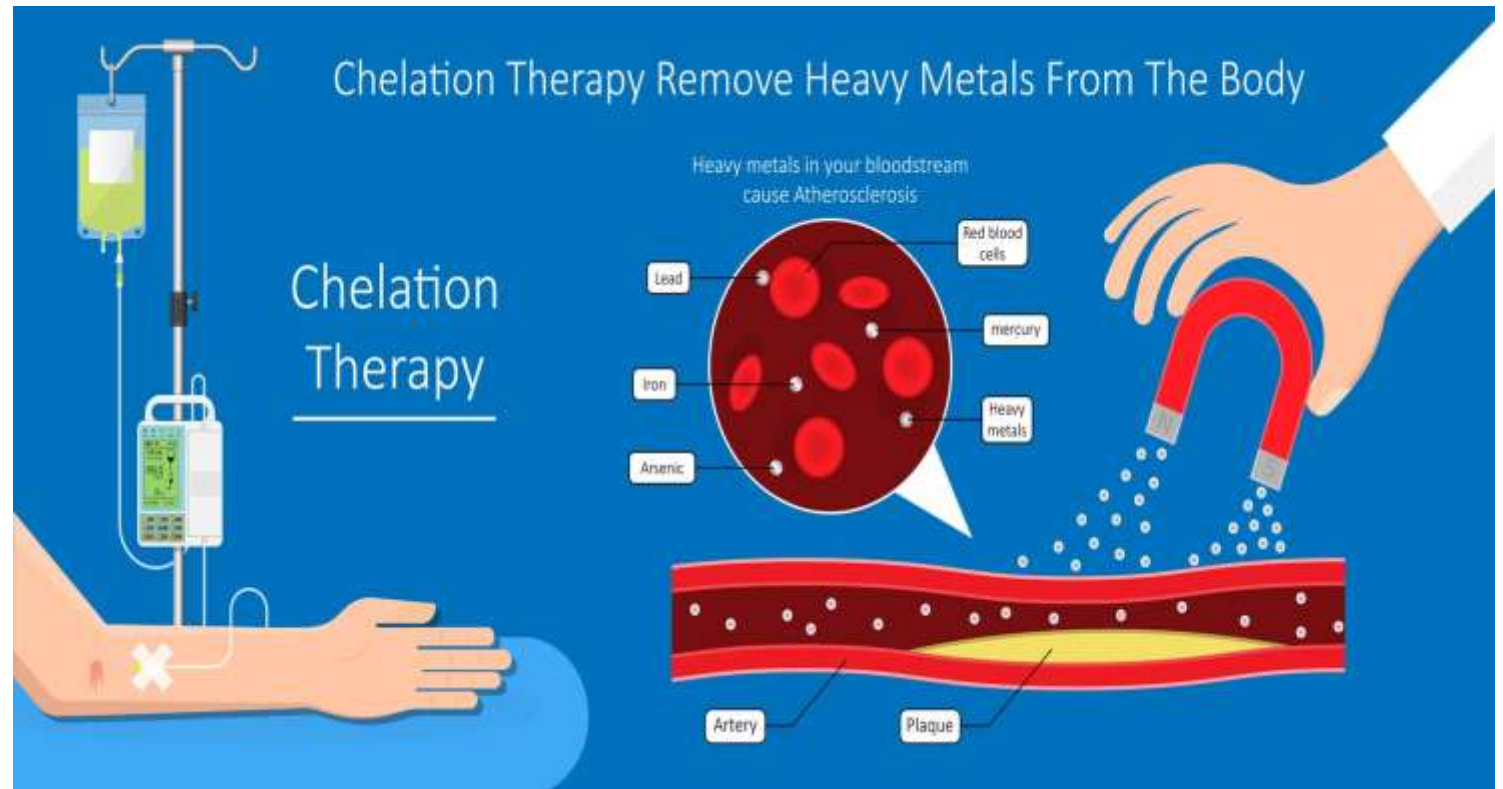
Treatments:-

succimer

penicillamine

dimercaprol

EDTA



Lead poisoning...

- Impairments induced by lead exposure during the lifespan,
- irreversible alterations in the structure and function of the CNS, reduce cytochrome Oxidase, and ATP depletion cause cellular death, Decreases GSH
- Chelation agents contain sulfhydryl groups that bind or chelate lead, and the resulting soluble complex is excreted either renally or hepatically.
- widespread use and exposure to different sources of this heavy metal, Pb intoxication is recognized as a public health problem in many countries (especially in Africa) ...Water supply ??

Lead poisoning

Cognitive and behavioral changes

Pb: 0.1–30 $\mu\text{g}/\text{dL}$ **Of Lead poisoning**

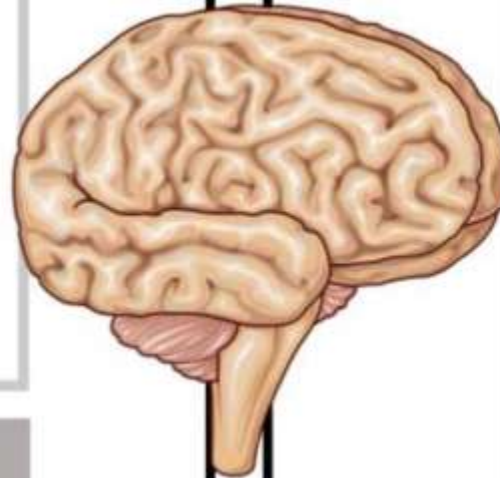
Prenatal and early life

- Sleep problems
- Motor alterations
- Cognitive dysfunction
- Aggressiveness
- Behavior disorders
- Criminal behavior
- IQ deficits
- Visual dysfunction
- Depression
- Attention disturbances
- $\downarrow$ verbal/language

Pb: 10–120 $\mu\text{g}/\text{dL}$

Adulthood

- Cognitive dysfunction
- Decreases short-term memory
- Motor control alterations



Morphologic, cellular and molecular alterations

- $\downarrow$ head circumferences reduction
- $\downarrow$ number and rate of neocortical synapses
- Swollen synapses
- $\downarrow$ reduced activity in cortical networks
- Oxidative environment
- Mitochondrial dysfunction
- Energy metabolism alteration
- Calcium signaling alteration
- Autophagy disruption
- Neurotransmission alterations
- Long-term potentiation impairment

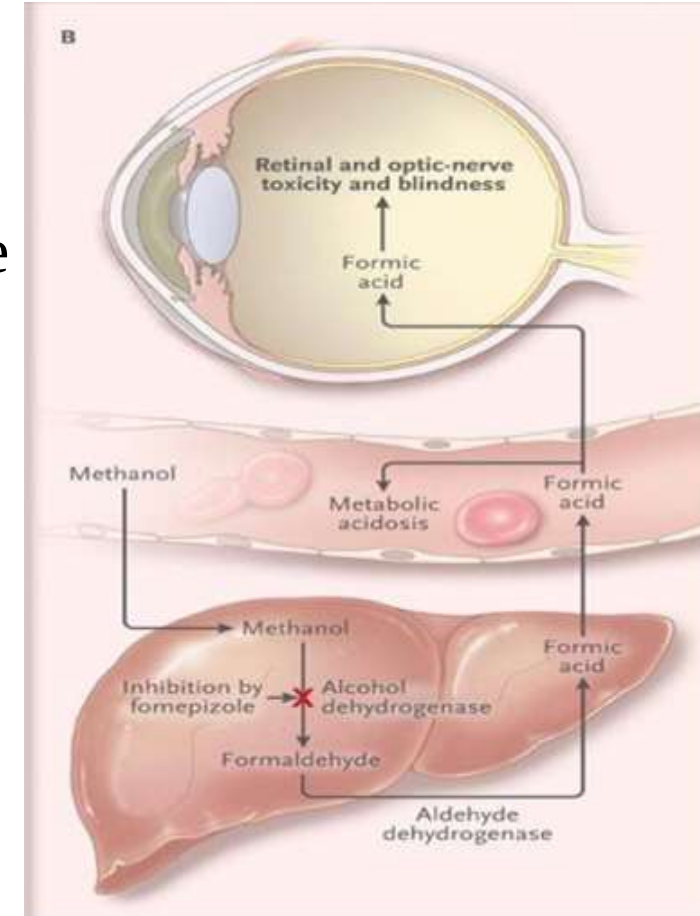
Methanol poisoning

- In humans Ethanol $\longrightarrow$ Aldehyde + Acetate

Alcohol Dehydrogenase

Methanol $\longrightarrow$ Formaldehyde

- Formaldehyde: causes tissue damage, blindness
- Therapy: intravenous infusion of ethanol
 - competes with methanol as a substrate for alcohol dehydrogenase
 - Slows down the formation of formaldehyde
 - Most methanol secreted harmlessly in urine

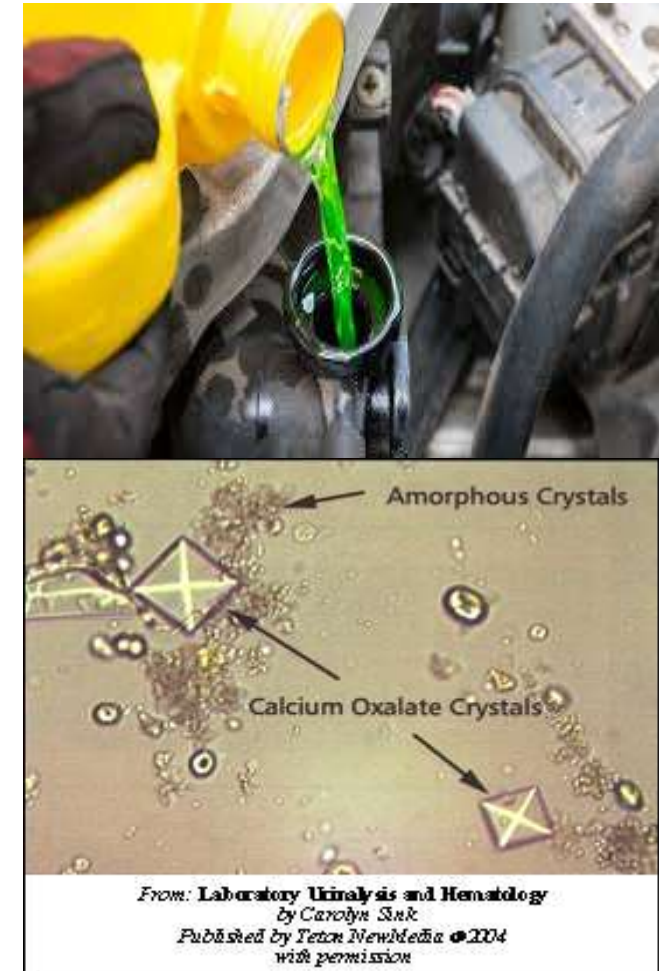
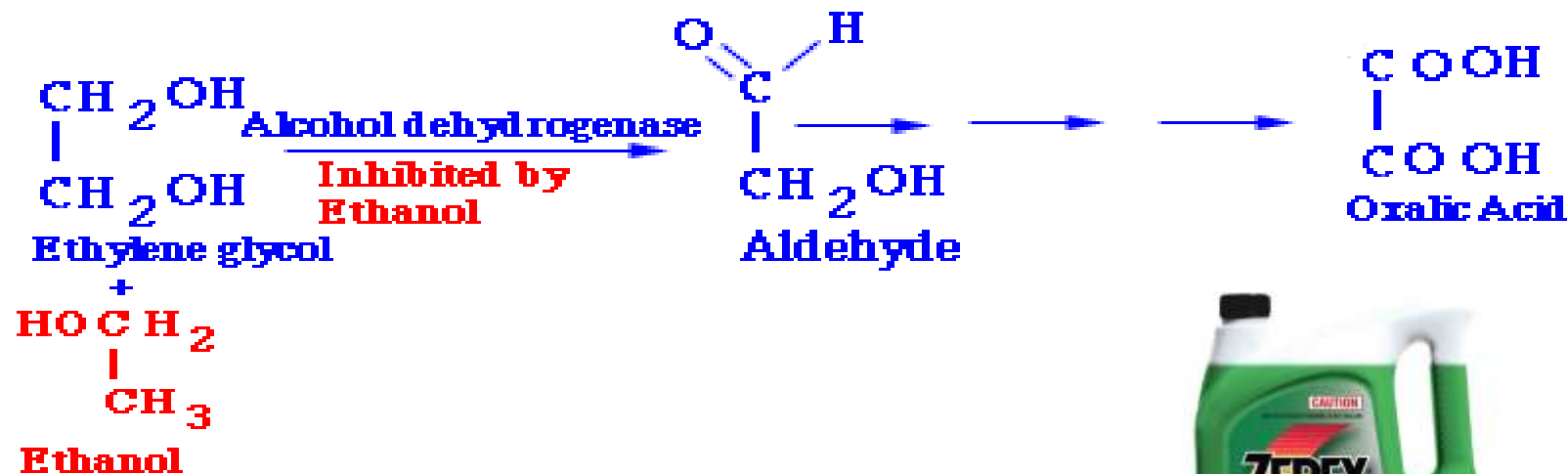


- Smelling of methanol causes blindness (very dangerous and toxic), and attacks proteins, due to its product formaldehyde
- IV administration of Ethanol is used to treat methanol poisoning, by competing for the common enzyme alcohol dehydrogenase
- Fomepizole (treatment) drug is an antidote for methanol and ethylene glycol poisoning
- Alcohol dehydrogenase catalyzes the oxidation of ethanol to acetaldehyde further oxidized to acetic acid(acetate ion), normal metabolite

Ethylene Glycol Poisoning

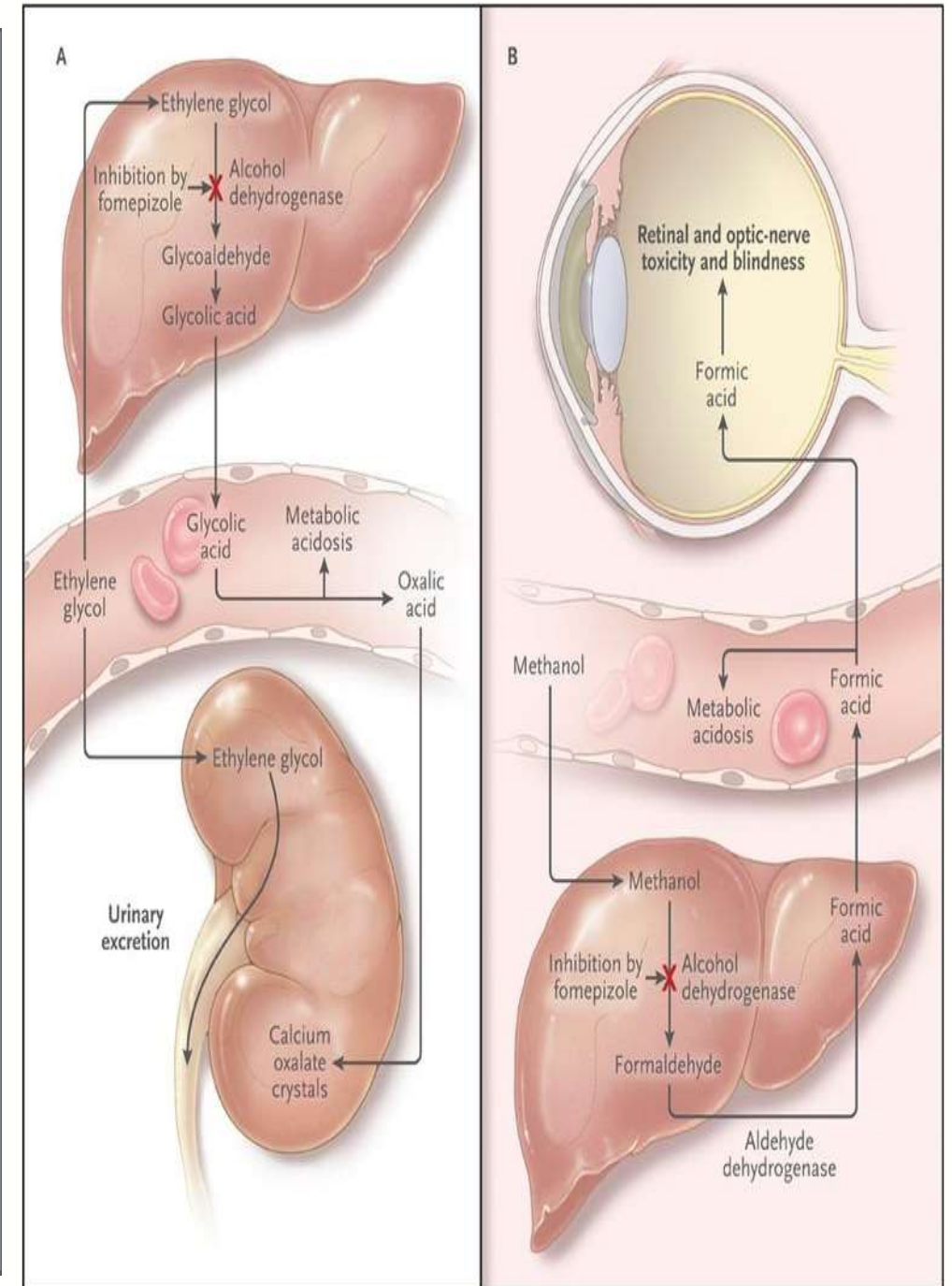
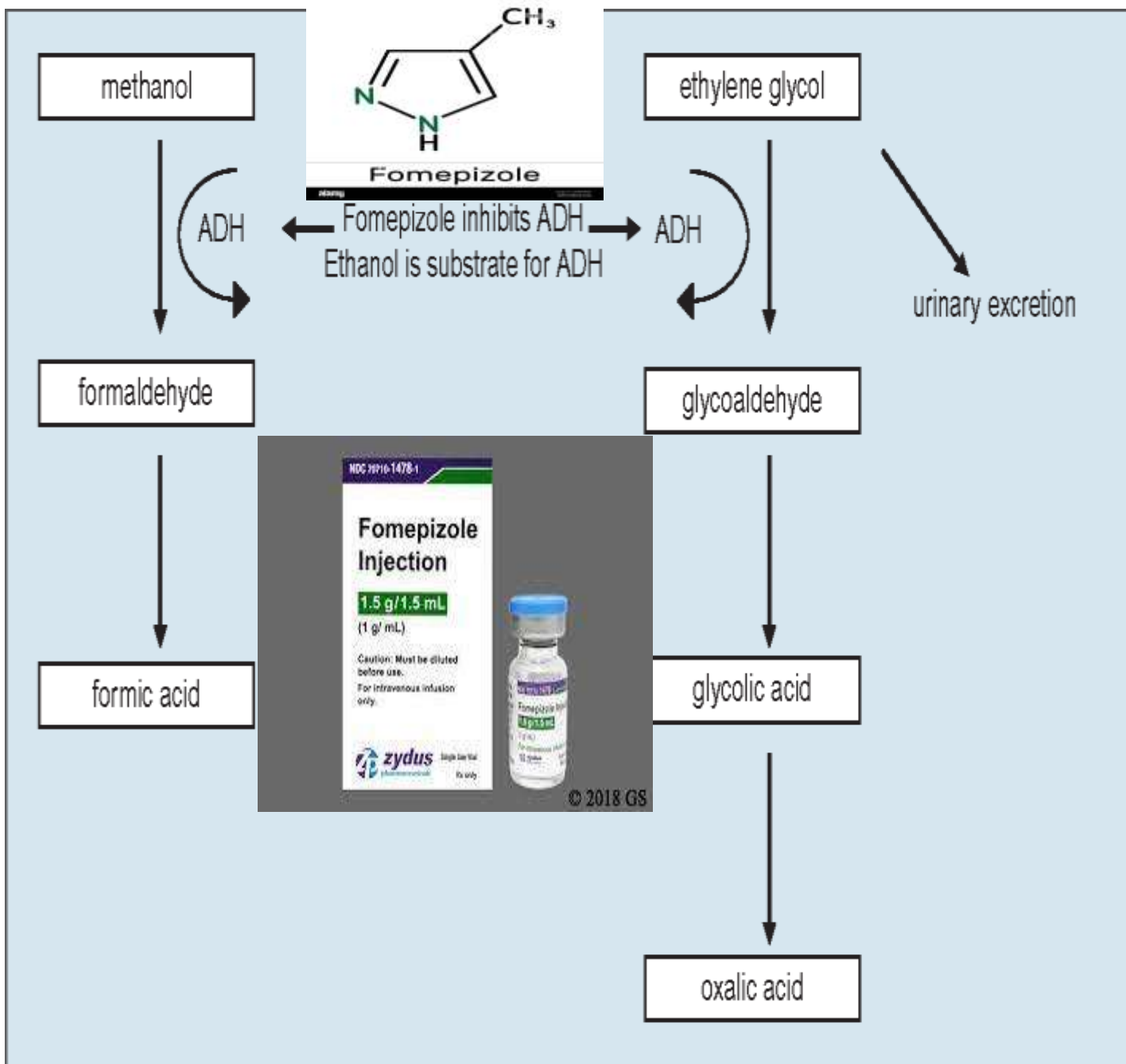
- People die from the ingestion of ethylene glycol because the kidneys are severely damaged by the deposition of its oxidation product of **oxalate crystals**

Formation of oxalic acid from ethylene glycol is inhibited by ethanol.



From: *Laboratory Urinalysis and Hematology*
by Carolyn Sisk
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- Ethylene glycol ($C_2H_6O_2$) is toxic alcohol, “sweet test”, found in industrial solvents (automobil, coolant) and as **an antifreeze** (lower freezing point of solvents), polyester fiber
- Poisoning caused by **drinking** propylene glycol (CNS) depressant (vomiting and abdominal pain are early symptoms)
- Oxalate diamond shape crystal in the urine (microscopy) “**kidney stone**”
- Treatment → Ethanol is also a good competitive inhibitor, of Ethylene glycol for ALDH (structural similarity), Fomepizole (antidote) Also called **Antizole**

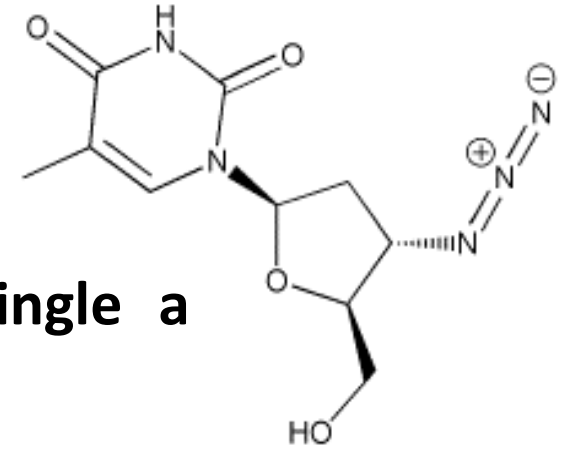


Antiviral drugs

- Many HIV-AIDS drugs act as enzyme inhibitors (integrase inhibitors, inhibitors of reverse transcriptase, protease inhibitors, fusion inhibitors, etc)
- Most of these drugs are designed to inhibit useful enzymes in the life cycle of the viruses
- Intervention in the life cycle brings about minimizing the replication and multiplication of the viruses

Reverse transcriptase inhibitors

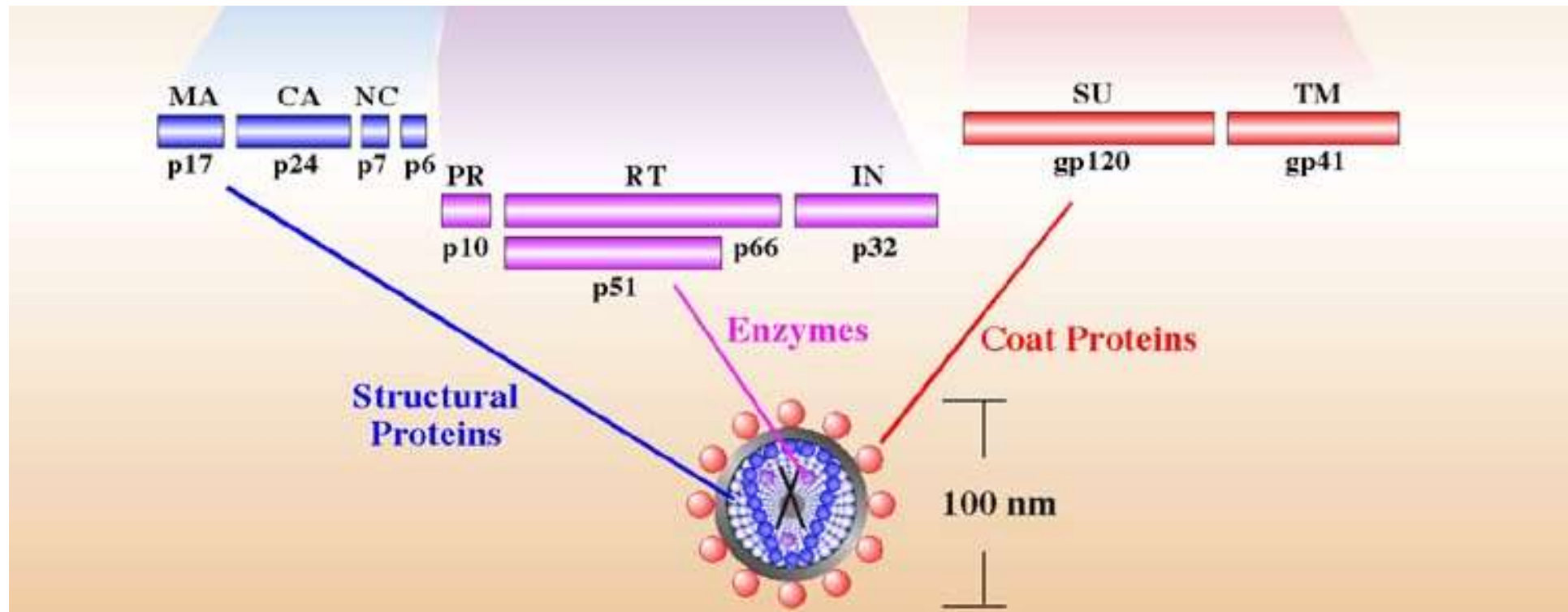
- **AZT**: analog of Thymidine (T) in DNA
- Zidovudine (ZDV), also known as **azidothymidine (AZT)** ...single a
membered ring with an azide group



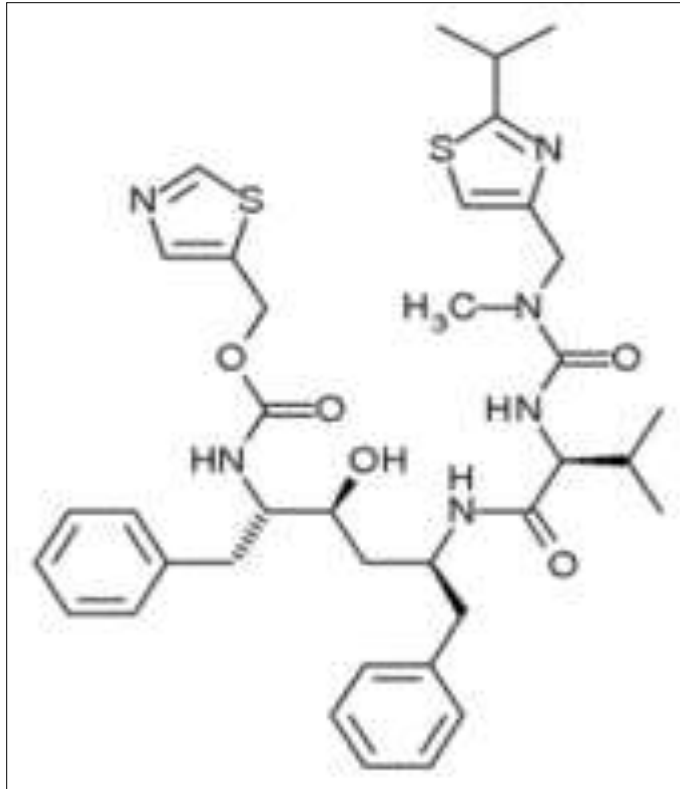
- Competes with T for incorporation in new viral DNA
- Jams up viral replication machinery (reverse transcriptase)
- 1st drug approved for the treatment of AIDS and HIV infection
- Side effect: can also bind to the human cell replication machinery
 - But more effective against viral replication
 - Because the virus needs to replicate much faster than human cells

Example: **HIV anti-viral drugs** **Protease inhibitors**

- All HIV proteins are synthesized as 3 long polypeptide chains
 - Protease (PR) releases itself and then cuts polypeptides into individual proteins

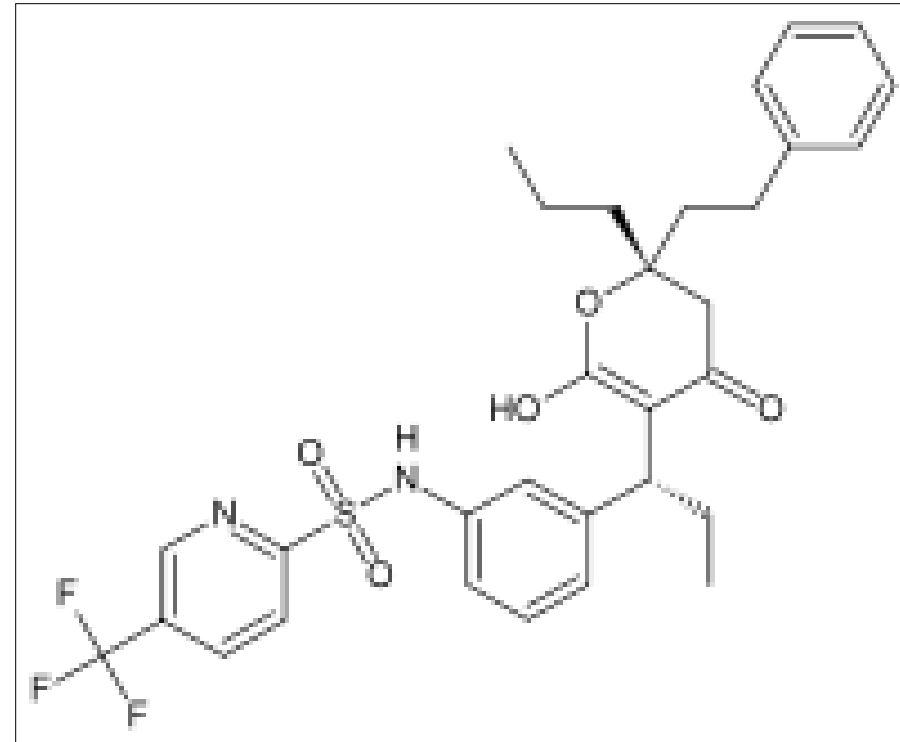


Protease inhibitors



Ritonavir

Peptide-based protease inhibitor



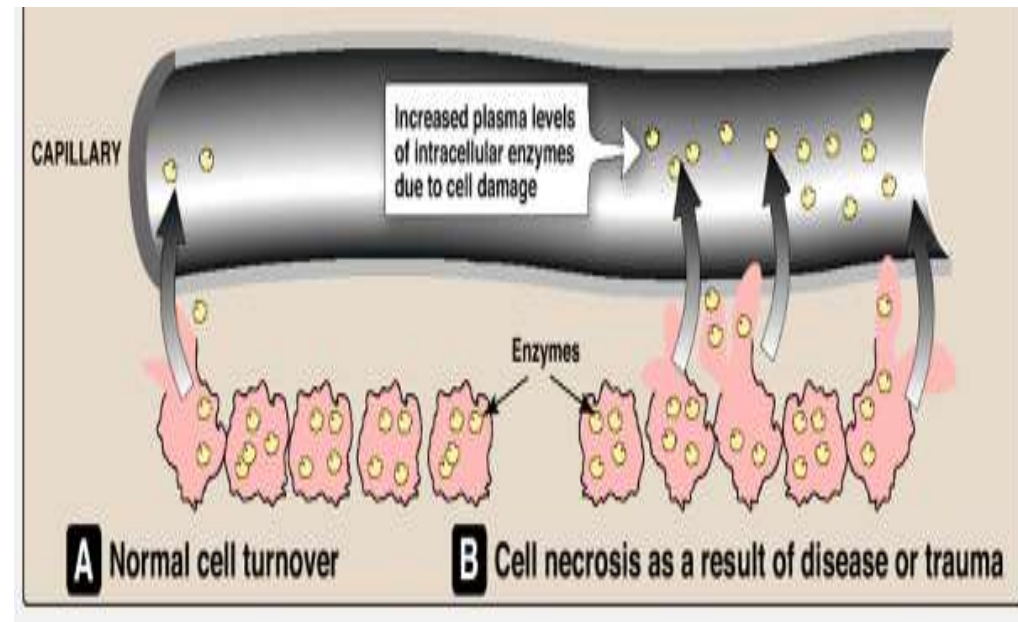
Tipranavir

Non-peptide based protease inhibitor

Protease inhibitors → inhibit viral protein synthesis (the virus use host machinery (ribosome) for translation)

Enzymes in Clinical Diagnosis

- The concentrations of tissue and plasma enzymes are maintained at some levels under normal conditions
- During tissue damage the enzymes that were functional in the tissue leak out increasing the plasma enzyme concentration
- This is used for the diagnosis of organ and tissue damage and diseases



Enzymes almost **always function** in the **tissues (intracellular)** and have no physiologic use in the plasma (**plasma nonfunctional**) steady-state

- In **healthy** individuals, the **levels of these enzymes are fairly constant** (the rate of **release** is **balanced**(equals) with the **rate of elimination**), **steady-state**
- **Tissue damage or necrosis** resulting from injury or disease is generally accompanied by **increases in the levels of several nonfunctional plasma enzymes**

Many diseases that cause tissue damage result in an **increased release of intracellular enzymes** into the plasma.

- **Lab diagnosis** (use a **serum** (plasma without clotting factor)

Plasma enzymes

plasma functional
Functional

plasma functional

present at all times in the circulation and perform a physiologic function in the blood (clotting factors, lipoprotein lipase to release fatty acid up taken by cell for energy or storage) effective in plasma (not intracellular)

****For example:-**

Non functional

Non- functional in plasma

No function in plasma (intracellular), released in a higher amount to plasma when tissue damage (diagnostic value)

1-Lactate dehydrogenase(LDH)

2-Creatine kinase(CK)



Isoenzymes

3-Alanine transaminase(ALT)

4-Aspartate transaminase(AST)

5-Alkaline phosphatase

6-Acid phosphatase

7-Amylase & Lipase enzymes

Isoenzymes (or isozymes) are a group of enzymes that catalyze the same reaction but have different enzyme forms and catalytic efficiencies (tissue location and kinetic). **LDH** has **5** Isoenzyme and **CK** has **3** isoenzyme

Information from enzyme measurement in serum:-

- Presence of disease
 - Organs involved
 - Etiology /nature of disease: differential diagnosis
 - Extent of disease-more damaged cells-more leaked enzymes in blood
- ✓ **Enzyme activity is expressed in an International unit (IU):** the amount of enzymes that catalyze the conversion of the substrate to a product per minute.

What are the clinically important enzymes?

- Liver enzyme:
 - Alanine aminotransferase (ALT; old name SGPT) is normally found largely in the liver, normal range 7 to 55 units per liter (U/L)
- Liver, heart, muscle, kidney, and brain:
 - Aspartate aminotransferase (AST; old name SGOT) is normally found in a diversity of tissues, N.R 8 to 48 U/L
- Bone enzyme:
 - Alkaline phosphatase (ALP), N.R 40 to 129 U/L
- Brain, heart, muscle:
 - Creatine kinase (CK; old name CPK), N.R 22 to 198 U/L
 - Lactate dehydrogenase (LD; old name LDH) N.R 140 to 280 U/L

Enzymes in Clinical Diagnosis...

Enzymes can be used in the diagnosis & treatment of certain diseases

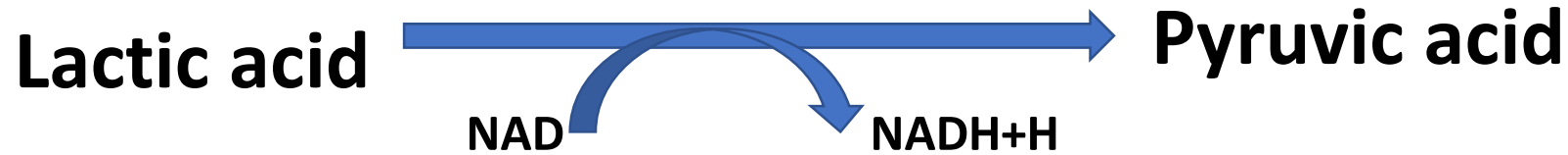
- Lactate dehydrogenase (LDH) is normally not found in high levels in the blood, as it is produced in cells
 - Increased levels of LDH in the blood indicate myocardial infarction (MI) (Heart attack)
 - Tissue plasminogen activator (TPA) activates the enzyme plasminogen that dissolves blood clots
 - Used in the treatment of MI

TPA is a protein involved in the **breakdown of blood clots**. Improvement in left ventricular function, administered by IV

TPA is a **serine protease** found in endothelial cells, the cells that **line the blood vessel**



Lactate dehydrogenase(LDH)



- LDH is elevated in **myocardial infarction**, and **blood disorders**.
- It is a **tetramer protein** and made of **two types of subunits** namely **H** = Heart, and **M** = skeletal muscle.
- It exists as **5 different iso-enzymes** with various combinations of H and M subunits.

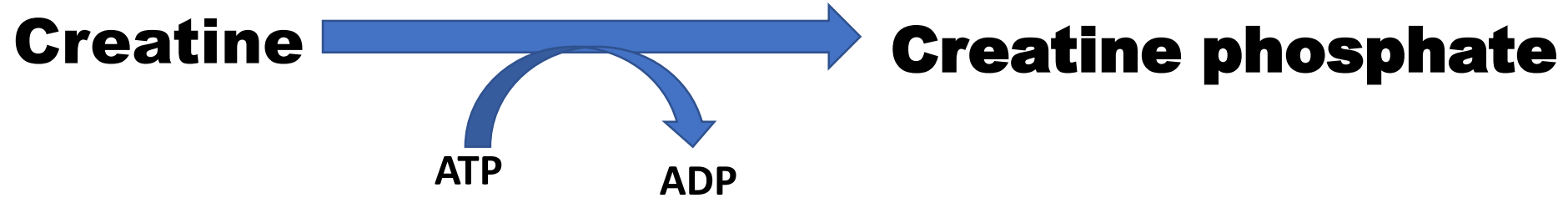
The heart requires the highest oxygen supply (for oxidative phosphorylation)

Heart Energy source → main source is fatty acid oxidation, lesser carbohydrates, under certain circumstances (amino acid, lactate, and ketone body even acetate) can be oxidized

❖ The composition of the enzyme includes Muscle and heart subunits

Isoenzyme name	Composition	Composition	Present in	Elevated in
LDH1	(H ₄)	HHHH	Heart, RBC	myocardial infarction
LDH2	(H ₃ M ₁)	HHHM	Heart, RBC	myocardial infarction
LDH3	(H ₂ M ₂)	HHMM	lungs and spleen	leukemia
LDH4	(H ₁ M ₃)	HMMM	lungs and spleen	viral hepatitis
LDH5	(M ₄)	MMMM	Skeletal muscle, Liver	Skeletal muscle and liver diseases

Creatine kinase (CK)



(Phosphocreatine – serves as energy reserve during muscle contraction)

- Creatine kinase is a dimer made of 2 monomers
- Skeletal muscle contains **M** subunit, Brain contains **B** subunits
- Three different isoenzymes are formed

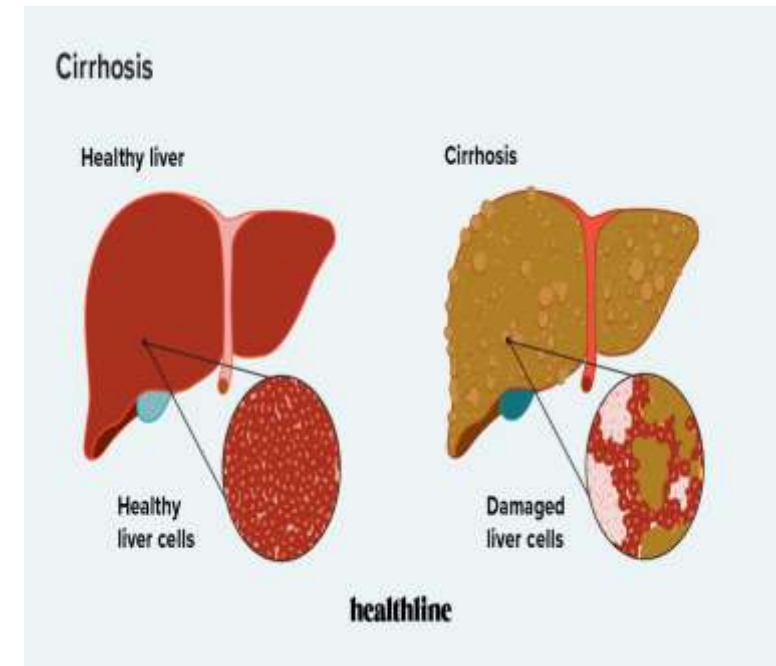
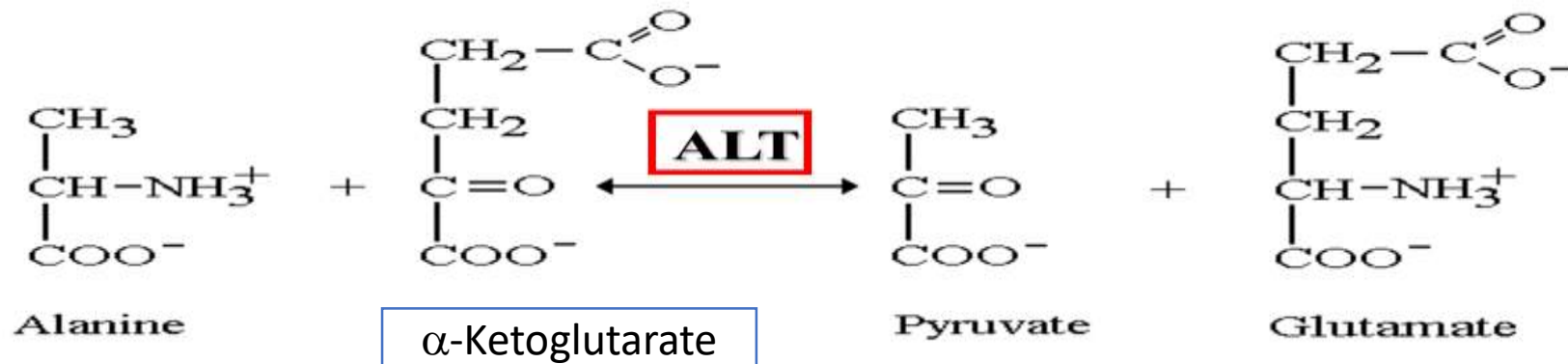
Iso enzyme name	Composition	Present in	Elevated in
CK-1	BB	Brain	CNS diseases brain tumors
CK-2	MB	Heart	Acute myocardial infarction
CK-3	MM	Skeletal muscle	Skeletal muscle diseases

Heart isoenzyme of CK MB and LDH(1 or 2) is useful for the diagnosis of heart disease in addition to ***Troponin*** measurement

Myoglobin, creatine phosphokinase (CK 3), and LDH 5 are the most important biomarkers for indicating muscle damage

Alanine transaminase(ALT)

- It is also called **Serum glutamic pyruvic transaminase(SGPT)**
- This enzyme is found predominantly in hepatocytes
- It increased **liver diseases**
- **Liver-specific biomarkers** (viral hepatitis, cirrhosis, alcohol-related)
- ALT is more specific for liver disease than AST



Alanine glucose cycle “ alanine from muscle gets transported to the liver for glucose formation” gluconeogenesis

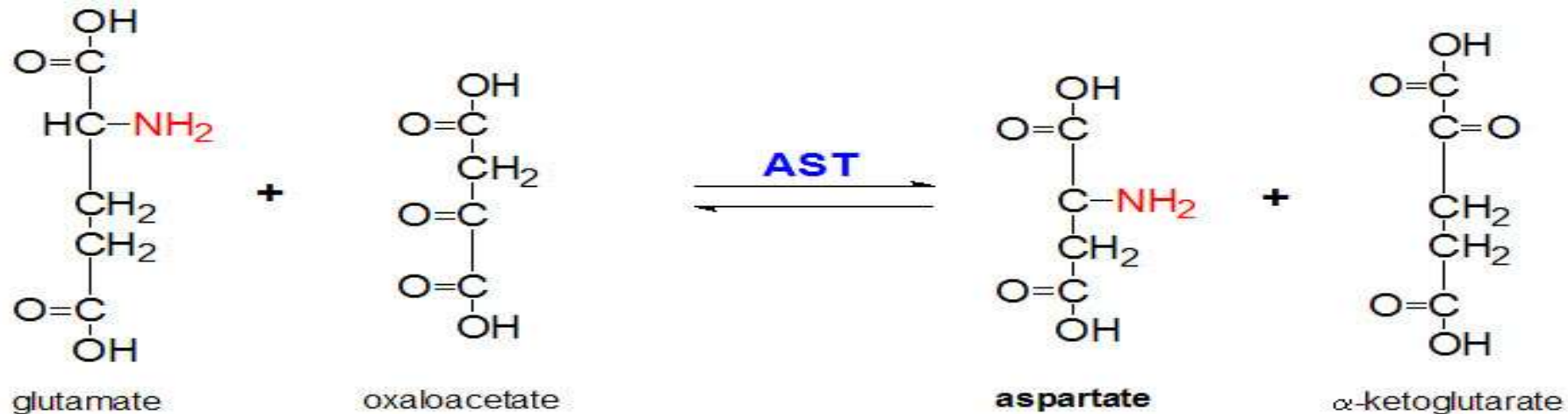
Aspartate transaminase(AST)

*It is also called

Serum glutamic oxaloacetic transaminase(SGOT)

*It increases **liver & heart diseases**

Found in many organs, should conduct along with ALT to assess liver damage



AST → used in amino acid metabolism O.G (TCA intermediate)

Alkaline phosphatase

**It is found in the bone, liver, intestine, placenta



**Increases mainly in bone and liver diseases

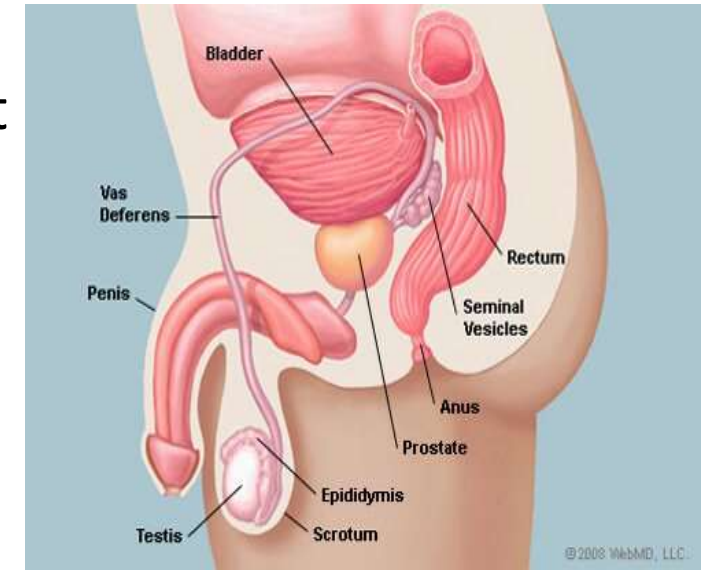
- ✓ Serum bone-specific **ALP** (bone formation), ALP rise in the obstructive liver (**Obstructive jaundice**)

Acid phosphatase

**is found in the prostate, red blood cells, bone, liver, and pancreas

- Mostly used to detect prostatic carcinoma

- ✓ Another most common biomarker for prostate cancer is **PSA (prostate-specific antigen)**, with higher sensitivity than acid phosphatase



**Amylase and lipase enzymes increase in diseases of the pancreas as acute pancreatitis

- ✓ **Pancreatic lipase** exhibits optimal activity under **alkaline conditions** and hydrolyzes **triglycerides** to **fatty acids** and **glycerol**,
- ✓ **Lipase** is more **specific** than **amylase** for the diagnosis of "**pancreatitis**"

Enzyme	Origin	Major diagnostic use
Lactate dehydrogenase (LDH)	Heart, liver, muscle, RBC	Myocardial infarction
Creatine kinase (CK)	Skeletal Muscle Heart Muscle Brain	Muscle disease Myocardial infarction
Alanine aminotransferase (ALT) Serum glutamate-pyruvate transaminase (SGPT)	Heart and Liver	Viral hepatitis
Aspartate aminotransferase (AST) Serum glutamate-oxaloacetate transaminase (SGOT)	Heart and Liver	Liver and heart disease
Alkaline phosphatase (ALP)	Hepatobiliary tree Bone, placenta	Liver and bone disease
Acid phosphatase (ACP)	Prostate, erythrocytes	Prostate cancer
Amylase	Pancreas, Salivary glands	Acute pancreatitis
Lipase	Pancreas	Acute pancreatitis

Biochemical markers for cellular organelles and membranes

- **Plasma membranes** → 5'-nucleotidase, Adenylyl cyclase,
- **Endoplasmic reticulum** → Glucose-6 Phosphatase (smooth),
 - ✓ Endoplasmic reticulum enzyme for glycogen breakdown
- **Golgi complex** → galactosyl transferase 4
- **Inner mitochondria membrane** → ATP synthase,
- **Cytoplasm** → lactate dehydrogenase
- **Lysosome** → cathepsin, **Peroxisome** → urate oxidase, cathepsin
 - ✓ Cathepsins are proteases with serine, cysteine, or aspartic acid residues as the nucleophiles, which are vital for digestion, coagulation, immune response, adipogenesis, hormone liberation, peptide synthesis,



Thank you